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Arbitrage-Free Implied Volatility Surface Construction

From Parametric Models to Neural Operators

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Declaration

The work contained in this thesis is my own unless otherwise stated. Generative AI tools were used for language editing, structural suggestions, and assistance with code debugging and pipeline implementation. They were not used to generate research results. I produced or independently verified all mathematical derivations, numerical results, figures, tables, and literature claims, and I take full responsibility for the content of this thesis.

Abstract

The implied volatility surface is the central object of equity derivatives markets, yet it is only ever observed as a sparse, noisy, irregular cloud of option quotes. Reconstructing a complete, smooth, arbitrage-free surface from these observations is an ill-posed inverse problem whose solution feeds pricing, hedging and risk management.

This thesis studies the problem across the full methodological spectrum on a single unified S&P 500 dataset (OptionMetrics, 2018–2025): the parametric benchmark of Gatheral–Jacquier (SVI per slice, SSVI surface-wide, with their complete no-arbitrage theory replicated and validated against the original paper), a derivative-constrained deep smoother in the spirit of Ackerer et al. and Hoshisashi et al. (an SSVI prior multiplied by a neural corrector, no-arbitrage enforced through exact automatic differentiation), and neural operators following Operator Deep Smoothing, which learn the smoothing map itself and apply to unseen days without retraining. All methods share one evaluation protocol: hold-out reconstruction error, a static-arbitrage audit built on the Durrleman condition, robustness to quote sparsity, and behaviour across market regimes.

The thesis's central contribution is methodological and transversal: *soft no-arbitrage penalties are enforced only where they are sampled.* We exhibit a collocation blind spot in which a calendar penalty is *exactly* zero at its training nodes while an independent audit grid reveals a 28% violation rate between them; we trace it to the interaction between exponential maturity spacing and the geometry of calendar arbitrage; we repair it with constraint-specific collocation grids, reducing material violations from 34.4% to 0.0% on a controlled stress test whose validity is itself verified by a two-condition gate; and we then show the same pathology, in a stronger, two-dimensional form that is blind between nodes *and* across days, in the neural operator, where the penalty-grid audit reports near-zero violations while the independent audit flags up to 356 of 386 out-of-sample days as materially butterfly-violating. The Gatheral–Jacquier conditions are used constructively: our SSVI prior is *certified* statically arbitrage-free on its working domain rather than luckily so. Finally, the audit is priced: through the identity $\sigma^2_{\text{loc}} = \partial_\tau w/g$, arbitrage violations become negative local variances (up to 50% of the operator's surface requires repair before a single Monte-Carlo path can be drawn) and negative risk-neutral density mass (up to 230% on some slices), and a residual-economics pipeline shows that surface dirtiness measurably degrades the information content of quote–model residuals. The conclusion is a change of question: not *which model fits best*, but *what a model's constraints actually guarantee, on which set, and at what economic cost when they silently fail.*

Acknowledgements

I am very proud to present this thesis as the culmination of my MSc studies. I would especially like to thank my parents, whose constant support, encouragement and sacrifices have made it possible for me to reach this point.

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All code, data pipelines and figures are available and fully reproducible from the six notebooks (NB01–NB06), available on GitHub.

Contents

Chapter 1

Introduction

1.1 The reconstruction problem

A common problem in equity derivatives is to reconstruct a complete volatility surface from observed option quotes. The market does not publish an implied volatility *surface*; it publishes a few thousand option quotes, with many quotes in some areas, few in others, noisy data, and a grid of strikes and maturities that changes every day. Everything downstream of those quotes, from marking books and computing Greeks to calibrating stochastic and local volatility models, stress testing portfolios, uses not the quotes themselves but a *smoothed and completed surface* built from them. So errors at this stage can affect every model that follows [11].

The construction is constrained because the surface must also satisfy basic no-arbitrage conditions. A model can fit the observed quotes very well and still produce prices that allow a risk-free profit. This is known as *static arbitrage*. The two main forms considered here are butterfly arbitrage across strikes and calendar arbitrage across maturities.

These violations also affect the models built from the surface. Through the Breeden–Litzenberger identity [3], butterfly arbitrage can lead to a negative risk-neutral density. Through Dupire's formula [9], arbitrage violations can also produce negative local variance [11]. Thus there is a clear trade-off between fit and structure. A flexible model can follow noisy market quotes closely but introduce arbitrage.

The literature offers three main families of models, differing by how much of the surface is specified in advance and how much is learned from data:

1. ***Parametric models***, such as SVI and SSVI [12, 14], fit a fixed functional form to each surface. This gives closed-form derivatives and, for SSVI, explicit no-arbitrage conditions, but the fixed form can limit how closely the model follows the observed quotes.
2. ***Per-surface neural smoothers*** [1, 16] keep a parametric model as a prior and add a neural correction. Arbitrage is controlled through soft penalties on surface derivatives.
3. ***Neural operators*** [29, 25, 20] learn the mapping from a day's quotes to the full surface instead of fitting each day separately. Once trained they can produce surfaces for new trading days without refitting.

1.2 The research gap:what do soft constraints guarantee?

The thesis started as a controlled comparison of the three model families on the same dataset and using the same evaluation protocol / audit. However, during the experiments a more specific issue became clear.

While training the derivative-constrained smoother, the calendar penalty was exactly zero on the collocation grid used for training. This suggested that the surface satisfied the calendar constraint at all points where the model was checked. But when the same surface was evaluated on a separate and finer audit grid, calendar violations appeared on 28% of the points.

Both results were correct. The penalty was zero where it was evaluated but violations appeared between those points. The reason was the choice of collocation grid. Maturities were spaced exponentially, which works well for the butterfly constraint, but left a large gap of about five months in a region where calendar arbitrage could occur. In other words the model was not being checked where the violation appeared.

Chataigner [4] already noted that soft penalties act “only on the grid.” Our results show exactly why this matters. A penalty can be zero at the points where it is evaluated while the surface still violates the constraint between them. So what is really at stake here is not only whether the penalty reaches zero, but where it is checked and what happens when violations affect pricing.

The thesis studies this problem in three steps:

1. **Diagnosis and repair** (Section 5.2): we identify the source of the blind spot. The two arbitrage constraints require different grid structures. Thus, we used a separate grid for each constraint. This allows to reduce material calendar violations from 34.4% to 0.0%.
2. **Extension to neural operators** (Section 5.3): the same issue appears again in the neural operator, but now in two dimensions. Violations can occur between grid points and on days that were not seen during training. Although the training penalty remains close to zero, an independent audit finds material butterfly violations on up to 356 of the 386 out-of-sample days.
3. **Pricing consequences** (Chapter 6): we then study what these violations mean in practice. They can lead to negative local variance, negative implied densities, and weaker information in model residuals. In the most affected cases, around half of the operator surface requires repair before it can be used for pricing.

Each problem also suggests its own solution, which we develop alongside the analysis. The prior is proved to be arbitrage-free, so its protective role is guaranteed rather than assumed. When this prior is built into the neural operator, the violations disappear without reducing accuracy.

1.3 Contributions

(i) **A controlled comparison of the three model families.** We implement parametric surfaces, per-surface neural smoothers, and neural operators on the same SP 500 dataset of 1,926 trading days from 2018–2025. All models are evaluated using the same protocol and the same arbitrage audit. To our knowledge, no previous study compares all three families under the same setting.

(ii) **Replication and validation of the Gatheral–Jacquier framework.** We reproduce the no-arbitrage conditions and calibration procedures of Gatheral and Jacquier, matching the results reported in the original paper to 10^{-7}. During this replication, we identify and correct a missing term in the published inverse of the natural-parameter transformation (Section 5.1, Remark 2.15).

(iii) **Identification and repair of the collocation blind spot.** We show that no-arbitrage penalties can miss violations between the points where they are evaluated. We explain the source of this problem and correct it by using a separate collocation grid for each constraint. We also introduce the node-vs-audit gap and a materiality threshold to measure these violations (Section 5.2).

(iv) **An arbitrage-free prior and a bound on the collocation gap.** We prove that the SSVI prior is statically arbitrage-free on the domain considered here (Theorem 2.25). This means that its protective role is guaranteed rather than observed only empirically. We also derive a node-gap bound (Theorem 3.2) that explains how violations can develop between collocation points and why the same grid cannot be used effectively for both constraints.

(v) **Extension to neural operators and a prior-based solution.** We study DeepONet and graph neural operator architectures over the full chronological sample, together with a synthetic pretraining experiment. We show that the collocation blind spot also appears in neural operators, both between grid points and on unseen trading days. We then embed the certified prior directly into the operator and let the network learn only a bounded correction. This removes material violations on all 386 out-of-sample days while also improving the fit (Section 5.3).

(vi) **Economic consequences of arbitrage violations.** We study how these violations affect Dupire local volatility, Breeden–Litzenberger densities, and the pricing and hedging of a path-dependent barrier option under an independent Heston benchmark. The most accurate unconstrained operator misprices the barrier by 60%. A static-arbitrage scanner on raw bid/ask quotes and a detection-power analysis further show that these violations are better interpreted as model risk rather than as trading opportunities (Chapter 6).

1.4 Related work and positioning

Parametric and arbitrage-free surfaces. SVI and SSVI [12, 14] remain standard parametric approaches to implied volatility modelling. They are popular because they are simple, easy to differentiate and SSVI come with explicit no-arbitrage conditions. Earlier work also addressed arbitrage-free smoothing directly in price space. Kahale [18] uses convexity-preserving interpolation, while Fengler [11] formulates the problem as a constrained spline optimisation. These approaches impose the constraints directly rather than through a training loss. Chapter 2 develops the SVI/SSVI framework used as the parametric benchmark in this thesis.

Neural smoothing and soft constraints. Neural networks have long been used for option pricing and surface approximation [17, 8]. More recent approaches enforce financial structure through derivatives of the fitted network, made available by automatic differentiation [2]. This idea is closely related to collocation methods used in differential-equation learning [21, 26], where a constraint is evaluated only at a finite set of points. For implied volatility surfaces, Ackerer [1] combine a parametric prior with a neural correction and impose no-arbitrage through soft penalties on an extended collocation grid. Hoshisashi et al. [16] use a related derivative-constrained approach with automatic differentiation. Chataigner et al. [4] make explicit an important limitation of this methodology: the constraints are enforced only on the grid where they are evaluated. Section 5.2 studies this issue directly and measures the gap between the training grid and an independent audit grid.

Neural operators. Neural operators extend standard neural networks by learning mappings between functions rather than fitting a single function at a time. DeepONet [25] and graph-based neural operators [23, 20] provide two main constructions. Operator Deep Smoothing [29] applies this idea to implied volatility by learning a map from a day's option quotes to the corresponding surface. Once trained, the model can therefore be applied to new trading days without fitting a new network. Section 5.3 studies this setting and asks whether the same collocation issue remains when the model must generalise across days.

Economic consequences. The no-arbitrage conditions are directly connected to quantities used in pricing. Breeden and Litzenberger [3] link option-price convexity to the risk-neutral density, while Dupire [9] links surface derivatives to local volatility. These relationships provide a natural way to assess whether arbitrage violations matter beyond the audit grid. Recent work also connects volatility-surface errors to trading and residual-based signals [7], while SANOS [28] considers non-parametric arbitrage-free smoothing on OptionMetrics data. Chapter 6 uses these links to study the pricing and economic consequences of the violations found earlier in the thesis.

Positioning. This thesis differs from the existing literature in three main ways. First, it compares parametric surfaces, per-surface neural smoothers, and neural operators on the same dataset and under the same evaluation protocol. Second, it measures the collocation blind spot directly by comparing the points used for training with an independent audit grid, and shows that the same problem appears both within a surface and across unseen trading days. Third, it evaluates the resulting violations through quantities that matter for pricing rather than reporting only the fraction of grid points that fail an arbitrage test. The aim is therefore not only to compare which model fits the data best, but also to understand what its no-arbitrage constraints actually guarantee.

Chapter 2

Static Arbitrage and Parametric Surfaces

A volatility smoother has to do two things: fit the observed quotes and produce reasonable prices between them. This chapter focuses on the second point.

We study butterfly and calendar arbitrage in total-variance coordinates (k, w), where they are captured by the signs of g and $\partial_\tau w$. These quantities are used throughout the thesis to audit fitted surfaces. Dupire's identity,

$$\sigma_{\text{loc}}^2 = \frac{\partial_\tau w}{g},$$

then links these conditions directly to local variance and to the pricing analysis of Chapter 6.

The chapter also introduces SVI and SSVI, which provide the parametric benchmark, the arbitrage-free prior used later in the neural models, and closed-form derivatives for validating automatic differentiation.

What is ours, and where the proofs are. Most results in this chapter come from the cited literature and are independently re-derived and checked numerically. Two results are original to this thesis: the correction to the natural-parameter inverse (Remark 2.15) and the endpoint-certification result (Theorem 2.25). Results from the literature are labelled *Proposition* or *Lemma*, while *Theorem* is reserved for original results. Longer verification proofs are given in Appendix A.

2.1 Market setting and static arbitrage

Fix a trading date t and let $S = (S_u)_{u \geq t}$ denote the underlying index. For a maturity $T > t$ write $\tau = T - t$, let $F = F(t, T)$ be the forward price for delivery at T and $D = D(t, T)$ the discount factor. European calls and puts of strike K and maturity T trade at prices $C(K, T)$ and $P(K, T)$.

Throughout, we use a standard *static* and *model-free* framework. The only structure imposed on the quoted (or fitted) price surface is the absence of ***static arbitrage***.

Definition 2.1 (Static arbitrage)**.** A call price surface $(K,T) \mapsto C(K,T)$ admits *static arbitrage* if there exist $n \in \mathbb{N}$, strikes $K_1, \dots, K_n$, maturities $T_1, \dots, T_n$, positions $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ in the corresponding calls, positions in forwards and cash, such that the portfolio's initial cost is strictly negative while its terminal payoff is nonnegative in every state. The surface is *free of static arbitrage* if no such portfolio exists.

Definition 2.1 is a statement about *portfolios*, whereas everything this thesis fits and audits is a *surface*. The bridge between the two deserves to be made explicit at the outset, because it is what turns an abstract condition into something a smoother can actually violate. A fitted surface is not a drawing. Through the Black formula it defines a complete system of option prices,

$$w(k,\tau) \;\longrightarrow\; \sigma_{\mathrm{IV}}(k,\tau) \;\longrightarrow\; C(K,T),$$

including at strikes and maturities where nothing was ever quoted. Those manufactured prices must satisfy the same inequalities as the quoted ones, and an interpolation that honours every observation point by point can still generate a system that does not.

Two failures are possible, and they correspond to the two directions of the surface. In the *strike* direction, fix a maturity and take three strikes $K_1 < K_2 < K_3$ with $K_2 = \lambda K_1 + (1-\lambda)K_3$. Convexity of the call price in K requires

$$C(K_2) \;\le\; \lambda\, C(K_1) + (1-\lambda)\, C(K_3),$$

and if the fitted surface breaks this inequality then the butterfly portfolio that is long the wings and short the body has a nonnegative payoff and a negative cost: free money. In the *maturity* direction, a longer-dated option must be worth at least as much as a shorter-dated one at the same moneyness, which in total-variance coordinates will read $\partial_\tau w \ge 0$; otherwise the calendar spread that is long the far leg and short the near leg has the same defect. The audit performed throughout this thesis is nothing more than checking these two families of inequalities on the surfaces our models produce, and the whole of Chapter 6 is the question of what it costs when they fail.

Why these two directions and not some longer list? This is what makes a two-constraint audit exhaustive rather than merely convenient. The remaining requirements are structural rather than substantive: total variance positive, correct behaviour as $\tau \downarrow 0$, the price bounds of Proposition 2.2, and the wing behaviour of Remark 2.7. Once they hold, the two *nontrivial* directions of static arbitrage are exactly these:

$$\text{no static arbitrage} \iff \begin{cases} \text{no butterfly arbitrage on every maturity slice,} \\ \text{no calendar-spread arbitrage across maturities.} \end{cases}$$

This decomposition is Lemma 2.1 of [14], resting on [6, 19]. Its sufficiency has a probabilistic reading worth stating once, because it explains why the two conditions are the right ones rather than two conditions among many. A butterfly-free slice is precisely a slice whose implied density is a genuine probability distribution, so each maturity contributes an admissible marginal law.

The calendar condition is precisely the statement that these marginals are correctly ordered, in convex order. And a family of marginals increasing in convex order is, by Kellerer's theorem [19], the marginal system of some martingale, which is in turn the absence of static arbitrage. Butterfly therefore governs coherence *within* a maturity, calendar governs coherence *across* maturities, and together they close the problem.

The classical characterisation, which we take as the organising principle of the thesis, is that freedom from static arbitrage is equivalent to the existence of a nonnegative *martingale measure* consistent with all quotes: there is a filtered probability space carrying a nonnegative martingale (M_u) with $M_t = 1$ such that

$$C(K,T) = D(t,T)\, F(t,T)\, \mathbb{E}\Big[\big(M_T - \tfrac{K}{F(t,T)}\big)^+\Big], \tag{2.1}$$

i.e. $M_T = S_T/F(t,T)$ is the maturity-T forward-normalised underlying. Sufficiency of (2.1) is immediate (linearity and positivity of expectations); necessity is the content of Kellerer's theorem [19] and its refinements for call surfaces [6, 27]: a system of call prices that is (per maturity) decreasing and convex in K with the correct bounds, and (across maturities) increasing in the sense of Section 2.4, is the marginal-expectation system of some martingale. We use this equivalence freely and now record the elementary per-maturity properties.

Proposition 2.2 (Shape of an arbitrage-free call slice)**.** *Fix T and suppose $C(\cdot,T)$ is free of static arbitrage. Write $c(K) = C(K,T)/D(t,T)$ for the* undiscounted *call, which strips out the interest-rate factor and equals the pure expected payoff. Then:*

(a) $K \mapsto c(K)$ is nonincreasing and convex;

(b) $(F-K)^+ \le c(K) \le F$ for all $K \ge 0$;

(c) if c is twice differentiable, $\partial_{KK} c(K) \ge 0$, and $\partial_{KK} c(K)$ is the (undiscounted) risk-neutral density of S_T at K (Breeden–Litzenberger [3]).

Proof. (a) **Monotonicity.** For $K_1 < K_2$, the call spread consisting of a long call at K_1 and a short call at K_2 has payoff $(S_T - K_1)^+ - (S_T - K_2)^+ \ge 0$. Therefore, its cost must be nonnegative: $c(K_1) - c(K_2) \ge 0$.

Convexity. Let $K_1 < K_2 < K_3$, with $K_2 = \lambda K_1 + (1-\lambda) K_3$. The *butterfly* portfolio consisting of a position $+\lambda$ at K_1, a position -1 at K_2, and a position $+(1-\lambda)$ at K_3 has payoff $\lambda(S_T - K_1)^+ - (S_T - K_2)^+ + (1-\lambda)(S_T - K_3)^+ \ge 0$, by convexity of the map $K \mapsto (S_T - K)^+$. Hence, its cost satisfies $\lambda c(K_1) - c(K_2) + (1-\lambda) c(K_3) \ge 0$.

(b) The upper bound follows from the dominating payoff S_T, whose cost is F and which can be replicated using a forward position together with cash. The lower bound follows from the dominated payoff $S_T - K$, which can be replicated using a forward position minus cash.

(c) Under (2.1), the undiscounted call is $c(K) = \mathbb{E}\big[(S_T - K)^+\big]$. Differentiating twice under the expectation gives $\partial_{KK} c(K) = \mathbb{E}\big[\delta_K(S_T)\big]$, which is the density of S_T evaluated at K. The positivity of this density is the infinitesimal counterpart of the convexity established in part (a). Indeed, the butterfly portfolio with strikes $K-\varepsilon$, K, and $K+\varepsilon$ has cost $\varepsilon^2 \partial_{KK} c(K) + o(\varepsilon^2)$.

$\square$

2.2 The Black model, implied volatility, and the (k, w) coordinates

2.2.1 The Black formula

Under the Black (forward) model, $M_T = S_T/F$ is lognormal: $M_T = \exp(\sqrt{v}\,Z - v/2)$ with $Z \sim \mathcal{N}(0, 1)$ and total variance $v = \sigma^2\tau$. The call price is

$$C(K, \tau; \sigma) = D\left[F\,N(d_+) - K\,N(d_-)\right], \qquad d_\pm = \frac{\log(F/K)}{\sigma\sqrt{\tau}} \pm \frac{\sigma\sqrt{\tau}}{2}, \tag{2.2}$$

with N the standard normal cdf. The derivation is standard and is reproduced in Appendix A.1.1.

Since the Black *vega* $\partial C/\partial\sigma = DF\varphi(d_+)\sqrt{\tau} > 0$, (2.2) is strictly increasing in σ and therefore inverts on the arbitrage-free price interval: the *implied volatility* $\sigma_{\mathrm{IV}}(K, \tau)$ of a market quote is the unique root of $C(K, \tau; \sigma) = C^{\mathrm{mkt}}$. The dependence of implied volatility on (K, τ) describes the volatility smile across strikes and the term structure across maturities, thereby defining the implied volatility surface.

2.2.2 Coordinates

Two changes of variable make the theory tractable. The *log-moneyness* $k = \log(K/F)$ centres the smile on the forward; the *total (implied) variance* is the natural vertical coordinate:

$$w(k, \tau) = \sigma^2_{\mathrm{IV}}(k, \tau)\,\tau \tag{2.3}$$

Both static no-arbitrage conditions, the risk-neutral density and Dupire's formula all take their simplest form in (k, w). Throughout, primes denote ∂_k and $\dot{w}$ or $\partial_\tau w$ the maturity derivative. We also write the normalized, undiscounted forward prices $c(k, w) = C/(DF)$, $p(k, w) = P/(DF)$, so that

$$c(k, w) = N(d_+) - e^k N(d_-), \qquad d_\pm(k, w) = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}, \tag{2.4}$$

and put-call parity reads $c - p = 1 - e^k$. Note the identity, used repeatedly below,

$$\varphi(d_+) = e^k\,\varphi(d_-), \tag{2.5}$$

which follows from $d_+^2 - d_-^2 = (d_+ + d_-)(d_+ - d_-) = (-2k/\sqrt{w})(\sqrt{w}) = -2k$.

2.2.3 The Black partial derivatives

All later derivations reduce to the following closed forms.

Lemma 2.3 (Black partials in (k, w))**.** *Let $c(k, w)$ be as in (2.4) and let subscripts denote partial* *derivatives at fixed* (k, w)*. Then*

$$c_k = -e^k N(d_-), \tag{2.6}$$

$$c_w = \frac{e^k \varphi(d_-)}{2\sqrt{w}} = \frac{\varphi(d_+)}{2\sqrt{w}} > 0, \tag{2.7}$$

$$c_{kk} - c_k = \frac{e^k \varphi(d_-)}{\sqrt{w}} = 2\sqrt{w}\, c_w, \tag{2.8}$$

$$c_{kw} = \Big(\frac{1}{2} - \frac{k}{w}\Big)\, c_w, \tag{2.9}$$

$$c_{ww} = \Big(-\frac{1}{8} - \frac{1}{2w} + \frac{k^2}{2w^2}\Big)\, c_w. \tag{2.10}$$

Proof. See Appendix A.1.2. □

The sign in (2.7) is the one fact from this lemma used qualitatively rather than computationally: $c_w > 0$ says the Black price is strictly increasing in total variance, which is what makes implied volatility well defined and what turns the calendar condition of Section 2.4 into a statement about w alone.

2.3 Butterfly arbitrage, the Durrleman function, and the implied density

We now take the two arbitrage directions in turn, starting with the strike direction. At a fixed maturity the question is whether the fitted smile corresponds to a valid probability distribution over the terminal underlying. Asking for a valid distribution is asking for a nonnegative density; by Breeden–Litzenberger that is asking for convexity of the call price in strike; and in the (k, w) coordinates that condition collapses to a single scalar built from the slice and its first two derivatives. That scalar is the Durrleman function g, and the entire butterfly audit of this thesis reduces to monitoring its sign. This section builds the chain and ends with the complete characterisation, which requires one boundary condition in addition to $g \geq 0$.

The decomposition recorded above splits the problem in two; this section treats the butterfly direction and Section 2.4 the calendar one. Both follow [14, 10, 27].

Definition 2.4 (Durrleman function and butterfly arbitrage)**.** For a C^2 total variance slice $w(\cdot) > 0$ at fixed maturity, the associated *Durrleman function* is defined by

$$g(k) = \Big(1 - \frac{k\, w'(k)}{2\, w(k)}\Big)^2 - \frac{w'(k)^2}{4}\Big(\frac{1}{w(k)} + \frac{1}{4}\Big) + \frac{w''(k)}{2}. \tag{2.11}$$

The slice is *free of butterfly arbitrage* if and only if the associated call prices $k \mapsto c(k, w(k))$ are convex in strike and vanish as $K \to \infty$.

The next proposition identifies g with the risk-neutral density and thereby with the convexity of the call slice. It is the precise statement of the slogan *a butterfly violation is negative probability mass.*

Proposition 2.5 (Implied density in log-moneyness)**.** *Let $X_T = \log(S_T/F_T)$ and let $w(\cdot)$ be a C^2 slice with implied call prices $c(k) = c\big(k, w(k)\big)$ as in (2.4). The risk-neutral density of X_T is*

$$q(k) = \frac{g(k)}{\sqrt{2\pi\, w(k)}} \exp\!\Big(-\tfrac{1}{2} d_-(k)^2\Big) = \frac{g(k)\, \varphi\big(d_-(k)\big)}{\sqrt{w(k)}}, \qquad d_-(k) = d_-\big(k, w(k)\big), \tag{2.12}$$

with φ the standard normal pdf. In particular $q \geq 0 \iff g \geq 0$.

Proof, first step; completed in Appendix A.1.3. **Density in log-moneyness.** By Proposition 2.2(c), the undiscounted density of S_T at K is $\partial_{KK}(C/D)$. Since $K = Fe^k$, then $q(k) = K\, \partial_{KK}(C/D)$. Using $C/D = Fc(k)$ and $\partial_K = K^{-1}\partial_k$,

$$\partial_{KK}(C/D) = \frac{1}{K}\partial_k \left[\frac{F}{K}\partial_k c\right] = \frac{F}{K^2}\big(\partial_{kk}c - \partial_k c\big),$$

hence

$$q(k) = e^{-k}\big(\partial_{kk}c - \partial_k c\big). \tag{2.13}$$

Expanding the total derivatives of $c(k, w(k))$ and substituting the Black partials of Lemma 2.3 into (2.13) recomposes the bracket of (2.11) exactly. That computation is carried out in Appendix A.1.3. ☐

Equation (2.13) is worth isolating, because it is used again in Section 2.5: it says the implied density is, up to the factor e^{-k}, the second-order strike behaviour of the call slice, which is why butterfly arbitrage and negative probability are the same statement read in two coordinate systems.

Positivity of q, i.e. $g \geq 0$, is however not the whole butterfly story: convexity of the call slice controls its *interior*, but the price must also decay at extreme strikes. The complete characterisation is the following.

Lemma 2.6 (Butterfly characterisation)**.** *A C^2 slice $w(\cdot) > 0$ is free of butterfly arbitrage if and only if*

(i) $g(k) \geq 0$ *for all* $k \in \mathbb{R}$, *and*

(ii) $\lim_{k\to+\infty} d_+\big(k, w(k)\big) = -\infty$ *(equivalently,* $c(k, w(k)) \to 0$ *as* $k \to +\infty$*).*

Proof. See Appendix A.1.4. ☐

Condition (i) is the interior condition audited on every surface in this thesis; condition (ii) is a constraint on the wings, and a nonvanishing call price at infinite strike is the classical arbitrage of selling arbitrarily far out-of-the-money calls at a price bounded away from zero.

Remark 2.7 (Lee's bound)*.* Condition (ii) is a constraint on the *wings*: by Lee's moment formula [22], any arbitrage-free slice has $w(k)/|k|$ bounded by 2 as $k \to \pm\infty$, so total variance grows at most linearly with asymptotic slope at most 2. The boundary case $w(k)/|k| \to 2$ is exactly $d_+ \not\to -\infty$, i.e. the borderline of Lemma 2.6(ii). The corresponding SVI wing condition $b(1 + |\rho|) \leq 2$ is derived in Section 2.6.

2.4 Calendar arbitrage and monotone total variance

Butterfly-freeness is a per-slice property, and it can hold at every single maturity while the family of slices remains mutually inconsistent. Even if each maturity is individually a valid distribution, the slices must be correctly ordered relative to one another, since an option cannot become cheaper by being granted more time. This section shows that the requirement is exactly monotonicity of total variance in maturity.

One structural detail deserves attention here rather than later, because the whole of Chapter 3 turns on it: the calendar condition involves a *first* derivative in τ, whereas the butterfly condition involves a *second* derivative in k. That asymmetry looks like a technicality at this point. Section 3.1 shows it is the mechanism behind the central finding of the thesis, because a constraint that can fail anywhere along an axis and one that concentrates at one end of it require incompatible sampling grids.

Definition 2.8 (Calendar arbitrage)**.** The surface is free of calendar arbitrage if and only if $\tau \mapsto w(k,\tau)$ is non-decreasing for every k. Equivalently, total-variance slices do not cross.

Proposition 2.9 (Calendar characterisation)**.** *Let $c(k,\tau) := c\big(k, w(k,\tau)\big)$ denote the normalised call surface. The following are equivalent:*

(a) $\tau \mapsto w(k,\tau)$ is non-decreasing for every k;

(b) $\tau \mapsto c(k,\tau)$ is non-decreasing for every k;

(c) the surface is consistent, in the calendar direction, with some *nonnegative martingale for the forward-normalised underlying, i.e. no calendar-spread portfolio (long the longer-dated call, short the shorter-dated call, at equal moneyness k) has negative cost and nonnegative value.*

Proof of (a) $\iff$ (b); the remaining implications are proved in Appendix A.1.5. By Lemma 2.3, $c_w > 0$, so at fixed k the Black price is strictly increasing in total variance. Hence $\tau \mapsto w(k,\tau)$ is non-decreasing if and only if $\tau \mapsto c(k,\tau)$ is non-decreasing. □

The equivalence with (c) is what gives the condition its meaning: $(c) \Rightarrow (b)$ is conditional Jensen applied to the convex payoff $x \mapsto (x - e^k)^+$, and $(b) \Rightarrow (c)$ is Kellerer's theorem, since butterfly-free slices increasing in the sense of (b) are marginals increasing in convex order. Both directions are given in full in Appendix A.1.5.

Remark 2.10. Proposition 2.9 is why the calendar penalty of every learning method in this thesis is a penalty on $\partial_\tau w$: it is the complete, coordinate-free content of the calendar constraint, and unlike the butterfly constraint, it involves only a *first* derivative. This asymmetry between the two constraints (first- vs second-order; anywhere on the axis vs concentrated at the short end) is developed in Section 3.1 and is the mechanism of the collocation blind spot.

2.5 Dupire's formula in (k, w): the identity that prices the audit

The two previous sections produced two audit quantities, g and $\partial_\tau w$, and so far they are only flags: a surface either satisfies them or does not. This section explains why they are considerably more than flags, and it is the reason the audit of this thesis can be given an economic denomination at all.

Dupire's insight is that an arbitrage-free surface of European prices already determines a diffusion that reprices every vanilla exactly, and that the diffusion's local variance is a ratio of derivatives of the surface. Written in our coordinates, that ratio turns out to be $\partial_\tau w$ over g: the calendar quantity divided by the butterfly quantity. In other words, the two things we audit are not incidental diagnostics but the numerator and the denominator of the object a desk actually simulates with. A calendar violation is a negative local variance; a butterfly violation is a vanishing or exploding one. We state the equation first, then change variables.

Proposition 2.11 (Dupire's equation)**.** *Work under zero rates and dividends (so $D \equiv 1$, $F \equiv S_t$, and the forward measure coincides with the risk-neutral one). Suppose the underlying follows the local-volatility dynamics $\mathrm{d}S_u = \sigma_{\mathrm{loc}}(u, S_u)\, S_u\, \mathrm{d}W_u$ and that the transition density $p(T, K)$ of S_T is $C^{1,2}$ with $\sigma^2_{\mathrm{loc}} K^2 p$ and its K-derivative vanishing as $K \to \{0, \infty\}$. Then the undiscounted call $C(K, T)$ satisfies*

$$\partial_T C(K, T) = \tfrac{1}{2}\, \sigma^2_{\mathrm{loc}}(T, K)\, K^2\, \partial_{KK} C(K, T). \tag{2.14}$$

Proof. See Appendix A.1.6. ◻

Dupire's insight [9] is to read (2.14) backwards: given an arbitrage-free call surface, *define* $\sigma^2_{\mathrm{loc}} = 2\, \partial_T C/(K^2 \partial_{KK} C)$; the resulting diffusion reprices every vanilla exactly. Changing coordinates turns this into the identity that organises the thesis.

Proposition 2.12 (Local variance as calendar over butterfly)**.** *Under the assumptions of Proposition 2.11, if the implied total-variance surface $w(k, \tau)$ is $C^{2,1}$ with $g > 0$, the Dupire local variance at (k, τ) is*

$$\sigma^2_{\mathrm{loc}}(k, \tau) \;=\; \frac{\partial_\tau w(k, \tau)}{g(k, \tau)}. \tag{2.15}$$

Proof. Substitute $C = F\, c\big(k, w(k, T)\big)$ into (2.14), with $k = \log(K/F)$ fixed as T varies (zero carry keeps F constant).

Left side. At fixed K (hence fixed k), the only T-dependence is through w: $\partial_T C = F\, c_w\, \partial_\tau w$, with $c_w = e^k \varphi(d_-)/(2\sqrt{w})$ by (2.7). (The Black "theta" in these coordinates is carried entirely by w.)

Right side. By (2.13) and Proposition 2.5,

$$K^2\, \partial_{KK} C = F\big(\partial_{kk} c - \partial_k c\big) = F\, e^k\, q(k) = F\, e^k\, \frac{g\, \varphi(d_-)}{\sqrt{w}}.$$

Equating,

$$F\, \frac{e^k \varphi(d_-)}{2\sqrt{w}}\, \partial_\tau w = \tfrac{1}{2}\, \sigma^2_{\mathrm{loc}}\, F\, e^k\, \frac{g\, \varphi(d_-)}{\sqrt{w}} \qquad \Longleftrightarrow \qquad \partial_\tau w = \sigma^2_{\mathrm{loc}}\, g,$$

which is (2.15). This is Gatheral's equation (1.10) [13] in our notation. ◻

Remark 2.13 (Why this identity organises the thesis). Equation (2.15) says that the two no-arbitrage constraints are not abstract flags: the *calendar* constraint is the numerator of the local variance and the *butterfly* constraint is its denominator. A calendar violation ($\partial_\tau w < 0$) is a negative local variance; a butterfly violation ($g \le 0$) is a non-positive or exploding one.

2.6 SVI: three parameterisations, exact conversions, and the wings

Everything so far has been a *test*: given a surface, decide whether it is arbitrage-free. Nothing so far tells us how to *build* one. The parametric approach answers that by restricting attention to a family small enough that the conditions of the previous sections can be checked, and in the best case proved, once and for all in terms of the parameters.

SVI is that family. It has survived two decades of practitioner use for reasons that are worth naming, because each of them is used somewhere in this thesis: its derivatives are available in closed form, which is what allows us to validate automatic differentiation to machine precision in Section 5.2; it admits three equivalent parameterisations, one convenient for calibration, one for the link to stochastic volatility, and one that traders can read off a smile; and its wing behaviour matches the model-free bound of Lee exactly. This section develops the three parameterisations and the exact maps between them. Those maps carry more weight than their bookkeeping appearance suggests: they are the vehicle of the butterfly-repair algorithm of Section 5.1, and round-tripping through them is what exposed an error in the published inverse (Remark 2.15).

2.6.1 The raw parameterisation and its derivatives

The *raw SVI* slice [12] at a fixed maturity t is

$$w(k;\chi_R) = a + b\Big(\rho\,(k-m) + \sqrt{(k-m)^2+\sigma^2}\Big), \qquad \chi_R = (a,b,\rho,m,\sigma), \tag{2.16}$$

with $b \ge 0$, $|\rho| < 1$, $\sigma > 0$, and $a + b\sigma\sqrt{1-\rho^2} \ge 0$ (which, as shown in Lemma 2.16 below, is exactly nonnegativity of the minimum of w). Its derivatives are closed-form:

$$w'(k) = b\Big(\rho + \frac{k-m}{\sqrt{(k-m)^2+\sigma^2}}\Big), \qquad w''(k) = \frac{b\,\sigma^2}{\big((k-m)^2+\sigma^2\big)^{3/2}} \;>\; 0. \tag{2.17}$$

These closed forms are used in Section 5.2 to validate automatic differentiation to machine precision. Note $w'' > 0$ everywhere: an SVI slice is strictly convex in k, yet convexity of w in k does not imply convexity of the call in K, i.e. does not imply $g \ge 0$.

2.6.2 Wing asymptotics and Lee consistency

From (2.16), as $k \to \pm\infty$,

$$w(k) = b\,(1 \pm \rho)\,|k| + O(1), \tag{2.18}$$

so the asymptotic total-variance slopes are $b(1+\rho)$ (right) and $b(1-\rho)$ (left). Lee's moment formula [22] bounds the asymptotic slope of total variance in any arbitrage-free slice: $\limsup_{k\to\pm\infty} w(k)/|k| \le 2$. Applying this to both wings gives the *wing condition*

$$b\,(1+|\rho|) \;\le\; 2, \tag{2.19}$$

with strict inequality required for condition (ii) of Lemma 2.6 ($w(k)/k \to 2$ is precisely $d_+ \not\to -\infty$). Equation (2.19) constrains the wings only; the interior condition $g \ge 0$ has no known tractable characterisation in terms of χ_R and must be checked (Section 5.1 does so on a fine audit grid).

2.6.3 The natural parameterisation and the corrected inverse map

The *natural* parameterisation writes the same family as a perturbation of a time-rescaled Heston-consistent form:

$$w(k;\chi_N) = \Delta + \frac{\omega}{2}\Big\{1+\zeta\rho\,(k-\mu)+\sqrt{\big(\zeta(k-\mu)+\rho\big)^2+1-\rho^2}\Big\}, \qquad \chi_N = (\Delta,\mu,\rho,\omega,\zeta). \tag{2.20}$$

The map between the two is elementary, and we give its derivation in full here rather than in the appendix: it is the derivation that supplies the term missing from the published inverse (Remark 2.15).

Lemma 2.14 (Raw $\leftrightarrow$ natural, cf. [14, Lemma 3.2])**.** *The parameterisations* (2.16) *and* (2.20) *describe the same slice under*

$$(a,b,\rho,m,\sigma) = \Big(\Delta+\tfrac{\omega}{2}(1-\rho^2),\ \tfrac{\omega\zeta}{2},\ \rho,\ \mu-\tfrac{\rho}{\zeta},\ \tfrac{\sqrt{1-\rho^2}}{\zeta}\Big), \tag{2.21}$$

with inverse

$$(\Delta,\mu,\rho,\omega,\zeta) = \Big(a-b\sigma\sqrt{1-\rho^2},\ m+\tfrac{\rho\sigma}{\sqrt{1-\rho^2}},\ \rho,\ \tfrac{2b\sigma}{\sqrt{1-\rho^2}},\ \tfrac{\sqrt{1-\rho^2}}{\sigma}\Big). \tag{2.22}$$

Proof. Set $v := \zeta(k-\mu)+\rho$. With $m = \mu-\rho/\zeta$ we have $k-m = v/\zeta$, so the raw form reads

$$a + b\Big(\rho\frac{v}{\zeta}+\sqrt{\frac{v^2}{\zeta^2}+\sigma^2}\Big) = a+\frac{b}{\zeta}\Big(\rho v+\sqrt{v^2+\zeta^2\sigma^2}\Big).$$

With $\sigma = \sqrt{1-\rho^2}/\zeta$, $\zeta^2\sigma^2 = 1-\rho^2$. The natural form expands as

$$\Delta+\frac{\omega}{2}\{1+\rho(v-\rho)\}+\frac{\omega}{2}\sqrt{v^2+1-\rho^2} = \Big(\Delta+\frac{\omega}{2}(1-\rho^2)\Big)+\frac{\omega}{2}\Big(\rho v+\sqrt{v^2+1-\rho^2}\Big).$$

Matching coefficients gives $b/\zeta = \omega/2$ and $a = \Delta+\omega(1-\rho^2)/2$, i.e. (2.21). Solving (2.21) for χ_N: $\zeta = \sqrt{1-\rho^2}/\sigma$, $\omega = 2b/\zeta = 2b\sigma/\sqrt{1-\rho^2}$, $\mu = m+\rho/\zeta = m+\rho\sigma/\sqrt{1-\rho^2}$, and

$$\Delta = a-\frac{\omega}{2}(1-\rho^2) = a-\frac{2b\sigma}{2\sqrt{1-\rho^2}}(1-\rho^2) = a-b\sigma\sqrt{1-\rho^2}.$$

□

Remark 2.15 (**A correction to the published inverse**). During the replication, we found that the natural-to-raw inverse reported in [14, Lemma 3.2] is missing the term $-b\sigma\sqrt{1-\rho^2}$ in Δ, or equivalently $-\alpha\sqrt{1-\rho^2}$ in the notation of the paper. This term appears directly in the derivation above.

The omission matters in practice because any calibration that passes through the natural parameterisation produces an incorrect value of Δ, and therefore an incorrect SVI slice. We identified the issue by testing 2,000 random parameter sets through the cycle raw→natural→JW→raw. With the corrected term, the round trip closes with a maximum error of 2.7×10^{-10}.

2.6.4 The Jump-Wings parameterisation

The raw and natural parameterisations are useful for calibration but they are not very intuitive: a set such as (a, b, ρ, m, σ) does not immediately tell us what the smile looks like. The *Jump-Wings* (JW) parameterisation [14] describes the same slice using five quantities that a trader can read directly from the smile. At maturity t, with $w_t := v_t, t$ the ATM total variance, these are the ATM variance v_t and ATM skew ψ_t, which describe the level and slope of the smile at the money; the left and right wing slopes p_t and c_t, which describe the steepness of the put and call wings; and the minimum implied variance $\tilde{v}_t$ may be shifted away from the money when $\psi_t \neq 0$.

Lemma 2.16 (JW from raw, [14, eq. (3.5)]). *For the raw slice* (2.16) *at maturity* t*,*

$$\begin{aligned} v_t &= \frac{a + b\big(-\rho m + \sqrt{m^2+\sigma^2}\big)}{t}, \quad \psi_t = \frac{1}{\sqrt{w_t}} \cdot \frac{b}{2}\Big(\rho - \frac{m}{\sqrt{m^2+\sigma^2}}\Big), \\ p_t &= \frac{b\,(1-\rho)}{\sqrt{w_t}}, \qquad c_t = \frac{b\,(1+\rho)}{\sqrt{w_t}}, \qquad \tilde{v}_t = \frac{a + b\sigma\sqrt{1-\rho^2}}{t}. \end{aligned} \tag{2.23}$$

Proof. See Appendix A.2.1. □

Lemma 2.17 (Raw from JW, [14, eq. (3.6)]). *Given* $(v_t, \psi_t, p_t, c_t, \tilde{v}_t)$ *with* $w_t = v_t t$*, set*

$$b = \frac{\sqrt{w_t}}{2}\,(c_t + p_t), \qquad \rho = 1 - \frac{p_t\sqrt{w_t}}{b} = \frac{c_t - p_t}{c_t + p_t}, \qquad \beta := \rho - \frac{2\psi_t\sqrt{w_t}}{b}, \qquad \alpha := \operatorname{sgn}(\beta)\sqrt{\frac{1}{\beta^2} - 1}, \tag{2.24}$$

(requiring $|\beta| \leq 1$ *for a real solution), and then*

$$m = \frac{(v_t - \tilde{v}_t)\,t}{b\,\big\{-\rho + \operatorname{sgn}(\alpha)\sqrt{1+\alpha^2} - \alpha\sqrt{1-\rho^2}\big\}}, \qquad \sigma = \alpha\, m. \tag{2.25}$$

The degenerate case $m = 0$ *occurs exactly when* $v_t = \tilde{v}_t$*, i.e. when the minimum of the slice sits at the money, so that the numerator of* (2.25) *vanishes and the formula* $\sigma = \alpha m$ *becomes indeterminate; it is handled separately by* $\sigma = \big((v_t - \tilde{v}_t)\,t\big)/\big(b\,(1 - \sqrt{1-\rho^2})\big)$*. Finally,* $a = \tilde{v}_t\, t - b\sigma\sqrt{1-\rho^2}$*.*

Proof. See Appendix A.2.2. □

2.6.5 The Vogt smile: arbitrage invisible to the eye

Example 2.18 (Vogt smile, [14, Ex. 3.1])**.** Consider the raw SVI slice at $t = 1$ with parameters

$$(a,\ b,\ m,\ \rho,\ \sigma) = (-0.0410,\ 0.1331,\ 0.3586,\ 0.3060,\ 0.4153).$$

This slice looks perfectly regular: total variance stays positive, the smile is convex, and the wing condition (2.19) is easily satisfied, with $b(1 + |\rho|) = 0.174 \ll 2$. Yet the Durrleman function becomes negative, with $\min_k g = -0.0329$ near $k \approx 0.88$, implying a negative density and therefore butterfly arbitrage. Section 5.1 reproduces the paper's Jump–Wings parameters and repair procedure. This example shows that a smooth-looking smile is not enough: arbitrage must be checked explicitly.

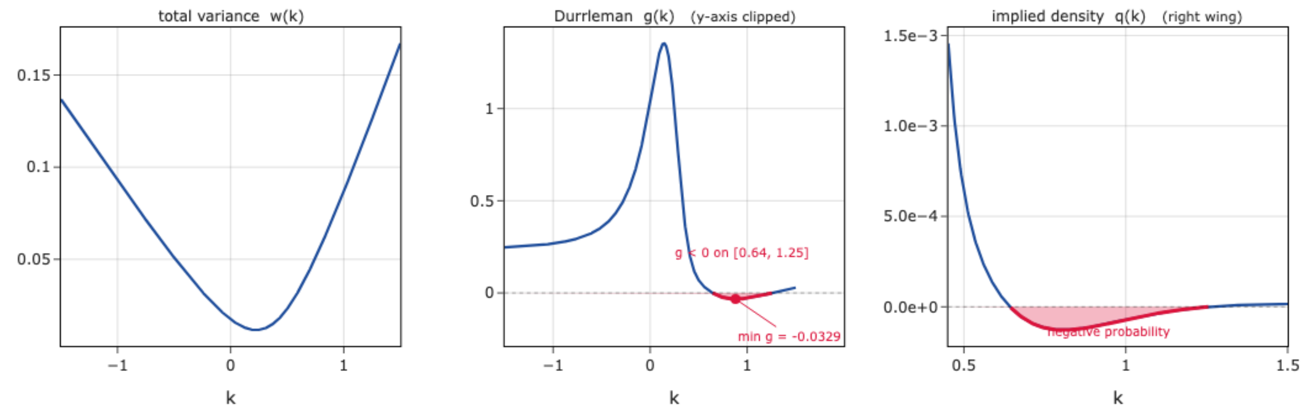

Figure 2.1: Vogt SVI slice. Left: total variance is positive and smooth. Centre: the Durrleman function becomes negative on $k \in [0.64, 1.26]$. Right: the implied density is negative on the same interval.

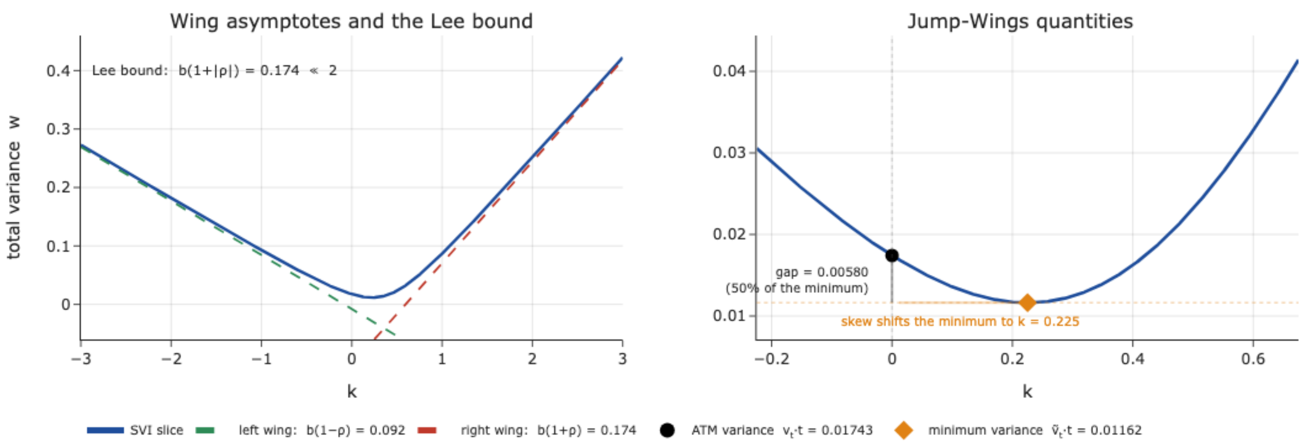

Figure 2.2: Vogt slice in Jump–Wings form. Left: asymptotic wing slopes $b(1 - \rho)$ and $b(1 + \rho)$ satisfy Lee's bound. Right: the ATM and minimum variances are shown near the trough, reproducing [14, Ex. 3.1].

2.7 SSVI and the Gatheral–Jacquier certificates

2.7.1 The surface

SVI describes one maturity at a time, and fitting each maturity independently gives excellent per-slice accuracy with no guarantee that adjacent slices are mutually consistent: nothing prevents two of them from crossing, which by Section 2.4 is calendar arbitrage. SSVI removes that risk by tying every slice to a single term structure, so the calendar direction is controlled by construction rather than repaired afterwards. The price is rigidity, and Section 5.1 measures it precisely.

What makes SSVI valuable here is not only the construction but what Gatheral and Jacquier prove about it: closed-form conditions on the parameters under which the surface is provably free of both arbitrages. That is a *certificate*, not an audit, and it is the only such object in the thesis. It is what the deep smoother of Section 5.2 uses as a prior, and it is why the ablation finding that the prior does the protective work rests on a theorem rather than on luck.

The *SSVI* (surface SVI) parameterisation [14] promotes SVI to a full surface driven by its ATM total-variance term structure $\theta_\tau := w(0,\tau)$:

$$w(k,\theta_\tau) = \frac{\theta_\tau}{2}\left(1 + \rho\,\varphi(\theta_\tau)\,k + \sqrt{\big(\varphi(\theta_\tau)k + \rho\big)^2 + 1 - \rho^2}\right), \tag{2.26}$$

where $\varphi : (0,\infty) \to (0,\infty)$ is a smooth curvature function and $|\rho| < 1$. Evaluating at $k = 0$ gives $w(0,\theta) = \theta$: the term structure is matched by construction. Throughout this thesis we use the *power law* $\varphi(\theta) = \eta\theta^{-\gamma}$ with $\theta_\tau = \alpha\tau^\beta$ ($\eta, \alpha, \beta > 0$, $\gamma \in (0,1)$).

Every SSVI slice is itself a raw SVI slice. This transports the whole SVI machinery (derivatives, g-audits, JW readings) to the surface, and it is the first step of the butterfly certificate.

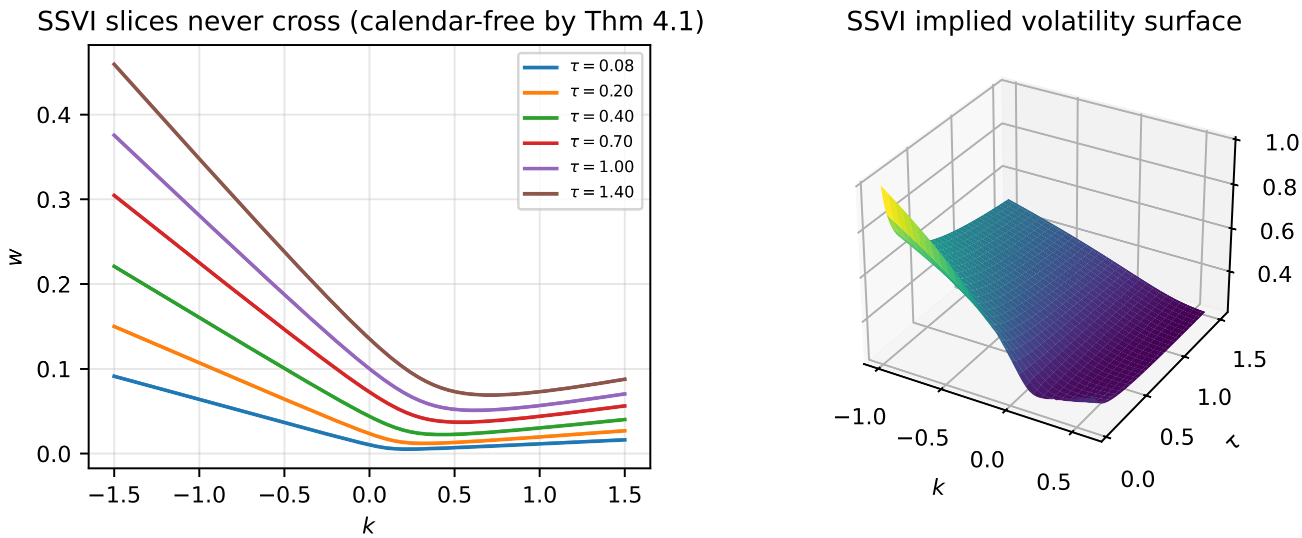

Figure 2.3: The SSVI surface (2.26) with power-law φ and $\theta_\tau = \alpha\tau^\beta$, on illustrative parameters ($\rho = -0.7$, $\eta = 0.8$, $\gamma = 0.45$, $\alpha = 0.10$, $\beta = 0.9$) satisfying (2.28) on the plotted range. Left: total-variance slices never cross, as guaranteed by Proposition 2.20. Right: the corresponding implied-volatility surface, whose skew is the visible signature of $\rho < 0$.

Lemma 2.19 (SSVI slices are SVI slices)**.** *For fixed $\theta > 0$, the slice $k \mapsto w(k,\theta)$ of (2.26) equals the raw SVI slice (2.16) with parameters*

$$(a, b, \rho, m, \sigma) = \Big(\frac{\theta}{2}(1-\rho^2),\ \frac{\theta\varphi(\theta)}{2},\ \rho,\ -\frac{\rho}{\varphi(\theta)},\ \frac{\sqrt{1-\rho^2}}{\varphi(\theta)}\Big). \tag{2.27}$$

Proof. See Appendix A.2.3. □

2.7.2 The calendar certificate

Proposition 2.20 (Calendar, [14, Thm. 4.1])**.** *The SSVI surface (2.26) is free of calendar arbitrage if and only if*

(i) $\tau \mapsto \theta_\tau$ is non-decreasing, and
(ii) $0 \le \partial_\theta\big(\theta\varphi(\theta)\big) \le \dfrac{1}{\rho^2}\Big(1+\sqrt{1-\rho^2}\Big)\varphi(\theta)$ for all $\theta > 0$.

Proof. See Appendix A.2.4. □

The two conditions separate cleanly. The first is the term structure moving in the right direction, which is what Definition 2.8 demands at the money. The second controls $\theta\varphi(\theta)$, which by (2.27) is twice the SVI slope b, and therefore governs how fast the slice tilts as θ grows: its lower bound keeps the wings from moving down with maturity, and its upper bound, whose constant is sharp, keeps the tilt from outrunning the level and so stops adjacent slices crossing away from the money.

For the family used, the second condition costs nothing.

Example 2.21 (Power law satisfies (ii) automatically)**.** For $\varphi(\theta) = \eta\theta^{-\gamma}$, writing $\psi(\theta) := \theta\varphi(\theta) = \eta\theta^{1-\gamma}$ gives $\psi'(\theta) = (1-\gamma)\varphi(\theta)$, so condition (ii) reads $0 \le 1-\gamma \le (1+\sqrt{1-\rho^2})/\rho^2$. The right-hand side is at least 1 for every ρ, while $1-\gamma \in (0,1)$, so the condition holds for all $\gamma \in (0,1)$ and all ρ: with a non-decreasing θ_τ, the power-law SSVI is calendar-free for free.

2.7.3 The butterfly certificate

The strike direction is where the real constraint sits.

Proposition 2.22 (Butterfly, [14, Thm. 4.2])**.** *The SSVI surface is free of butterfly arbitrage if, for all $\theta > 0$,*

$$\theta\varphi(\theta)\,(1+|\rho|) < 4 \qquad \text{and} \qquad \theta\varphi(\theta)^2\,(1+|\rho|) \le 4. \tag{2.28}$$

Together with Proposition 2.20 this yields freedom from static *arbitrage ([14, Cor. 4.1]).*

Proof. See Appendix A.2.5. □

Each condition controls one feature of the slice. The first is the Lee wing bound (2.19) rewritten for SSVI, since $b = \theta\varphi/2$ by (2.27); it governs the *wings* and delivers condition (ii) of Lemma 2.6.

The second bounds $\theta\varphi^2$, the slice's natural *curvature scale*, which is exactly the combination through which the terms amplified by $1/w$ in (2.11) can drive g negative.

We do not merely quote these two propositions: both are implemented as *executable certificates* and verified against the paper's own published condition values (Section 5.1).

Example 2.23 (φ families)**.** Two other families admit a global certificate: the Heston-like $\varphi(\theta) = \frac{1}{\lambda\theta}\big(1 - \frac{1-e^{-\lambda\theta}}{\lambda\theta}\big)$ for $\lambda \geq (1+|\rho|)/4$, and the modified power law $\varphi(\theta) = \eta\theta^{-\gamma}(1+\theta)^{\gamma-1}$ with $\gamma \in (0, \frac{1}{2}]$ and $\eta(1+|\rho|) \leq 2$. All three, including the plain power law used here, are implemented and verified in NB02 against the paper's published condition values.

Remark 2.24 (Why the plain power law cannot be certified globally; [14, Rem. 4.4])*.* The plain power law cannot satisfy (2.28) on all of $(0, \infty)$: $\theta\varphi = \eta\theta^{1-\gamma} \to \infty$ as $\theta \to \infty$, and for $\gamma > \frac{1}{2}$ also $\theta\varphi^2 \to \infty$ as $\theta \downarrow 0$. No choice of (η, γ) escapes both. But this is the wrong question to ask of a smoother, which never evaluates more than a *compact* range of θ — and on a compact range the certificate collapses to two endpoint checks.

2.7.4 Endpoint certification on the working range

Theorem 2.25 (Endpoint certification)**.** *Let $\varphi(\theta) = \eta\theta^{-\gamma}$ with $\eta > 0$, $\gamma \in (0, 1)$, and let $0 < \theta_{\min} \leq \theta \leq \theta_{\max}$ be the range attained by $\theta_\tau = \alpha\tau^\beta$ on the working maturities $[\tau_{\min}, \tau_{\max}]$ $(\alpha, \beta > 0)$. Then*

$$\sup_{\theta\in[\theta_{\min},\theta_{\max}]} \theta\varphi(\theta) = \eta\,\theta_{\max}^{1-\gamma}, \qquad \sup_{\theta\in[\theta_{\min},\theta_{\max}]} \theta\varphi(\theta)^2 = \eta^2 \max\big(\theta_{\max}^{1-2\gamma},\, \theta_{\min}^{1-2\gamma}\big).$$

Consequently, checking (2.28) at the two endpoints certifies the SSVI surface free of static arbitrage on the entire working domain $[\tau_{\min}, \tau_{\max}] \times \mathbb{R}$.

Proof. $\theta\varphi(\theta) = \eta\theta^{1-\gamma}$ is increasing for $\gamma < 1$, so its supremum is at $\theta_{\max}$. $\theta\varphi(\theta)^2 = \eta^2\theta^{1-2\gamma}$ is increasing iff $\gamma \leq \frac{1}{2}$ (supremum at $\theta_{\max}$) and decreasing iff $\gamma > \frac{1}{2}$ (supremum at $\theta_{\min}$); the displayed maximum covers both. Monotonicity of $\theta_\tau = \alpha\tau^\beta$ maps $[\tau_{\min}, \tau_{\max}]$ onto $[\theta_{\min}, \theta_{\max}]$, so the working θ-range is exactly this interval; conditions (2.28) on the range then follow from the two endpoint evaluations, and Proposition 2.22 plus Example 2.21 plus Proposition 2.20(i) give static-arbitrage-freeness on $[\tau_{\min}, \tau_{\max}] \times \mathbb{R}$. □

Remark 2.26. Theorem 2.25 turns the Gatheral–Jacquier conditions from an *audit* into a *constraint*: the prior calibrator imposes (2.28) at the endpoints during the search, hard-projects η afterwards, and prints a certificate. Anything below $\tau_{\min}$ remains uncertified extrapolation, flagged as such downstream, which is why the Monte-Carlo time grids of Chapter 6 are floored at $\tau_{\min}$. The gain is not only theoretical: an earlier calibrator clipped $\gamma \leq \frac{1}{2}$ so as to certify on $(0, \theta_{\max}]$, but SPX data prefer $\gamma > \frac{1}{2}$, and the clip cost about 0.5 vp of prior accuracy. The endpoint form recovers it at no loss of guarantee.

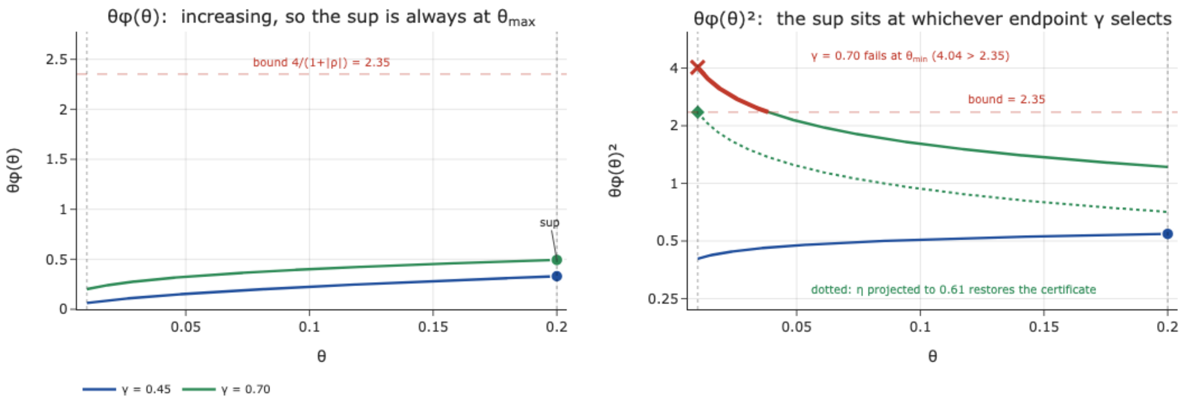

Figure 2.4: Endpoint certification (Theorem 2.25) with $\eta = 0.8$, $\rho = -0.7$ on the working range $[\theta_{\min}, \theta_{\max}] = [0.01, 0.20]$ (dotted). Left: $\theta\varphi(\theta)$ is increasing for every $\gamma \in (0,1)$, so its supremum (marker) always sits at $\theta_{\max}$, here far below the bound $4/(1+|\rho|)$ (dashed red). Right, on a log scale: $\theta\varphi(\theta)^2$ is increasing for $\gamma = 0.45$ and decreasing for $\gamma = 0.70$, so the supremum moves to the opposite endpoint. The $\gamma = 0.70$ family breaches the bound at $\theta_{\min}$ (4.04 against 2.35, red), which is precisely what the two endpoint checks detect; projecting η down to 0.61 (dotted) restores the certificate on the whole range.

2.7.5 Interpolation, extrapolation, and crossedness

Two further Gatheral–Jacquier results are used by the parametric benchmark, one for filling gaps between calibrated slices and one for measuring the damage when slices cross.

Between two calibrated slices w_1, w_2 at ATM total variances $\theta_1 < \theta_2$, butterfly-free and non-crossing, [14, Lem. 5.1] construct an *arbitrage-free interpolation* at any intermediate $\theta_t \in [\theta_1, \theta_2]$. The trick is to interpolate in *price* space rather than directly on w: with the $\sqrt{\theta}$-weight

$$\alpha = \frac{\sqrt{\theta_2} - \sqrt{\theta_t}}{\sqrt{\theta_2} - \sqrt{\theta_1}} \in [0, 1], \tag{2.29}$$

one takes the convex combination of the two normalised call slices and inverts it back through the Black map (2.4),

$$w(k, \theta_t) = c^{-1}\Big(k,\ \alpha\, c\big(k, w_1(k)\big) + (1-\alpha)\, c\big(k, w_2(k)\big)\Big), \tag{2.30}$$

where $c^{-1}(k, \cdot)$ is Black inversion at fixed k, unique since $c_w > 0$ (Lemma 2.3). A convex combination of convex prices stays convex, so the interpolated slice is butterfly-free by construction. Beyond the last calibrated maturity we extrapolate by the dominating shift $w_{\text{ext}}(k) = w_2(k) + (\theta_{\text{far}} - \theta_2)$, which lifts the whole slice and so introduces no crossing. Both are implemented and checked numerically: the interpolated slice attains $\min_k g = 0.52$, and the extrapolated slice dominates the last calibrated one everywhere.

When fitted slices *do* cross, the crossing is quantified by the *crossedness* of [14, Def. 5.1]: for adjacent slices χ_1, χ_2 with intersection points $\tilde{k}_1, \dots, \tilde{k}_n$, it is the maximum over these points of $\max\big(0,\ w(\tilde{k}; \chi_1) - w(\tilde{k}; \chi_2)\big)$. A penalty on total crossedness is the calendar-repair device of Section 5.1.

Chapter 3

Constrained Learning of Volatility Surfaces

The previous chapter considered SSVI, for which the no-arbitrage conditions can be guaranteed analytically on the working domain. This guarantee comes with the restriction of a five-parameter functional form, and Section 5.1 will show what it costs in fit.

This chapter turns to more flexible model classes for which no comparable analytical guarantee is available. A neural network can represent essentially any smooth surface, so the rigidity disappears; but no theorem attaches to it, and the arbitrage conditions must instead be imposed by penalising violations at a finite set of points during training. The choice of these points is central to this chapter.

Section 3.1 develops the main theoretical results used later in the thesis: what minimising a penalty to zero actually proves, how far that statement extends beyond the points where it was checked, why the butterfly and calendar constraints demand incompatible sampling grids, and when a measured violation is large enough to matter economically. Sections 3.2 and 3.3 then supply the approximation-theoretic background for the two learned families studied empirically: the per-day neural smoother, and the neural operator, which changes the object being learned from a surface to the map that produces surfaces. Section 3.4 closes by stating the reconstruction problem the rest of the thesis solves and scores, including the one criterion by which our framing departs from the literature.

3.1 Soft constraints, sampling sets, and materiality

Parametric models can sometimes provide analytical certificates, while neural smoothers generally cannot. They cannot impose $g \geq 0$ and $\partial_\tau w \geq 0$ pointwise, and instead penalise violations on a finite *collocation set* $\hat{X}$ [1, 16]:

$$\mathcal{L} = \underbrace{\frac{1}{p}\sum_i \omega_i \big(w_\theta(k_i,\tau_i) - w_i\big)^2}_{\text{fit}} + \lambda_{\mathrm{b}}\, \overline{\mathrm{ReLU}(-g_\theta)^2}\Big|_{\hat{X}_{\mathrm{b}}} + \lambda_{\mathrm{c}}\, \overline{\mathrm{ReLU}(-\partial_\tau w_\theta)^2}\Big|_{\hat{X}_{\mathrm{c}}} + \lambda_{\mathrm{w}}\, \overline{(\partial_{kk} w_\theta)^2}\Big|_{\hat{X}_{\mathrm{wing}}}, \tag{3.1}$$

where overlines denote averages over the indicated collocation grids and ω_i are fit weights. Everything that follows in this section concerns one question about (3.1): when the penalty terms reach zero, what exactly has been proved?

Remark 3.1 (A penalty is a statement about $\hat{X}$). Minimising (3.1) to λ-term zero proves $g_\theta \geq 0$ *on* $\hat{X}_\mathrm{b}$ and $\partial_\tau w_\theta \geq 0$ *on* $\hat{X}_\mathrm{c}$. Whether that extends between the nodes depends on the interaction between the network's inductive bias and the geometry of the constraint. Sections 5.2–5.3 show empirically that it can fail, exactly, and at scale.

3.1.1 The node-gap bound

Remark 3.1 identifies the region between nodes as the potential source of violations, but does not quantify their size. The following elementary result answers that: between nodes, the only control a node-wise constraint gives is through a regularity bound on the model class, and the violation budget scales *linearly* with the node gap.

Theorem 3.2 (Node-gap violation bound)**.** *Fix k and let $f(\tau) := \partial_\tau w_\theta(k,\tau)$ on an interval containing adjacent collocation nodes $\tau_1 < \tau_2$ with gap $h = \tau_2 - \tau_1$. Suppose $f(\tau_1) \geq 0$, $f(\tau_2) \geq 0$ (the penalty is satisfied at the nodes) and f is Lipschitz with $|f'(\tau)| = |\partial_{\tau\tau} w_\theta(k,\tau)| \leq L$ on $[\tau_1, \tau_2]$. Then*

$$f(\tau) \;\geq\; -\frac{L\,h}{2} \qquad \text{for all } \tau \in [\tau_1, \tau_2],$$

and the bound is sharp: for every L, h there is an f with $|f'| \leq L$, $f(\tau_{1,2}) = 0$ and $\min f = -Lh/2$. *Consequently, the deepest calendar violation a node-satisfying surface can hide between nodes is proportional to the node gap, with constant the curvature budget of the model class.*

Proof. For $\tau \in [\tau_1, \tau_2]$, let $\tau^* \in \{\tau_1, \tau_2\}$ be the nearer node, so $|\tau - \tau^*| \leq h/2$. By the Lipschitz bound, $f(\tau) \geq f(\tau^*) - L|\tau - \tau^*| \geq -Lh/2$. Sharpness: take $f(\tau) = L\,|\tau - \frac{\tau_1+\tau_2}{2}| - \frac{Lh}{2}$ (or a C^1 mollification thereof), which vanishes at both nodes, has $|f'| = L$, and attains $-Lh/2$ at the midpoint. □

Remark 3.3 (Reading the bound). Two consequences drive Section 5.2. First, *without* an a-priori bound L on $\partial_{\tau\tau} w_\theta$, node constraints control nothing between nodes, and a trained network under fit pressure has no useful such bound: the fit term of (3.1) can then improve accuracy by creating a violation between nodes without increasing the penalty. Second, *with* whatever L the trained network does exhibit, the violation budget is proportional to the node gap h. Shrinking the calendar grid's maximal gap from 0.41y, the exponential grid's long-end hole, to 0.017y, the repaired uniform grid, shrinks the worst admissible hidden violation by a factor of about 24 at equal curvature. This is exactly the repair adopted in Section 5.2, and the reason the repaired node and audit penalties agree to grid noise. Figure 3.1 illustrates the extremal f, and Figure 3.2 shows the two grids on the thesis's actual maturity axis.

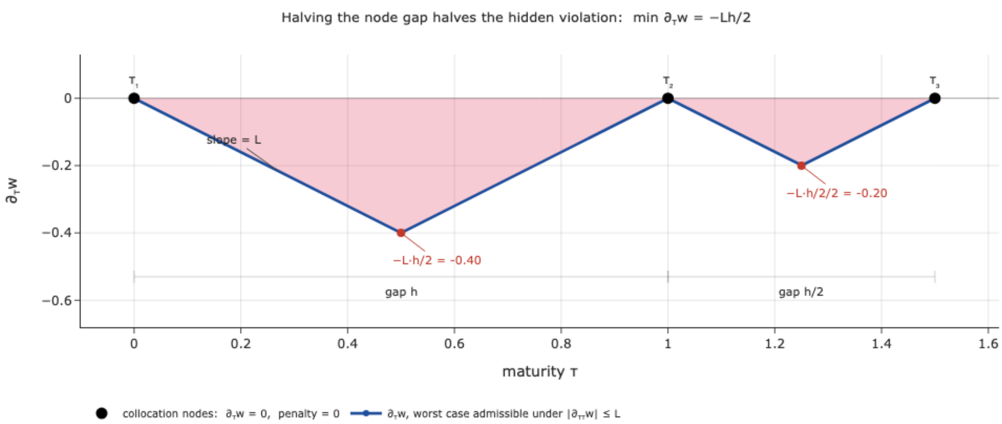

Figure 3.1: The node-gap bound (Theorem 3.2). The calendar penalty is satisfied exactly at every collocation node (dots, $\partial_\tau w = 0$), yet $\partial_\tau w$ dips to $-Lh/2$ between them (shaded), the branches falling at the maximal admissible slope L. Halving the gap halves the violation: the right-hand pair, spaced $h/2$, can hide only $-Lh/4$. That proportionality is what the repair of Section 5.2 exploits.

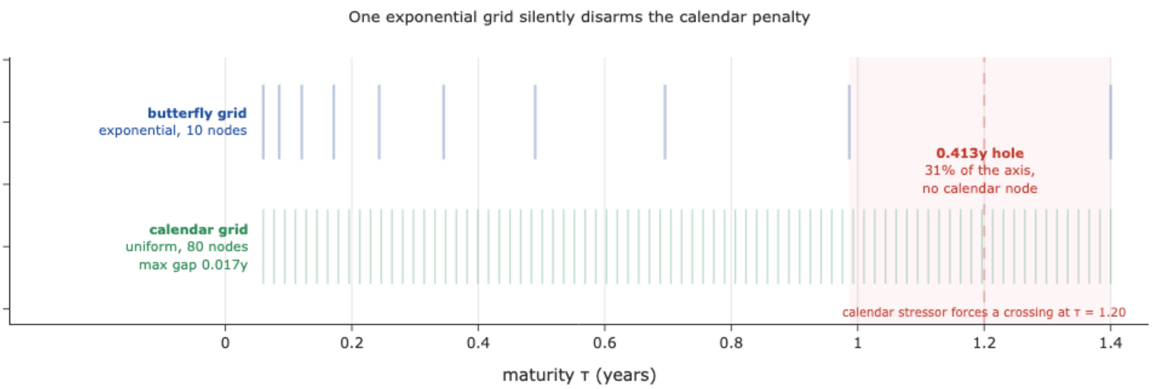

Figure 3.2: Constraint-specific collocation on the maturity axis $[0.06, 1.4]$. The literature's exponential grid (top, ten nodes) crowds the short end, where the butterfly constraint bites, but leaves a 0.413-year gap at the long end, 31% of the axis with no calendar node (shaded). The calendar stressor of Section 5.2 forces its crossing inside that gap (dashed), so the penalty reads zero on a surface that is violating. Giving the calendar constraint its own uniform grid (bottom, 80 nodes) brings the maximal gap down to 0.017 years.

3.1.2 Why one grid cannot serve both constraints

The bound explains how violations can occur between nodes. The next question is why such large gaps arise in practice. The answer is that the two constraints of Definition 2.4 and Definition 2.8 require different sampling patterns along the maturity axis:

Constraint	**Where it bites**	**Required grid in τ**
Butterfly $g \geq 0$	Short end, where small w amplifies the term $\frac{(w')^2}{4w}$ in (2.11)	Dense near the short end, for example an exponential grid
Calendar $\partial_\tau w \geq 0$	Anywhere along the maturity axis, since it involves a τ-derivative	Uniformly bounded node gaps, by Theorem 3.2

The asymmetry in the table is the same one flagged in Remark 2.10: butterfly is a second-derivative condition concentrated at one end of the axis, calendar a first-derivative condition that can fail anywhere on it. What matters for the calendar penalty is therefore the *largest* node gap, not the average spacing, and an exponential grid puts its largest gap at the long end. Using the same exponential grid for both constraints therefore leaves the calendar condition least controlled at the long end. This is Chataigner et al.'s caveat [4] given a concrete mechanism, and it is the collocation blind spot exhibited in Section 5.2.

3.1.3 Regularity of the model class: activations

The sampling grid is not the only source of failure. The choice of model class also matters.

The butterfly penalty needs w''_θ, so the corrector's activation must be twice differentiable, and with a curvature the training can actually shape. Recall the two canonical choices:

$$\mathrm{ReLU}(x) = \max(x, 0), \qquad \tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}. \tag{3.2}$$

Their second derivatives behave very differently. ReLU is piecewise affine, so $\mathrm{ReLU}''(x) = 0$ almost everywhere, and therefore $\partial_{kk}\mathrm{MLP}_\theta = 0$ almost everywhere [15, 2]. In contrast, tanh is smooth, with $\tanh''(x) = -2\tanh(x)\big(1 - \tanh^2(x)\big)$, so it provides non-zero curvature.

The implication for (2.11) is straightforward. With a ReLU corrector, w''_θ is, at almost every collocation node, just the second derivative of the SSVI prior. The penalty can still affect the corrector through the level and slope terms in g, but not through curvature, which is the part it is meant to control.

This does not mean that tanh is required. Any C^2 activation with non-zero curvature can be used, including GELU and SELU. The neural operators in Section 3.3 use GELU for this reason. We keep tanh in (3.4) because its bounded range helps keep the multiplicative correction stable near initialisation. The main restriction is therefore to avoid piecewise-affine activations, since they make the butterfly penalty ineffective through the curvature term.

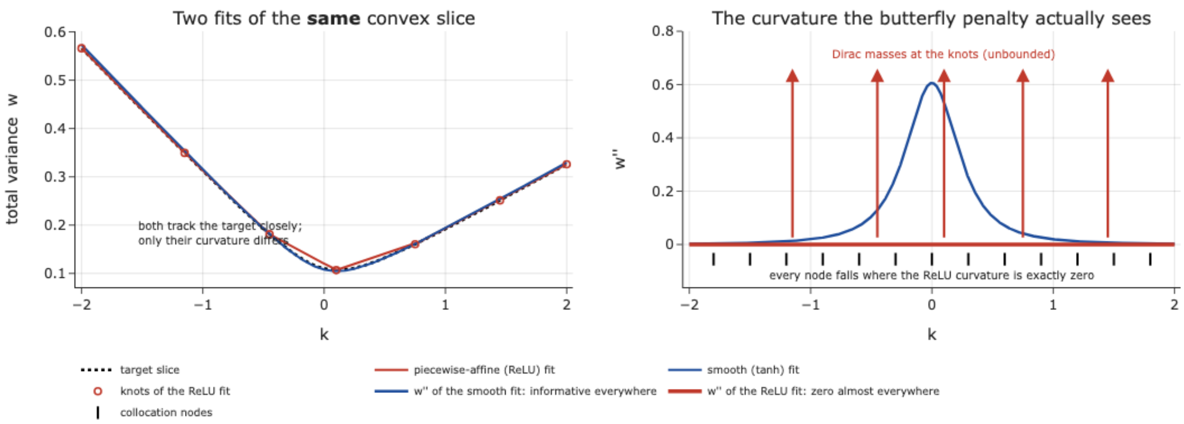

Figure 3.3: Curvature seen by the butterfly penalty. Left: ReLU and tanh networks both fit the same convex slice closely. Right: ReLU has $w'' = 0$ almost everywhere, while the smooth fit has non-zero curvature throughout. The penalty therefore sees no ReLU curvature at ordinary collocation nodes.

3.1.4 Materiality

The next question is how large a detected violation must be to be economically meaningful. An audit that flags every sign change of $\partial_\tau w$ or g, however small, is useless: a soft-constraint method leaves tiny numerical residuals everywhere, and calling them "violations" would make it difficult to distinguish numerical residuals from economically relevant violations. We need a threshold below which a violation is indistinguishable from quote noise and above which it is economically meaningful.

Definition 3.4 (Materiality threshold)**.** A violation is *material* if it exceeds $\mathrm{tol}_{\mathrm{mat}} = 10^{-3}$ in surface units: w-units for g-type quantities (through their effect on w), and $\partial_\tau w$-units for calendar.

The value 10^{-3} is not arbitrary; it is calibrated to the noise in the quotes themselves. Multiplicative implied-volatility noise of relative size ε propagates to total variance by the delta method, $\delta w/w = 2\,\delta\sigma/\sigma \approx 2\varepsilon$. With $\varepsilon \approx 0.5$–0.75% (half the 1–1.5% multiplicative band observed on w) and $w \approx 0.03$ at the long end, the noise band on w is about 4×10^{-4}. A calendar violation must therefore sustain $|\partial_\tau w| \gtrsim 10^{-3}$ over half a year to move w beyond this band. Below that, a "violation" is a numerical residual of the soft-constraint approach, not a monetisable opportunity; Hoshisashi et al. make the same observation for their DCNN [16].

Two consequences hold throughout the thesis. Every violation rate is reported twice, *strict* ($< -10^{-8}$, any sign change at all) and *material* ($< -\mathrm{tol}_{\mathrm{mat}}$, economically real), so the reader can always separate the two. And blind-spot detection compares the node grid and the audit grid *on the same material scale*: an earlier, magnitude-inconsistent detector false-alarmed on residuals four orders of magnitude below materiality (Section 5.2), which the material-scale comparison eliminates.

3.1.5 The audit-grid methodology

Remark 3.1 and Theorem 3.2 together say that a soft penalty certifies its constraint only on the finite set where it is sampled, and that the violation it can hide between samples grows with the gap. A penalty reporting zero is therefore not evidence of an arbitrage-free surface; it is evidence about the sampling set. For this reason, each model is also evaluated on an independent audit grid that is not used during training. Four rules implement this.

(A1) **Two grids, never one.** Every penalty is evaluated twice: once on its own collocation grid, which is what the training loss sees and minimises, and once on an *independent audit grid* that is finer than, and disjoint from, every collocation grid. The training grid measures how well the constraint was *imposed*; the audit grid measures whether it actually *holds*. Only their disagreement is informative, which is why the two must not share nodes.

(A2) **Compare on the material scale.** Node and audit statistics are compared at the materiality tolerance of Definition 3.4, not in raw units, so that numerical residuals are not mistaken for arbitrage. A *blind-spot flag* is raised precisely when the nodes are materially clean while the audit is materially violating, i.e. when the training loss would report success on a surface that an honest audit rejects.

(A3) **For operators, audit unseen days too.** A per-day smoother is audited between its nodes; an operator must additionally be audited on *days* disjoint from training, because its penalty is a statement about training-day surfaces only, with nothing guaranteeing that the constraint transports to a new day (Section 5.3). The blind spot thus acquires a second axis, space and time, for operators.

(A4) **Gate every stress and invariance claim.** Before an arbitrage comparison on a stressed or subsampled input is read, a fit-validity gate certifies that the model actually fitted that input. Without the gate, a model that simply failed to fit could be mistaken for one that fitted cleanly, and a collapsed operator's trivially perfect invariance could be reported as a genuine property (Sections 5.2 and 5.3).

3.2 Neural networks as surface models

With the theory of soft constraints in place, we turn to the model classes they are meant to constrain, starting with the simpler of the two.

3.2.1 From parametric forms to learned functions

Here, we view a neural network simply as a parameterised family of functions used to approximate the volatility surface. The basic building block, the *multilayer perceptron* (MLP), alternates affine maps and a fixed nonlinearity. With activation $\phi_a : \mathbb{R} \to \mathbb{R}$, depth D and widths $n_0, \dots, n_D$, it is the map $x \mapsto z_D(x)$ defined recursively by

$$z_0 = x, \qquad z_\ell = \phi_a\big(W_\ell z_{\ell-1} + b_\ell\big) \quad (\ell < D), \qquad z_D = W_D z_{D-1} + b_D, \tag{3.3}$$

with parameters $\theta = (W_\ell, b_\ell)_{\ell \le D}$ and ϕ_a applied componentwise. In our setting the input is a point of the surface, $x = (k, \tau)$, and the output is a scalar. Figure 3.4 shows the object.

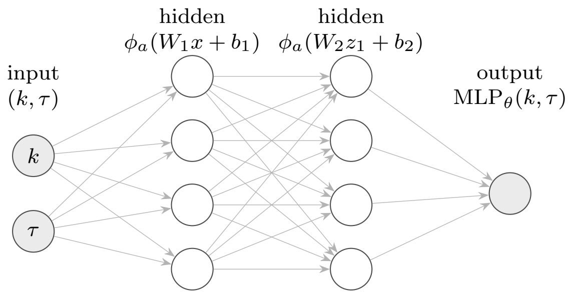

Figure 3.4: A multilayer perceptron (3.3): it maps *one point* of the surface to *one number*. Each arrow is a learned weight; each circle applies the activation ϕ_a. To obtain a whole surface, the network is evaluated point by point, and the parameters θ are fitted to one day's quotes.

The classical justification for this class is that it is dense in the space of continuous functions and, for our purposes, dense in derivatives as well.

Proposition 3.5 (Universal approximation; Hornik [15]). *If ϕ_a is continuous, bounded and non-constant, then for every compact $K \subset \mathbb{R}^d$ the set of one-hidden-layer MLPs is dense in $C(K)$ for the uniform norm. If, in addition, $\phi_a \in C^m$, density holds in the C^m norm on compacts: the network approximates the target* and *its derivatives up to order m simultaneously.*

The C^m statement is the relevant one here, and not a technicality. The quantities penalised in (3.1) are w_θ'' and $\partial_\tau w_\theta$, so approximation power must reach the derivatives, not only the level; and by the curvature argument above, the activation must be at least C^2 for the butterfly channel to be trainable at all.

3.2.2 Exact derivatives by automatic differentiation

The approximation result concerns the existence of these derivatives. In practice, they must also be computed efficiently during training.

Automatic differentiation (AD) computes derivatives *exactly*, to floating-point precision, rather than by finite differences: the program defining w_θ is a composition of elementary operations whose local derivatives are known in closed form, and the chain rule is applied mechanically along the computational graph [2]. The idea is not native to finance. It is the mechanism underlying physics-informed neural networks [26, 21], where a differential equation is imposed by penalising its residual at sampled points, which is only meaningful because the residual can be evaluated exactly.

Two consequences matter. First, the surface derivatives entering (3.1), namely w_θ' and w_θ'' in k and $\partial_\tau w_\theta$ in τ, are exact functionals of θ; the penalties are therefore differentiable in θ and their gradients are themselves exact, a nested application of AD. Second, exactness is *testable*: the AD derivatives can be checked against the closed-form SVI derivatives (2.17) in k and against central differences in τ, which is how a silent wiring error in a derivative path would be caught. The neural operators of Section 3.3, following the reference design [29], instead resolve their penalties by finite differences on a grid; the gap between what that grid sees and what an independent audit sees is itself one of this thesis's measurements.

3.2.3 The multiplicative prior-corrector architecture

Used alone, an MLP would discard everything the parametric theory provides. The architecture of [1], adopted in Section 5.2, instead keeps the certified surface and learns only the deviation from it:

$$w_\theta(k,\tau) = \underbrace{w_{\mathrm{SSVI}}(k,\tau)}_{\text{certified prior}} \times \underbrace{\exp\big(\mathrm{MLP}_\theta(\tilde{k},\tilde{\tau})\big)}_{\text{positive corrector}}, \tag{3.4}$$

where the SSVI prior is fitted to the day's quotes and certified arbitrage-free by Theorem 2.25, and the corrector is a tanh MLP on rescaled inputs, initialised so that the exponential is ≈ 1 and training therefore *starts at the prior.*

Three properties of (3.4) are used repeatedly. *Positivity* of w_θ holds by construction, since a product of a positive prior and an exponential cannot vanish or change sign. *Additivity in log-space*, $\log w_\theta = \log w_{\mathrm{SSVI}} + \mathrm{MLP}_\theta$, means the corrector models the *relative* deviation from the prior, which also explains a failure mode observed in Section 5.2: a wrong prior is worse than a flat one, because the corrector must first spend capacity undoing it. *Inheritance*: wherever the corrector is flat, all derivatives of w_θ, hence g and $\partial_\tau w$, are the prior's, and the prior carries a theorem.

3.3 Neural operators

3.3.1 What is being learned: functions versus operators

The models above learn a *function*: given the numbers (k, τ), return a number. Their parameters are fitted to one day's quotes, and a new day requires a new optimisation from scratch. Neural operators change the object being learned. Instead of one surface, they learn the *map that produces surfaces.*

The distinction is one of level. A function maps numbers to numbers, for instance $(k, \tau) \mapsto w$. An operator maps *functions to functions*: its input is not a number but an entire object, and so is its output; differentiation is the canonical example, sending f to f'. In our problem, the operator of interest is

$$\mathcal{G} : \underbrace{a}_{\text{a day's quote cloud}} \longmapsto \underbrace{w(\cdot,\cdot)}_{\text{the whole smoothed surface}}, \qquad a = \big\{(k_i, \tau_i, \sigma_{\mathrm{IV},i})\big\}_{i \le p}, \tag{3.5}$$

which takes an entire configuration of quotes and returns an entire surface, evaluable at any (k, τ) we care to query. Formally a is a discrete measure, a discretisation of an element of a function space, and $\mathcal{G}$ acts between function spaces. Figure 3.5 contrasts the two levels.

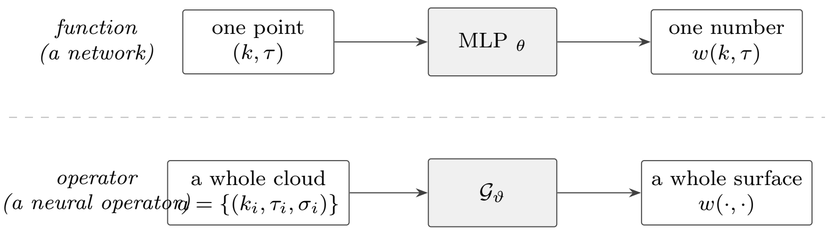

Figure 3.5: The two levels of learning. Top: a network maps *numbers to numbers*; its parameters encode one surface, and a new day requires refitting. Bottom: an operator maps a *whole quote configuration to a whole surface*; its parameters encode the smoothing *procedure*, so a new day is a single forward pass.

The main practical advantage is that the operator is trained once on the historical sample and can then be applied to new days with a single forward pass. Two structural properties distinguish a genuine operator from a plain network fed with a flattened input. It must accept inputs of *any size and layout*, since the number of quotes and their locations change every day, which requires invariance under permutation of the index i and tolerance of variable p. And it aspires to *discretisation invariance*: refining the sampling of the input should not change the answer.

Definition 3.6 (Discretisation invariance, after [20])**.** A parametric map $\mathcal{G}_\vartheta$ on point-cloud inputs is discretisation-invariant at u if, for any sequence of discretisations $a^{(n)}$ of u refining to u, the outputs $\mathcal{G}_\vartheta(a^{(n)})$ converge to a common limit $\mathcal{G}_\vartheta(u)$, uniformly on compacts. Empirically one tests a finite version: the output surface must be stable under subsampling of the input quotes. Section 5.3 gates this test on fit validity, because a collapsed operator that outputs a constant is trivially invariant.

We consider two architectures for learning the output surface: DeepONet and graph neural operators.

3.3.2 DeepONet: a learned global basis

The first answer writes the output function in a *basis*. DeepONet [25] represents the operator in separated form,

$$\mathcal{G}_\vartheta(a)(k,\tau) = \sum_{j=1}^{p^*} \underbrace{b_j\big(E(a)\big)}_{\text{coefficients: depend on the quotes}} \underbrace{t_j(k,\tau)}_{\text{basis functions: fixed}} + b_0. \tag{3.6}$$

Three pieces do the work. The *encoder* E compresses the irregular quote cloud into a fixed-length vector and must be permutation-invariant, for which we use the Deep Sets form

$$E(a) = \rho_E\Big(\sum_{i\le p} \phi_E(k_i,\tau_i,\sigma_i)\Big), \tag{3.7}$$

the general shape of a continuous permutation-invariant map [30]: the inner sum discards the ordering of the quotes, and the outer network restores expressiveness. The *branch* network $b:\mathbb{R}^{d_E}\to\mathbb{R}^{p^*}$ turns that summary into p^* coefficients. The *trunk* network $t:\mathbb{R}^2\to\mathbb{R}^{p^*}$ produces p^* smooth functions of the query point, and it does *not* see the quotes at all. The day's information therefore enters only through the coefficients, while the shapes available are fixed once training is over.

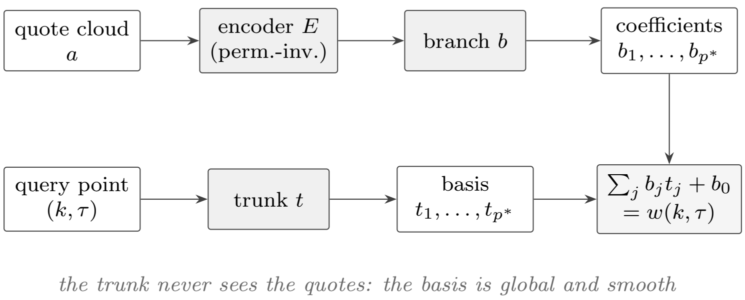

Figure 3.6: DeepONet (3.6). The quotes determine only the *coefficients* (top path); the *shapes* come from a trunk basis that is the same for every day (bottom path). The output is their weighted sum.

The construction is backed by an operator version of universal approximation.

Proposition 3.7 (Chen–Chen [5]; Lu et al. [25])**.** *Let $\mathcal{G}$ be a continuous operator from a compact subset of a Banach space of functions to $C(K)$, with $K\subset\mathbb{R}^d$ compact. Then for every $\varepsilon>0$*

there exist networks b, t, and an encoder acting on point evaluations of the input, such that the separated form (3.6) *approximates $\mathcal{G}$ uniformly to within ε.*

The structural consequence exploited in Sections 5.3 and 6 is visible directly in (3.6): for *every* input, the output lies in the span of the same p^* trunk functions, a global, input-independent, smooth basis. Whatever the quotes do, the surface inherits the trunk's regularity, and fine-scale curvature is simply not expressible. This is why the DeepONet turns out to be largely immune to butterfly violations, its residual weakness being a thin calendar blind spot, at the price of visibly coarser fits.

3.3.3 Graph neural operators: a learned local kernel

The second answer builds the output *locally* from nearby data rather than from a global basis. The neural-operator programme [23, 24, 20] models $\mathcal{G}$ as a composition of *kernel integral layers*,

$$v_{\ell+1}(x) \;=\; \phi_a\Big(W_\ell\, v_\ell(x) \;+\; \int \kappa_\ell(x,y)\, v_\ell(y)\, \nu(\,\mathrm{d}y)\Big), \tag{3.8}$$

with learned kernels κ_ℓ and ν the input measure. The integral is what makes this an operator rather than a network: the value at x depends on the input *everywhere*, weighted by a learned kernel, instead of on a fixed-length vector of features.

On an irregular quote cloud the integral cannot be computed exactly, so it is discretised by *message passing* over a graph. For each query point x one keeps its m nearest quotes $N_m(x)$ and forms

$$\int \kappa_\ell(x,y)\, v_\ell(y)\, \nu(\,\mathrm{d}y) \;\approx\; \frac{1}{m} \sum_{y \in N_m(x)} \kappa_\ell\big(x, y, a(y)\big)\, v_\ell(y), \tag{3.9}$$

a Monte-Carlo quadrature of (3.8) localised to the neighbourhood. Operator Deep Smoothing [29] instantiates exactly this design for implied volatility and is the reference for the graph-neural-operator (GNO) variant of Section 5.3.

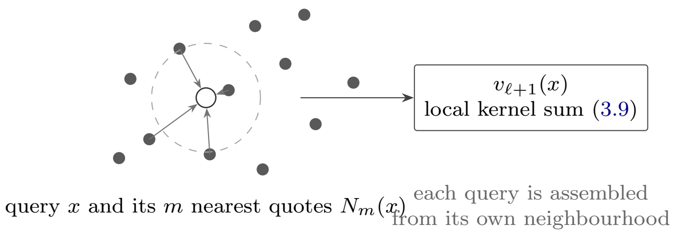

Figure 3.7: Graph neural operator (3.9). The surface value at a query point x is assembled by message passing from the m nearest quotes, a discretisation of the kernel integral (3.8). Neighbourhoods change as x moves, which buys local adaptivity and, as Remark 3.8 explains, costs global control of curvature.

Remark 3.8 (Inductive bias and the audit)*.* The local structure of the GNO improves flexibility, but it also makes global curvature harder to control. The output at x is assembled from the nearest quotes, so the operator follows local smile features closely and leads the accuracy table of Section 5.3. At the same time, nothing global constrains the assembled surface's second derivative. Neighbourhoods change discretely as x moves and each carries its own kernel response, so the output exhibits fine-scale curvature at the scale of the quote spacing. A butterfly penalty resolved by finite differences on a coarse grid, here the 22×9 grid of the reference design with maximal τ-gap 0.176y, is blind to curvature at finer scales in exactly the sense of Theorem 3.2; an independent 61×31 audit is not. The empirical counterpart, that the accuracy leader is also the most arbitrage-dirty model in the study, is Section 5.3's central exhibit.

3.3.4 Why this matters here

Table 3.1: The three model classes compared. The last row is what Chapters 5–6 measure.

	per-day MLP smoother	DeepONet	graph neural operator
learns a	function (one surface)	operator	operator
new day costs	a full refit	one forward pass	one forward pass
output shapes	free	span of a global basis	locally assembled
regularity	from prior and penalty	by construction	none imposed
derivatives for penalties	exact (AD)	exact (AD)	finite differences
expected failure mode	between nodes	coarse fits	fine-scale curvature

Table 3.1 sets the two operator families side by side with the per-day smoother. Read across the last row, it also states what each is expected to get wrong, and the empirical chapters test exactly those predictions.

Both operator families are trained with soft penalties, and both therefore inherit the question of Section 3.1: on which set is the constraint actually enforced? For a per-day smoother the answer concerned the space between collocation nodes. For an operator a second gap opens, because the penalty is minimised on *training days* and nothing transports that property to a day the model has never seen. Section 5.3 measures both gaps.

3.4 The reconstruction problem

We can now state the reconstruction problem used in the rest of the thesis, and make explicit the one criterion by which its framing departs from the literature.

Given a single day's quotes $\{(k_i, \tau_i, \sigma_{\mathrm{IV},i})\}_{i=1}^{p}$, sparse, irregular and noisy, the task is to construct a surface $\widehat{w} : \mathbb{R} \times (0, \infty) \to (0, \infty)$ that

(i) **fits** the observed quotes;
(ii) **is arbitrage-free**, i.e. satisfies Definitions 2.4 and 2.8 and hence, by Lemma 2.6 and Proposition 2.9, admits no static arbitrage;
(iii) **generalises** to quotes held out from the fit; and, the criterion this thesis adds,
(iv) **is honest about its own guarantee**: it states the set on which (ii) is actually enforced, and the economic size of any residual violation off that set.

Criteria (i)–(iii) are the standard trio, and every method in Chapters 5–6 is scored on them identically. Criterion (iv) is the one this thesis argues cannot be omitted: a method can satisfy (i)–(iii) while failing (ii) between its samples, or, for an operator, on days it never saw. Good performance on these three criteria does not by itself guarantee that the surface is arbitrage-free away from the sampled points. We therefore also report where the no-arbitrage condition is enforced and the size of violations outside that set.

Each criterion is scored by a specific instrument, all shared across families. The unified protocol of Chapter 4 scores (i)–(iii). The audit-grid methodology (A1)–(A4), together with the materiality calculus of Definition 3.4, scores the guarantee half of (iv). And Chapter 6 scores its cost half, turning a violation into a repair rate, a negative density mass, or a degraded signal through the identity $\sigma^2_{\mathrm{loc}} = \partial_\tau w/g$ of Proposition 2.12. With the problem and its scoring fixed, the benchmark can begin.

Chapter 4

Data and Methodology

4.1 Data

All experiments use end-of-day SPX option quotes from OptionMetrics IvyDB US, accessed through WRDS under secid `108105`. The sample runs from 2 January 2018 to 29 August 2025 and contains 1,926 trading days, spanning several distinct market regimes: the February 2018 volatility spike known as Volmageddon, the COVID-19 crash, the 2022 monetary tightening cycle, and the 2024–2025 period.

Five OptionMetrics tables enter the pipeline: the option-quote table, the zero-coupon yield curve, the forward-price table, the proprietary volatility surface, and a historical-volatility series. The proprietary surface is used only as an industry benchmark and never as training input. The CBOE `VIX_History` file provides the published VIX series used for the external validation in Section **??**.

NB01 transforms the raw tables into a single reproducible Parquet dataset consumed by every subsequent notebook. Each cleaning rule is stated explicitly and enforced by an automated test before the dataset is written to disk.

4.1.1 Coordinates and feature engineering

For each quote, we define the maturity $\tau = \frac{\text{DTE}}{365}$ using the ACT/365 convention. We also compute the mid price and quoted half-spread as

$$\text{mid} = \frac{\text{bid} + \text{ask}}{2}, \qquad \text{halfspread} = \frac{\text{ask} - \text{bid}}{2}.$$

Log-moneyness is defined by $k = \log\left(\frac{K}{F}\right)$, where F denotes the forward associated with the quote's expiry. Strikes are rescaled from the OptionMetrics convention using $K = \frac{\texttt{strike\_price}}{1000}$.

A quote is classified as *out of the money* when $k \geq 0$ for a call or $k < 0$ for a put. Only the OTM side is retained, since it is the more liquid and informative side of the smile and is the object every model in this thesis reconstructs.

4.1.2 Cleaning rules

Cleaning is implemented as a lazy query plan so that filters reduce the data before materialisation.

Quality filters. A quote is retained only if its bid is strictly positive and its market is not crossed: $0 < \text{bid} \leq \text{ask}$. Maturity must lie between 7 and 730 calendar days. Shorter maturities are strongly affected by microstructure noise, while longer maturities are comparatively illiquid. The call and put indicator must belong to $\{\mathrm{C}, \mathrm{P}\}$.

Missing implied volatility. Approximately 5.7% of the raw rows have no reported implied volatility. Delta and vega are also missing on exactly these rows, so the Greeks cannot be used to recover the volatility. About 99.98% of the affected observations are deep in-the-money options. Their prices are almost entirely intrinsic value, which makes a reliable Black inversion difficult relative to the quoted spread.

Of the 8.5 million raw rows, only about 118 of those lacking an implied volatility are OTM. Since the thesis reconstructs the OTM surface and this residual count is negligible, all rows with missing implied volatility are removed rather than repaired.

4.1.3 Forwards and discount factors

The quote-level forward is retained whenever it is available. Missing values are filled from the OptionMetrics forward table through a left join on (`secid`, `date`, `expiry`, `settlement`). Every retained quote therefore has a valid forward for the construction of the log-moneyness.

The zero-curve rate at the nearest available horizon is then attached through an as-of join for each trading date. The corresponding discount factor is $D = e^{-r\tau}$.

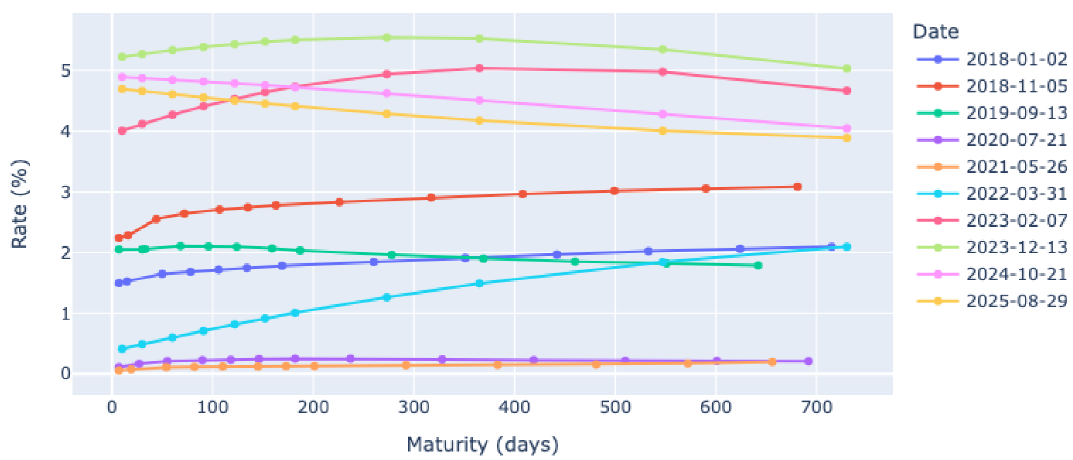

Figure 4.1: OptionMetrics zero-coupon yield curves on ten trading dates spanning the sample, one per year. The near-zero curves of 2020–2021, the steeply rising 2022 curve, and the 4–5% levels of 2023–2025 trace the monetary regimes the dataset spans; these rates enter the discount factor $D = e^{-r\tau}$ used for quote validation.

4.1.4 One expiry, one contract

Standard SPX options are generally AM-settled, while SPXW weekly options are PM-settled. The two families can share an expiration date while trading at slightly different prices. Retaining both would create two implied volatilities for the same approximate (k, τ) location.

Each collision is therefore resolved deterministically in favour of the AM-settled SPX contract. A diagnostic verifies that no pair $(\texttt{date}, \texttt{expiry})$ retains more than one contract type.

Contract identity is preserved throughout the procedure. Expiries are not merged by rounded values of τ because this produces artificial sawtooth patterns in the term structure.

4.1.5 Independent implied-volatility validation

The transformation from raw quotes to (k, τ, w) depends on the forward, discount factor, coordinate change, and implied-volatility inversion. A silent error in any of these components would contaminate the complete dataset.

We therefore recompute implied volatility independently for every eligible quote using vectorised bisection on the Black–76 price. The result is compared with the OptionMetrics value $\mathrm{iv}_{\mathrm{OM}}$. Inversion is attempted only when the observed price satisfies the relevant no-arbitrage bounds. This provides an initial quote-level arbitrage screen.

The median difference between the recomputed volatility and $\mathrm{iv}_{\mathrm{OM}}$ is small enough to validate the forward, the discount factor, the coordinate change and the inversion code jointly.

4.1.6 Reproducibility guardrails

All diagnostics are converted into blocking tests before the clean Parquet file is written. The notebook stops rather than producing a dataset when any required assumption fails. The tests verify that only the expected SPX secid remains, that every maturity satisfies the DTE filter, and that every bid is positive. They also exclude crossed markets and require complete, finite values of k and the OTM indicator. Finally, they verify that no expiry retains multiple contract types.

Once all tests pass, the clean dataset is written to disk. Figure 4.2 reports the number of retained OTM quotes through time. Figure 4.3 illustrates the sparse and irregular geometry of the resulting input on a representative trading day.

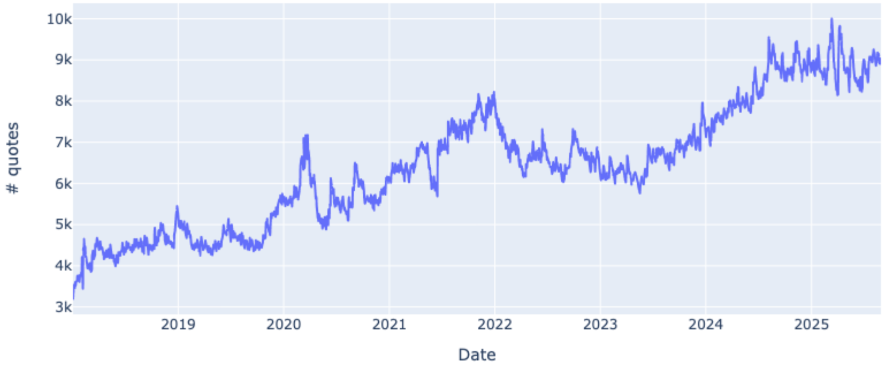

Figure 4.2: Number of OTM quotes retained on each trading day after application of the NB01 cleaning pipeline.

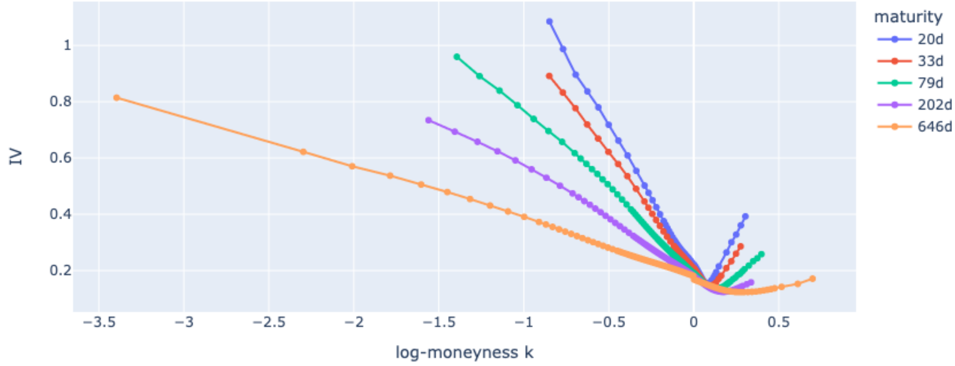

Figure 4.3: Cleaned OptionMetrics smile cross-section for one trading day across all retained expiries. The observations form the sparse, irregular, and noisy input consumed by the reconstruction methods.

4.2 Evaluation domains and unified protocol

Two evaluation domains are used throughout the thesis. The *full quoted domain* retains the complete observed range and therefore includes deep OTM puts with log-moneyness close to $k = -2.2$. The *paper domain* is restricted to $k \in [-0.6, 0.4]$, DTE ≥ 10. This second domain facilitates comparison with the existing literature.

For each trading day, 20% of the observations are held out within each expiry slice under a deterministic seed, so that every model family is scored on exactly the same quotes. The construction is given in Section 4.2.1.

Arbitrage is audited both on the quoted domain and on a capped extension. The extended range is twice the observed k-range and is clipped to $|k| \leq 1.5$. This restriction prevents economically irrelevant strikes from inflating the measured violation rates. For example, auditing at $k = -4.4$ would correspond to a strike near one percent of spot.

The audit follows rules (A1) to (A4) of Section 3.1: independent grids, blind-spot flags, and separate strict and material statistics with $\mathrm{tol}_{\mathrm{mat}} = 10^{-3}$. Stress and invariance conclusions are reported only when their validity gates pass.

Aggregate arbitrage statistics are also conditioned on fit validity through the `gated_*` columns and `n_fit_ok`. A day on which a model fails to fit is excluded from its arbitrage comparison. Otherwise, numerical non-convergence could be misreported as an economically meaningful arbitrage violation.

4.2.1 Data splits

Two distinct splitting schemes are used, because the two model families answer different questions. Both are stated here rather than in the results chapters.

Per-day models. The parametric benchmark of Section 5.1 and the deep smoother of Section 5.2 are fitted independently on each trading day, so there is no temporal split to make: generalisation is measured *within* the day. On every day, 20% of the quotes are held out per expiry slice under a deterministic seed crc32(date), and the remaining 80% are used for fitting. Because the seed depends only on the date, every model family in this thesis is scored on exactly the same held-out quotes, and staging and production runs remain comparable day by day.

Operators. An operator is trained once across many days, so the split must be temporal, and it is *chronological* rather than random: the 1,926 trading days are divided in order into 1,155 training days, 385 validation days and 386 test days. The test set is therefore the final year and a half of the sample, and no training day post-dates a test day. Every operator number reported in Section 5.3 is consequently a temporal generalisation claim rather than an in-sample one. The per-day 20% hold-out is applied *in addition*, so that quote-level and day-level generalisation are measured separately. Validation days are used only for monitoring and for the choice of training length; no architecture or hyperparameter is selected on the test set.

Synthetic data. The synthetic bank used for pretraining is generated independently of the real sample and never enters validation or test. Its role is to answer a transfer question, whether certified synthetic surfaces can substitute for data scale in the academic-data regime, and it is reported as such.

4.2.2 Compute infrastructure and execution

The production runs are too long for a laptop and are executed on a shared multi-core Linux compute server at Télécom Paris, accessed over SSH through the institutional gateway. The environment is a dedicated conda environment with a CPU build of PyTorch; no GPU is used anywhere in this thesis, which is worth recording because it constrains the operator training budgets discussed in Section 5.3.

Parallelisation by year. The dominant cost is the per-day deep smoother of Section 5.2, which fits one network per trading day and is therefore embarrassingly parallel across days. The 2018–2025 sample is sharded by calendar year into eight independent jobs, one per year, executed concurrently on eight cores, with a ninth process handling the synthetic experiments. Each shard writes to its own output namespace, so no two processes contend for the same file, and shards can be re-run individually after a failure without repeating the others. This reduces a sequential run of roughly thirty hours to a wall-clock time of a few hours.

Detached execution and reproducibility. Notebooks are executed non-interactively with `nbconvert`, so the executed notebook including all outputs is preserved as an artefact alongside the Parquet files. The whole pipeline is launched inside a `tmux` session, which detaches the computation from the SSH connection: the run survives disconnection, and progress is monitored by tailing per-shard log files. Results are transferred back with `rsync`. The practical consequence for reproducibility is that every reported number is traceable to an executed notebook and a log file, both retained, rather than to an interactive session that no longer exists.

With the data cleaned and the scoring protocol fixed, the benchmark can begin.

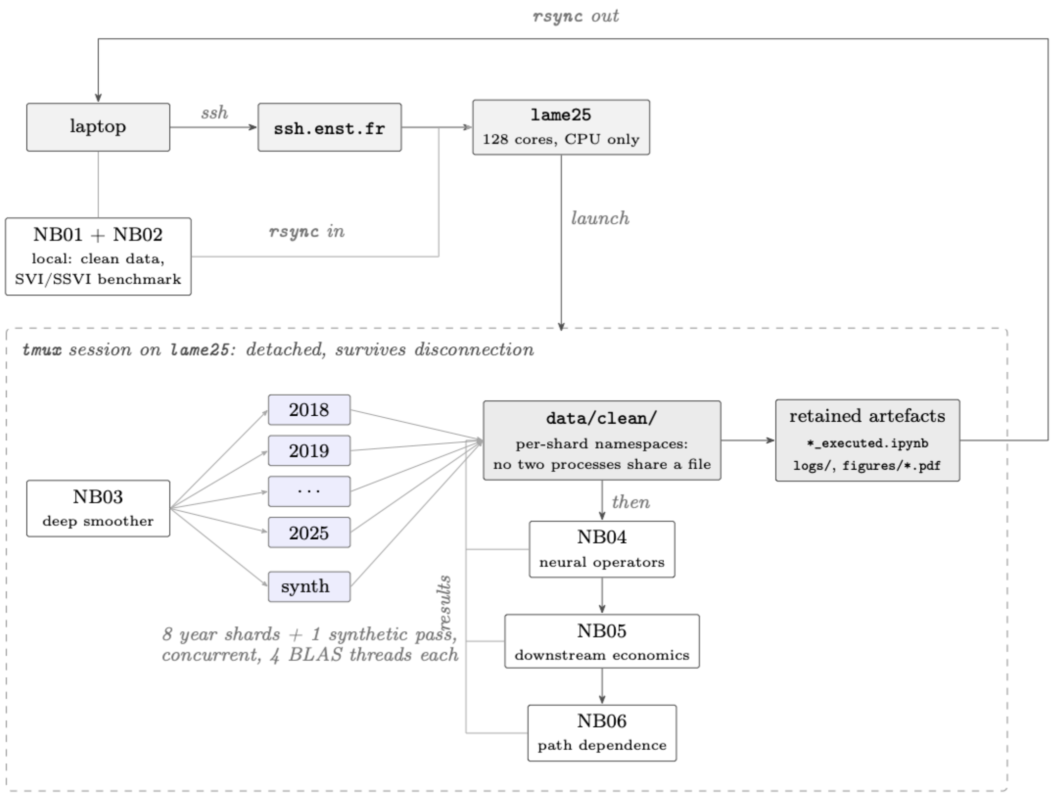

Figure 4.4: Execution of the production pipeline. NB01 and NB02 run locally and their outputs are transferred once; NB03 to NB06 then run on a single CPU node inside a `tmux` session, so the computation survives the SSH connection being closed. NB03 fits one network per trading day and is the dominant cost, so it is sharded by calendar year into eight independent jobs, plus a ninth process handling the synthetic stress experiments; each job caps its BLAS thread count and writes to its own output namespace, so no two processes contend for a file and a failed year can be relaunched alone. NB04, NB05 and NB06 consume the shared `data/clean/` directory and run sequentially afterwards. Every reported number is traceable to a retained executed notebook and its log file.

Chapter 5

Method Families and Empirical Results

This chapter implements and evaluates the three method families under the unified protocol of Chapter 4: the parametric benchmark (Section 5.1), the derivative-constrained deep smoother and the collocation blind spot (Section 5.2), and the neural operators (Section 5.3).

5.1 The parametric benchmark: SVI and SSVI

This section establishes the benchmark used to evaluate the later methods, and validates the machinery that the rest of the thesis reuses against the source paper itself.

5.1.1 Replication of Gatheral–Jacquier

The raw, natural, and Jump–Wings parameterisations in Lemmas 2.14–2.17 are implemented directly in closed form. As a consistency check, we round-trip 2,000 randomly generated parameter sets through raw→natural→JW→raw. The largest reconstruction error is 2.7×10^{-10}. This test also revealed the missing $-b\sigma\sqrt{1-\rho^2}$ term in the published inverse transformation discussed in Remark 2.15: without this term, the round trip does not recover the original parameters.

As a further validation, we apply the transformations to the Vogt smile of Example 2.18. The resulting Jump–Wings parameters reproduce the published values

$$(v_t, \psi_t, p_t, c_t, \tilde{v}_t) = (0.01742625,\ -0.1752111,\ 0.6997381,\ 1.316798,\ 0.0116249) \tag{5.1}$$

to the reported precision *(Notebook 02, figure "JW of Vogt".)*. We then reproduce the repair procedure proposed in the paper, keeping (v_t, ψ_t, p_t) fixed while moving $(c_t, \tilde{v}_t)$ to the boundary of the arbitrage-free region. This gives $c_t' = 0.3493158$ and $\tilde{v}_t' = 0.0154818$, exactly as reported. The repair raises the minimum Durrleman value for the slice in Figure 2.1 from $\min_k g = -0.0329$ to $+0.2704$, thereby removing the region of negative implied density *(Notebook 02, figure "Vogt repair".)*.

The slice calibration routine follows the quasi-explicit decomposition of [14]. An outer Nelder–Mead optimisation is performed over (m, σ); conditional on each candidate pair, the remaining three parameters are obtained from a constrained linear least-squares problem, exploiting the fact that w is linear in those parameters once (m, σ) are fixed. On a synthetic test slice the procedure recovers the generating surface to $\mathrm{RMSE}(w) = 9.9 \times 10^{-5}$, and the recovered slice stays comfortably butterfly-arbitrage free: $\min_k g = 0.49$ over the quoted range and 0.32 over the wider interval $k \in [-2, 2]$. The wide-range minimum is attained at the edge of the audit interval rather than inside it, so the margin reported is the one this audit range can certify, not a global bound. Beyond the last quote the calibrated slice separates from the ground truth, as it must, since the fit is extrapolating there; the wide-range audit checks the behaviour of that extrapolation.

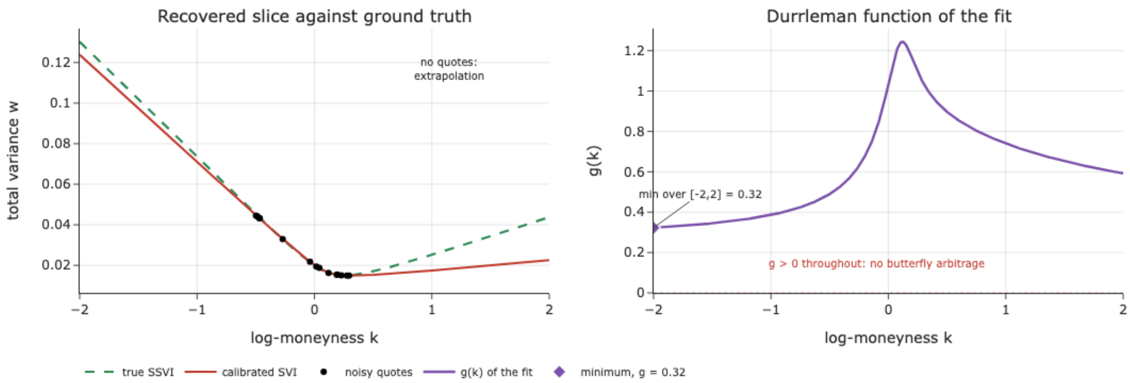

Figure 5.1: Quasi-explicit SVI calibrator validated on synthetic data. Left: the recovered slice against the SSVI ground truth; the shaded bands carry no quotes, so the two curves are free to separate there. Right: the Durrleman function of the fit on the wide audit range $k \in [-2, 2]$, positive throughout, with its minimum marked. Source: Notebook 02, cell 13.

The conditions in Propositions 2.20–2.22 are implemented as executable numerical certificates. We verify them on the three φ families considered in Example 2.23, recovering the published condition values in each case, and we confirm numerically that no two adjacent slices cross. The interpolation and extrapolation constructions in (2.29)–(2.30) are likewise checked: the interpolated slice inherits butterfly-freeness from the convex combination of prices, and the extrapolated slice dominates the last calibrated one at every strike. Finally, we implement the crossedness penalty of [14, Def. 5.1] on a deliberately stressed synthetic term structure, where one short-maturity slice is inflated so as to force a crossing. The penalty reduces total crossedness from 0.422 to 0.000, eliminating the calendar violation.

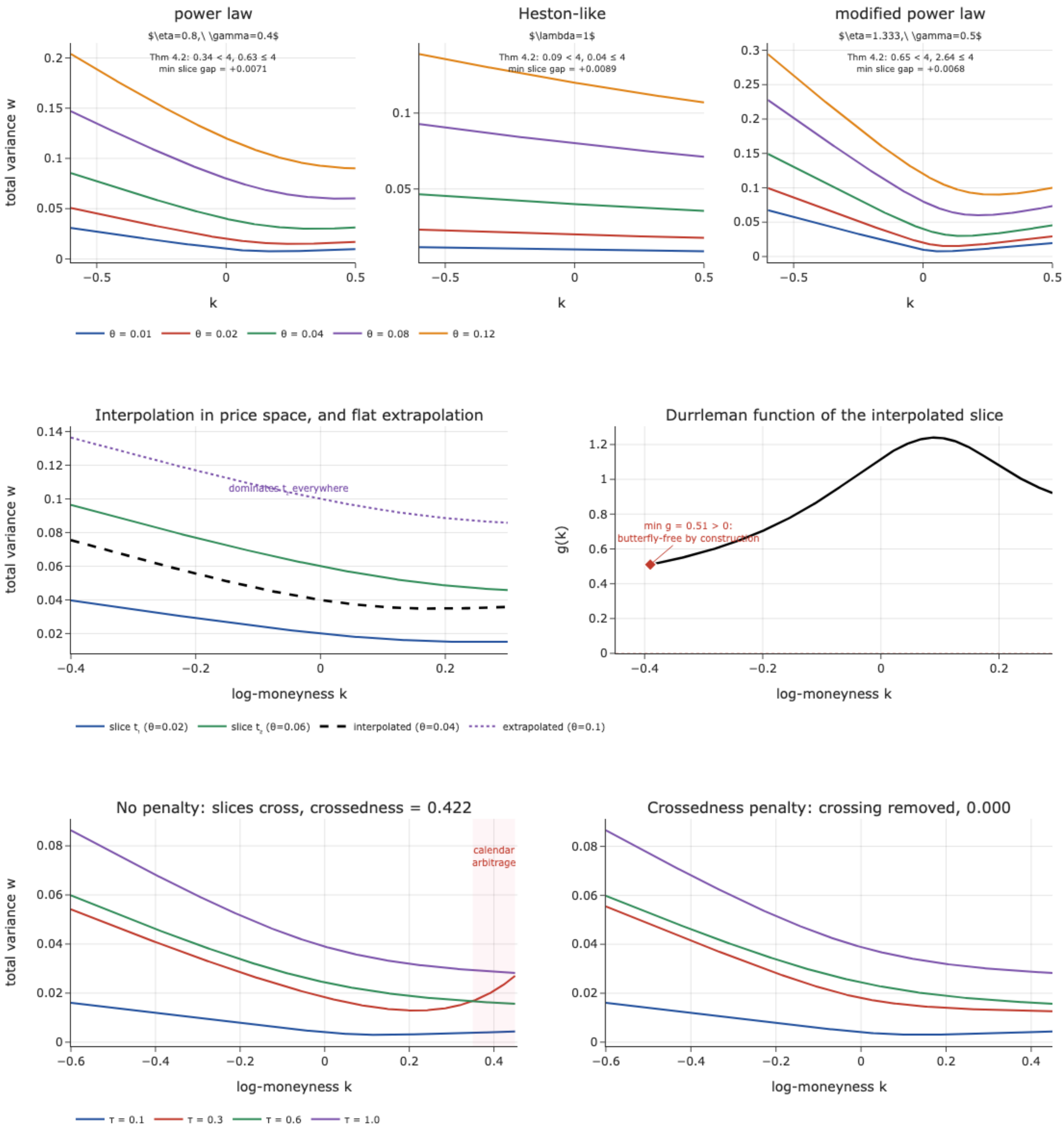

Figure 5.2: SSVI construction and its arbitrage-free operations. **Top:** the three φ families of Example 2.23, each annotated with its Theorem 4.2 condition values and with the smallest gap between adjacent slices, positive in every case (Proposition 2.20). **Middle:** arbitrage-free interpolation in price space and flat extrapolation; the right panel shows the Durrleman function of the interpolated slice, positive throughout with its minimum marked. **Bottom:** calendar repair by a crossedness penalty (paper §5.2). Without the penalty (left) a shorter slice rises above a longer one over the shaded interval; with the penalty (right) the crossing is gone.

5.1.2 Results on real data

For each trading day we fit a *per-slice* SVI surface and a *surface-wide* SSVI, both with the same inverse-spread and IV-target weights, and put each day through the full protocol: crc32 hold-out, g-audits on the quoted and wide grids, slice crossedness, and the paper's two repair procedures applied to the real fits.

Table 5.1: Parametric benchmark under both evaluation domains (8 days; 258 slices on the full quoted domain, 246 on the paper domain $k \in [-0.6, 0.4]$, DTE ≥ 10). *(Notebook 02, figure "benchmark summary table".)*

	Full domain		Paper domain	
	SVI	SSVI	SVI	SSVI
In-sample RMSE (vp)	0.635	1.957	0.548	1.810
Hold-out RMSE (vp)	0.640	1.913	0.522	1.730
Butterfly-violating slices	12.0%	0%	7.7%	0%
Median crossedness	$2.07 \times 10^{-2\,\dagger}$	0	$1.14 \times 10^{-2\,\dagger}$	0

Table 5.2: Full-domain results by maturity bucket. Violations concentrate almost entirely at the ultra-short end, where w is small and the $1/w$ term of (2.11) amplifies the curvature of the fit, and within those slices they sit exclusively in the *core* moneyness region: all 31 minimisers of g lie in $k \in [-0.6, 0.4]$, none in either wing. Fit quality, by contrast, improves monotonically with maturity. *(Notebook 02, figure "fit and violations by maturity bucket".)*

Maturity bucket	Median RMSE (vp)	Violations	Slices
7–14 d	0.649	82.8%	29
15–60 d	0.658	6.4%	109
61–180 d	0.554	0.0%	56
>180 d	0.110	0.0%	64

Three findings organise the section.

The results show a clear fit-arbitrage trade-off. Per-slice SVI reaches 0.64 vp on held-out quotes, the best fit in the study, but produces butterfly-arbitrage violations on 12% of slices and carries non-zero crossedness between adjacent slices. SSVI, certified arbitrage-free by Theorem 2.25, pays roughly three times the error for that guarantee. The interval [0.64, 1.91] vp provides the reference range for the learning methods: more accurate than SSVI, cleaner than per-slice SVI.

The difficulty is concentrated at short maturities. Violations are overwhelmingly an ultra-short-maturity phenomenon: 82.8% of 7–14 day slices violate, against 6.4% between two weeks and two months and none at all beyond, and every one of the 31 violations sits in the *core* moneyness region rather than in a wing. Fit quality moves the other way, improving steadily with maturity from 0.649 to 0.110 vp, so the slices with the largest fitting errors are not those with the most arbitrage violations: the ultra-short end is difficult in the *arbitrage* direction rather than the accuracy one. Restricting the evaluation domain to the paper's $k \in [-0.6, 0.4]$ with

DTE ≥ 10 reduces the violation rate from 12.0% to 7.7% without eliminating it, and improves the hold-out fit from 0.640 to 0.522 vp. Filtering reduces the problem but does not eliminate it.

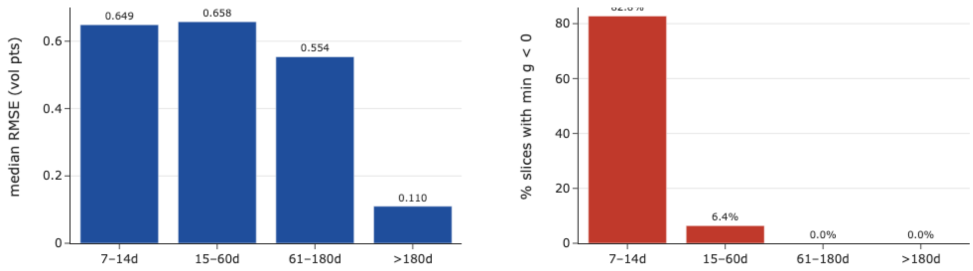

Figure 5.3: Median fit error and butterfly-violation rate by maturity bucket (Table 5.2); violations are confined to the ultra-short end. Source: Notebook 02, cell 37.

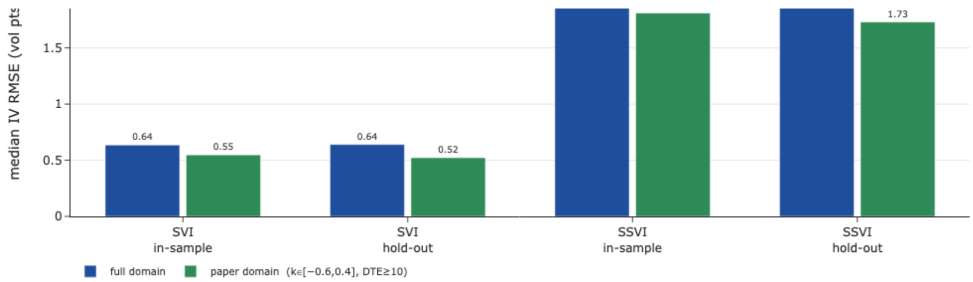

Figure 5.4: Median IV RMSE under the full and paper evaluation domains, in-sample and hold-out. Restricting to $k \in [-0.6, 0.4]$ with DTE ≥ 10 improves the hold-out fit ($0.640 \rightarrow 0.522$ vp for SVI, $1.913 \rightarrow 1.730$ vp for SSVI) and cuts the violation rate from 12.0% to 7.7%, but does not remove it. Source: Notebook 02, cell 39.

The benchmark matches the industrial reference. Evaluating our per-day SVI fits on the proprietary OptionMetrics grid gives a median absolute gap of 0.93 vp over 2,958 points (90th percentile 3.13 vp), so the median difference from the commercial smoother is below one volatility point. Repairing the real fits is by contrast expensive: a median +4.22 vp for butterfly elimination, which cures all 31 violating slices, and +0.15 vp for the calendar refit, which is applied to every slice of a day carrying material crossedness rather than only to the offending ones. Because ex-post repair can cost several volatility points, it is preferable to incorporate the constraints during training, which is what Section 5.2 attempts.

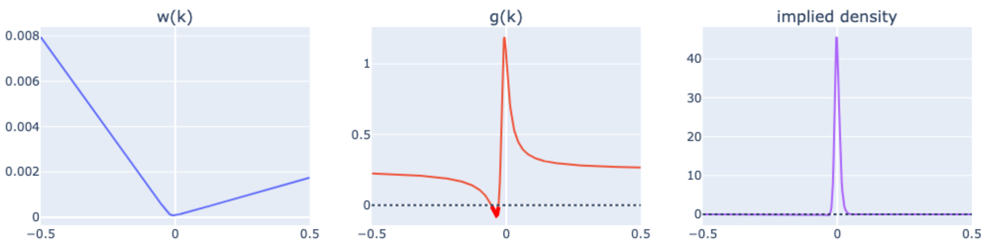

Figure 5.5: The worst real slice in the sample (2018-01-08, $\tau = 0.022$, $\min_k g = -0.072$): the Vogt phenomenon in live SPX data, and the ex-ante stress candidate used in Section 5.2. Source: Notebook 02, cell 45.

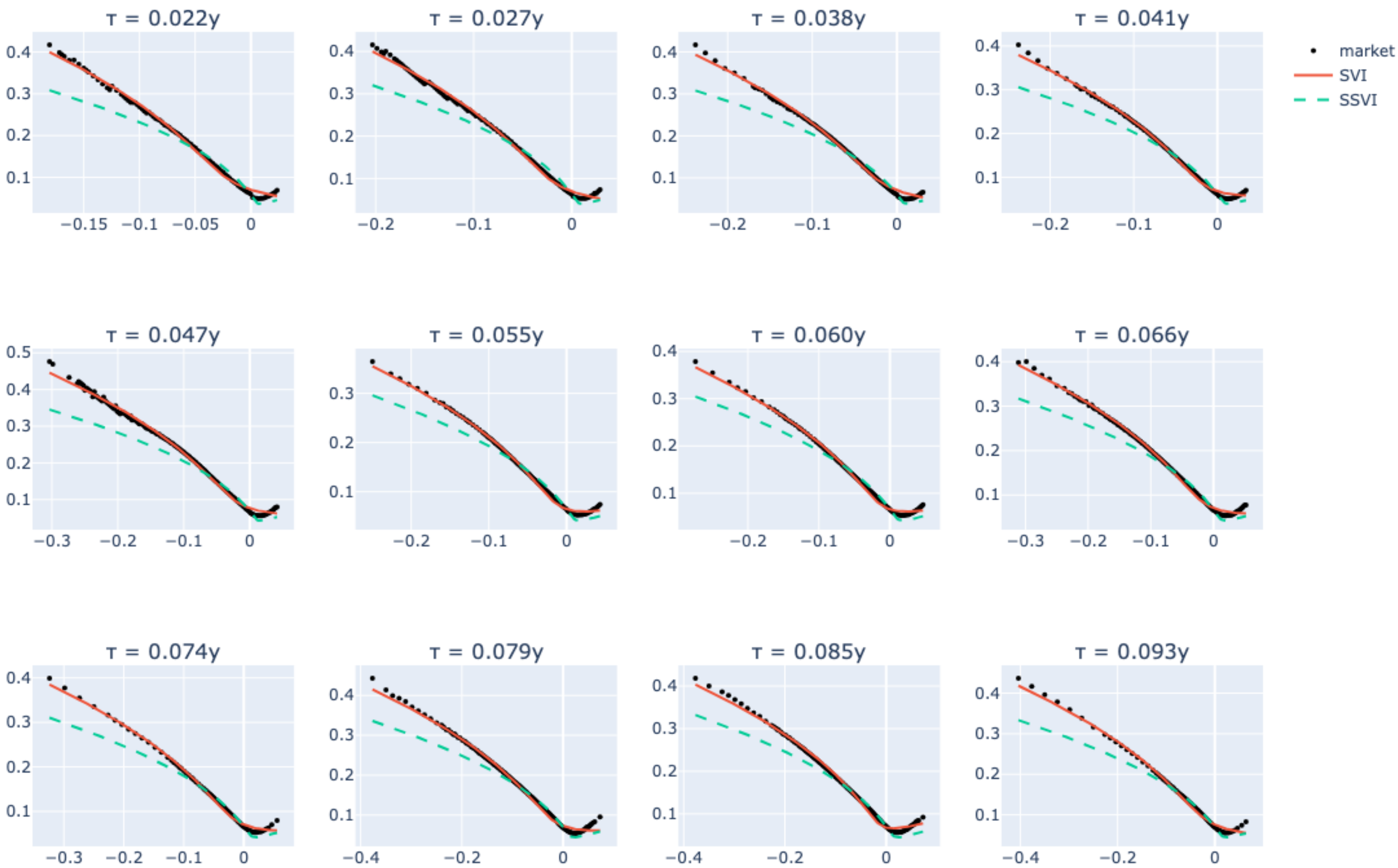

Figure 5.6: One representative trading day: market quotes (dots) with the per-slice SVI fit (solid) and the surface SSVI fit (dashed), the analogue of the paper's Fig. 4. Source: Notebook 02, cell 41.

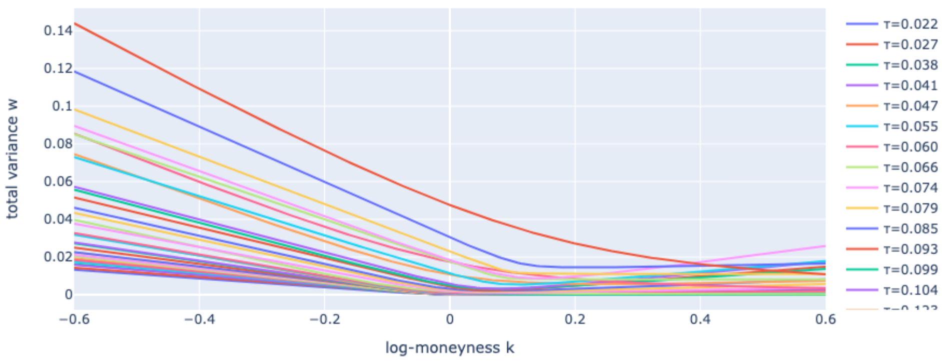

Figure 5.7: The same day's fitted SVI slices in total variance; any crossing is a calendar arbitrage between adjacent per-slice fits, the incoherence SSVI avoids by construction. Source: Notebook 02, cell 43.

5.2 Derivative-constrained deep smoothing and the collocation blind spot

5.2.1 Method

Following [1], the surface is the prior-corrector model (3.4): an SSVI prior with power-law term structure fitted to the day's quotes, multiplied by a positive corrector (2×64 tanh units, ≈ 1 at initialisation so that training *starts at the prior*). The prior is calibrated inside the Gatheral–Jacquier region, so by Theorem 2.25 it carries a printed certificate; on the synthetic experiments it reads $\theta_{\max} = 0.062$ with condition values $0.28 < 4$ and $0.83 \leq 4$.

The loss is (3.1) with $\lambda_{\mathrm{b}} = \lambda_{\mathrm{c}} = 10$, $\lambda_{\mathrm{w}} = 0.1$, and IV-targeted fit weights $\omega_i \propto 1/(4w_i\tau_i)$, the same weights as the calibrators of Section 5.1 so that the cross-family comparison is symmetric. All penalty derivatives are obtained by exact automatic differentiation and validated against closed forms.

5.2.2 Equal-epoch ablations on synthetic ground truth

Ground truth is a known SSVI surface; the quotes are sparse (54 over six maturities), irregular and noisy; and the prior is refit *from the quotes*, so it is deliberately misspecified. Four variants train for the same number of epochs with persistent optimiser state.

Table 5.3: Equal-epoch ablation (800 epochs). RMSE is measured against the dense truth at the quoted maturities; the audit uses the independent extended grid. *(Notebook 03, figure "equal-epoch dynamics".)*

Variant	RMSE (vp)	$\min g$ (audit)	bfly / cal violations
full (prior + penalties)	0.096	+0.310	0% / 0%
no constraints	0.094	+0.311	0% / 0%
no prior (+ penalties)	1.060	+0.268	0% / 0%
no prior, no constraints	1.060	+0.268	0% / 0%
prior only (no corrector)	0.154	n/a	0% / 0%

Two features of Table 5.3 are worth noting. The two *no-prior* rows are identical to the last digit, and their training traces coincide exactly (Figure 5.8). This equality is expected rather than a copy-paste error: on clean data $g_\theta > 0$ holds throughout training, so $\mathrm{ReLU}(-g_\theta)^2$ and its gradient vanish *exactly*, and the penalised and unpenalised runs follow the same trajectory. Second, the certified prior carries the accuracy: 0.154 vp for the prior alone, 0.096 vp with the corrector, against 1.06 vp with no prior. On clean data the prior provides most of the protection, while the penalties remain inactive. To find out whether the penalties do anything at all, we therefore need a stress case that induces arbitrage.

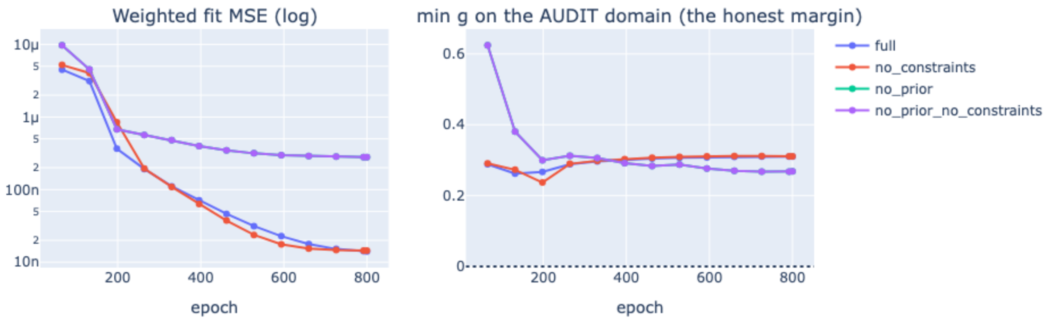

Figure 5.8: Equal-epoch training dynamics. Only three of the four traces are visible: `no_prior` lies pixel for pixel beneath `no_prior_no_constraints`, since with $g_\theta > 0$ throughout training the penalty gradient vanishes identically and the two runs are the same computation. The near-coincidence of `full` and `no_constraints` is by contrast only approximate, the residual gap being what the penalties charge on data that never demands arbitrage. Source: Notebook 03.

5.2.3 A stress test that is guaranteed, fittable and gated

We deflate the last maturity slice by a factor 0.45. Since $0.45 < \theta(0.9)/\theta(1.4) \approx 0.657$, the deflated quotes sit strictly below those of the previous maturity, so *any* surface fitting both must have $\partial_\tau w < 0$ somewhere between them; the data-level crossing is $+0.0290$ in w. The stressor is therefore verified analytically to force the pathology rather than assumed to. Deflation is chosen over inflation because the IV-target weights *raise* the stressed slice's weight and the corrector feature required is low-frequency. A synthetic *butterfly* stressor is deliberately absent: at short maturities $g < 0$ is driven by $w'^2/(4w)$ with tiny w, and would require distortions sharper than a smooth prior-times-corrector can produce at fittable widths. The butterfly penalty is instead ablated on real quotes in Section 5.2.6.

A stress comparison is interpretable only if the model actually *fits* the stressor: if it does not, apparent arbitrage violations may simply reflect poor convergence. Validity is therefore verified by a two-condition gate on the unconstrained model: the deflated-slice RMSE must be below 1 vp, *and* the fitted surface must reproduce at least half the data-level crossing. The first alone would be insufficient, since a surface can graze the deflated slice from above without ever crossing. The gate reads `PASS/PASS`, with 0.069 vp and a fitted crossing of $+0.0296$ against $+0.0290$ in the data.

5.2.4 The blind spot: exhibit, mechanism, repair

Exhibit. On this gated stress, an early version of the smoother produced, for the constrained model ($\lambda = 10$), the following unexpected result:

$$\texttt{pen\_cal}\big|_{\hat{X}_c} \approx 0, \qquad \text{while} \qquad \text{audit-grid calendar violations} = 28\%.$$

Both were true. Diagnostics excluded the optimiser, since the constrained run had the *lower* total loss under its own objective, and excluded automatic differentiation, since the τ-path gradient was 0.25 at a violating point and the τ-derivative validated to 10^{-11}. What remained was the grid.

Mechanism. The collocation followed the literature's prescription: exponentially spaced maturities, dense at the short end where the butterfly constraint bites [1]. With ten nodes on $[0.06, 1.4]$ there is *no node at all* between 0.987 and 1.400, a 0.41-year hole covering 30% of the axis (Figure 3.2), exactly where the stressor forces the crossing. The optimiser satisfied the penalty at every node and let the surface dip between them, which is precisely the extremal configuration of Theorem 3.2; the penalty therefore remained zero despite violations between grid points. Using the same exponential grid for both constraints leaves the calendar penalty poorly resolved at long maturities, which is [4]'s caveat in its most literal form. This was detectable only because the audit grid is independent of, and finer than, the collocation grid (Remark 3.1).

Repair. The butterfly penalty keeps its ATM-dense grid (maximal gap 0.10y); the calendar penalty, which costs only one first derivative per node, gets its own uniform grid of 80 nodes with maximal gap 0.017y. By Theorem 3.2 that is roughly a 24-fold reduction of the worst admissible hidden violation at equal curvature. A resolution assertion verifies, before any stress run, that the calendar grid can actually see the stressed interval.

Table 5.4: Calendar stress, λ-sweep (final run, gated). "mat" denotes material violations at $\mathrm{tol}_{\mathrm{mat}} = 10^{-3}$ (Definition 3.4). Node and audit penalties now agree to grid noise. *(Notebook 03, figure "lambda sweep log".)*

λ	deflated fit (vp)	fitted crossing	cal. viol. strict (mat)	pen. nodes / audit
0	0.069	+0.0296	34.7% (34.4%)	1.35e-3 / 1.29e-3
1	1.991	+0.0000	4.4% (0.0%)	1.3e-9 / 1.2e-9
10	1.963	−0.0002	1.6% (0.0%)	8.4e-11 / 1.2e-10

Result. The unconstrained model follows the deflated quotes down, with a fitted crossing of +0.0296 and material violations on a third of the audit domain; the constrained model does not follow the deflated quotes, with a fitted crossing of −0.0002, at a measured price of about 1.9 vp on the deflated slice. Material violations fall from 34.4% to 0.0%. The residual *strict* rate of 1.6%, whose deepest residual is $\partial_\tau w = -1.3 \times 10^{-4}$, is a residual effect of using soft rather than hard constraints: reduced but not abolished, and now *sized* rather than merely counted.

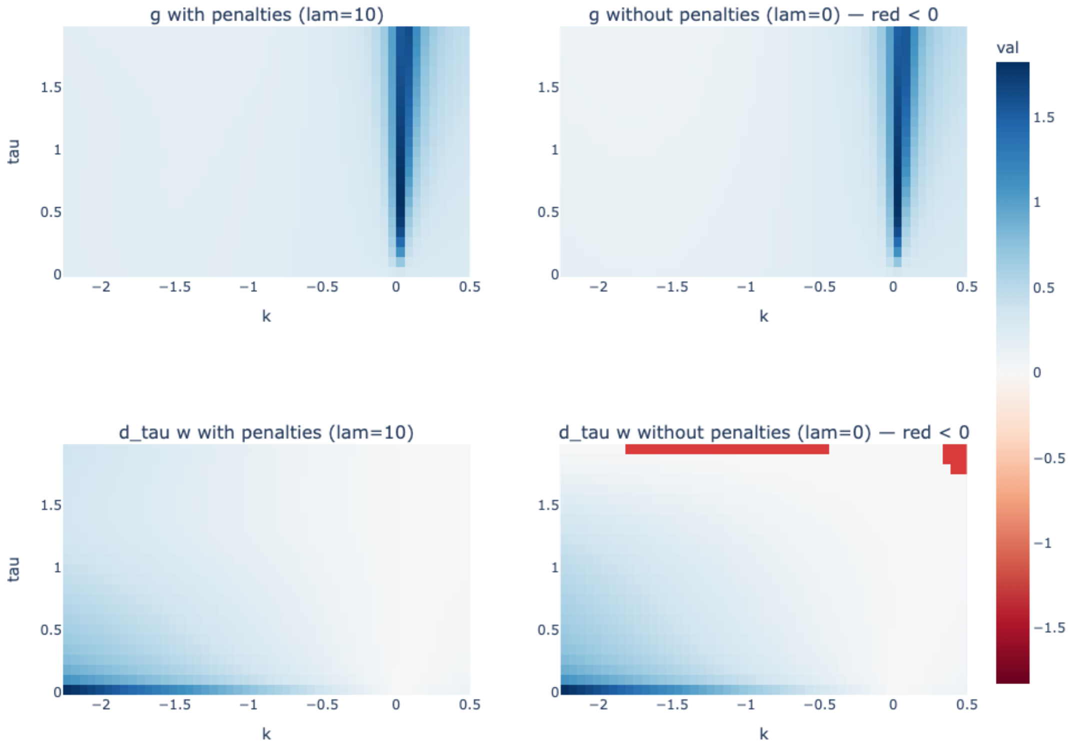

Figure 5.9: The calendar stress under both regimes. The upper panels show g and the lower panels $\partial_\tau w$, with and without penalties; red marks negative values, on a colour scale shared by both quantities. Without penalties (lower right) $\partial_\tau w$ turns negative in a band at the long end, at and beyond the last collocation node $\tau = 1.4$, where the exponential grid places no node at all and the penalty therefore has nothing to enforce; with penalties (lower left) the surface holds $\partial_\tau w \geq 0$ across the whole audit domain. The butterfly direction stays clean in both regimes. The 1.9 vp cost of that constraint is read off the deflated slice in Table 5.4.

Finding 5.1 (The collocation blind spot)**.** *A soft no-arbitrage penalty is a statement about its sampling set. A grid optimal for one constraint can be structurally blind to another, the failure is invisible to the training loss exactly, and it is detectable only by an audit grid independent of every collocation grid, compared on a materiality scale. Constraint-specific collocation closes it.*

5.2.5 Robustness to sparsity

Subsampling the quotes from 100% to 50% and then 25%, against per-slice SVI, exposed in sequence two failures of *our own* experimental design. Both are reported because each produced an implausible result that revealed a problem in the experimental design.

The first was a data leak: the prior, collocation and input scaling were calibrated on the *full* quote set inside the subsampling loop, so the deep model kept seeing quotes SVI was denied, and its RMSE *fell* as 74% of the data was removed. The second was an estimator failure: one 50% draw exploded to 14.2 vp, because the naive ATM interpolation clamps on one-sided slices and corrupts $\theta(\tau)$, and for a multiplicative corrector a *wrong* prior is worse than a *flat* one (14.2 against 1.7 vp), exactly as the log-decomposition of (3.4) predicts. The robust nearest-ATM calibrator fixed it, and the study became multi-seed with the prior-only RMSE printed for every draw, the diagnostic that separates prior failures from corrector failures.

The deep smoother then degrades gracefully, 0.096 → 0.22 → 0.34 vp, bounded below by the prior-only curve. SVI first loses accuracy and then *coverage*: at 25% no slice retains the six quotes a five-parameter SVI needs. Since SVI's RMSE is conditional on a calibrability set that itself changes with the subsampling fraction, the comparison is *paired* on the same slices, draw by draw; the deep smoother wins 4/4, including 0.137 against 1.615 vp. One 25% draw still degrades to 1.68 vp, and its prior-only column of 1.21 vp indicates that the degradation comes mainly from the prior rather than the corrector.

Finding 5.2 (Sparsity robustness is inherited)**.** *The deep smoother's robustness to sparse quotes is inherited from the calibration robustness of its prior. The prior-times-corrector architecture moves the failure point; it does not remove it.*

5.2.6 Real SPX days and the penalty ablation

The per-day driver reuses the NB02 protocol and was staged on early-January 2018 days together with an *ex-ante* stress candidate selected from Section 5.1's audit, 2018-01-08, which carries six butterfly-violating SVI slices with $\min_k g = -0.071$. On these days the staging run returns a hold-out of 1.70–1.81 vp, zero strict violations on both domains, no blind flags, and a valid certificate every day. The $\gamma \leq \frac{1}{2}$ episode of Remark 2.26 was diagnosed here, the per-day prior-only RMSE being what localised it, and endpoint certification recovers the free fit: hold-out improves to 1.15–1.31 vp.

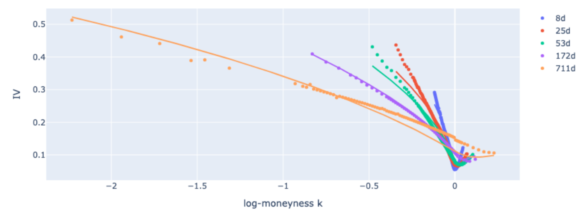

Figure 5.10: The deep smoother on real SPX quotes, 2018-01-08: market quotes with the fitted surface, on the day carrying the worst butterfly violation of the parametric benchmark. Source: Notebook 03.

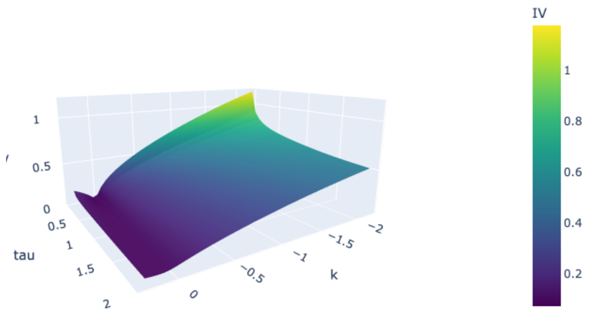

Figure 5.11: The corresponding deep-smoothed implied-volatility surface for 2018-01-08. Source: Notebook 03.

Which constraint binds on real data? Refitting the stress-candidate day with $\lambda_b = \lambda_c = 0$, at identical seed and epoch budget, answers the question. The butterfly penalty is *identically zero at both* values of λ: with the certified prior the surface never goes butterfly-negative even when unconstrained, so NB02's 82.8% ultra-short violation rate is a statement about raw per-slice SVI, not about this architecture. The binding real constraint is the *calendar* one: on 2018-01-03, $\lambda = 0$ leaves 19.1% of the extended domain violating with $\min \partial_\tau w = -0.25$, against 0.0% at $\lambda = 10$; while on the most butterfly-stressed day the calendar penalty is also inactive and the prior alone suffices. The hypothesis that real ultra-short quotes are the natural butterfly stressor is thereby downgraded to a question for the full five-year run, with the Volmageddon and COVID weeks the natural suspects.

5.3 Operator deep smoothing: one model for every surface

Section 5.2 fits one network per day; this section learns the smoothing *map* itself. Following [29], an operator $\mathcal{G}_\vartheta$ consumes a day's quote cloud and returns $\widehat{\sigma}_{\mathrm{IV}}$ at any query point, for any day, with no per-day optimisation. Three questions organise the section: what the operator programme delivers in the *academic-data* regime, one index and 1,926 days, against the reference paper's decade of proprietary multi-asset data; whether synthetic pretraining closes the gap; and whether the operator's soft constraints survive the audit of Finding 5.1.

5.3.1 Design

The real bank holds the 1,926 cleaned days, split chronologically into 1,155/385/386 as specified in Section 4.2.1, so every test number below is a temporal generalisation claim. The synthetic bank holds 1,000 SSVI surfaces with certified parameters and irregular quote configurations, an unlimited arbitrage-free pretraining source.

We train a DeepONet-class operator (3.6) (67,009 parameters) and a graph-neural-operator variant (3.9) (34,697 parameters), each under three regimes: **R1** real-only, **R2** synthetic-only with zero-shot transfer, and **R3** synthetic pretraining followed by real finetuning. Penalties are resolved by finite differences on a 22×9 grid. That coarse grid is *deliberately kept* as in the reference design, because the gap between what it sees and what an independent audit sees is itself one of this thesis's measurements.

Two engineering failures are worth reporting. The DeepONet trained from scratch on real data collapsed to a constant: the training loss sat at exactly 1.0000, the value of the relative loss for a zero prediction, and the audit returned $\min g = 0.9995$, the value $g \equiv 1$ takes for a surface flat in k. Real SPX days are dominated by far-OTM puts whose near-zero vegas explode the normalised weights, driving the pre-activation far negative until the softplus gradient vanishes; output-bias initialisation, a weight cap and a lower learning rate cure it. The second failure follows from the first: the collapsed model displayed *perfect* discretisation invariance, because a constant is trivially invariant to its inputs (Definition 3.6). The invariance study is therefore *gated* on fit validity, the same epistemic move as the stress gate of Section 5.2.3.

5.3.2 Accuracy and transfer

On the 386 out-of-sample days the graph operator is the accuracy leader, reaching 1.62 vp in R1 and 1.26 vp in R3, while the DeepONet depends on the synthetic curriculum: 6.47 vp zero-shot in R2 against 2.20 vp in R3. The transfer question therefore receives a qualified answer. Pretraining on certified synthetic surfaces closes most of the zero-shot gap, 6.47 to 2.20 vp, so certified synthetic surfaces do carry usable structure into the real problem; and unlike the DeepONet, the graph operator does improve on its real-only run under the curriculum, 1.62 to 1.26 vp. But zero-shot synthetic-to-real remains roughly three times worse than finetuned. The sim-to-real gap is real and finetunable rather than absent.

Gated on fit validity, discretisation invariance holds well: the graph operator's dense-surface shift under subsampling stays below 0.16 vp with only 25% of inputs retained. The amortisation argument is quantified directly: 0.8–0.9 ms per surface for the DeepONet and 39–413 ms for the graph operator on CPU, against roughly 80–100 s per day for the Section 5.2 smoother and about 1 s for SVI.

Figure 5.12: Hold-out IV RMSE across families and transfer regimes on the 386 out-of-sample days, the same metric reported in Table 5.5. Synthetic pretraining followed by real finetuning (R3) improves both families, and rescues the DeepONet from its zero-shot R2 performance; the graph operator leads on fit in every regime. Source: Notebook 04, cell 9.

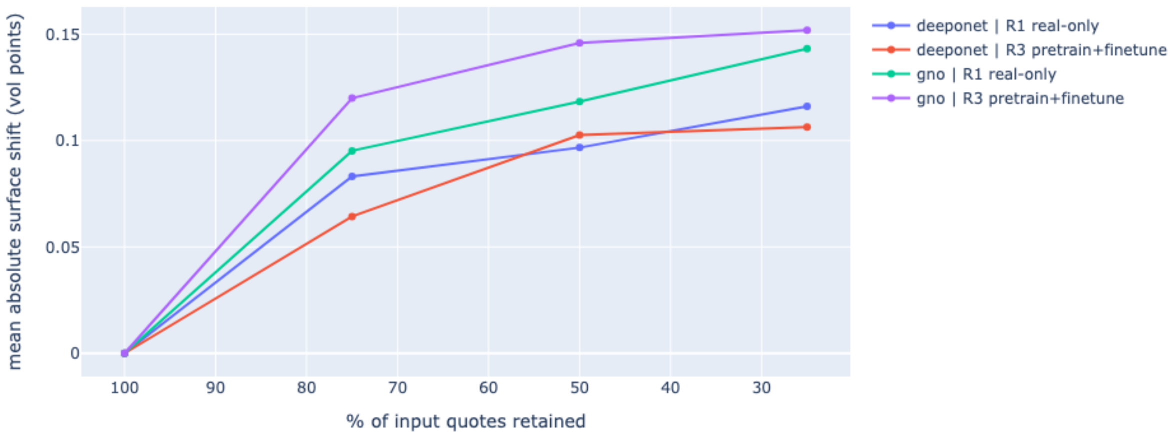

Figure 5.13: Discretisation invariance, gated on fit validity: dense-surface shift under input subsampling. Only the real-only and finetuned regimes are shown. The zero-shot synthetic-only DeepONet is omitted because it passes the fit gate on 2 days out of 386, and the invariance of a surface that does not fit the market is vacuous.

5.3.3 The audit: the blind spot is transversal

Every penalty is evaluated twice: on the *penalty grid* (22×9, maximal τ-gap 0.176y, the grid the optimiser saw) and on an *independent audit grid* (61×31, uniform, disjoint nodes), with strict and material rates and per-day blind flags. Restricting the audit to the convex hull of the quoted domain, which covers 77% of the box, does not rescue the result: the graph operator's material butterfly rate is 22.1% hull-restricted in R1 and 21.9% in R3, essentially unchanged.

Table 5.5: Operator audit on the 386 out-of-sample days: material violation rates on the independent 61×31 grid, blind-spot days (materially clean on the penalty grid, materially violating on the audit grid), the worst audit g, and the number of days passing the 5 vp fit gate. Penalty-grid rates, which are what the training loss reports, are near zero throughout. *(Notebook 04, figure "operator summary table".)*

Model	Regime	Hold-out	Audit material viol.		Blind days /386		min g	Fit gate
		(vp)	bfly	cal	bfly	cal	(audit)	/386
DeepONet	R2 synth-only	6.47	0.000%	0.000%	0	0	+0.241	2*
DeepONet	R1 real-only	2.69	0.000%	0.056%	1	49	−0.003	383
DeepONet	R3 pre+fine	2.20	0.008%	0.000%	25	0	−0.029	383
GNO	R2 synth-only	4.79	23.35%	0.412%	303	275	−35.3	218
GNO	R1 real-only	1.62	21.17%	0.033%	356	79	−30.6	386
GNO	R3 pre+fine	**1.26**	21.75%	0.067%	355	86	**−32.6**	386

* The synthetic-only DeepONet passes the fit gate on 2 days out of 386, so its zero violation rates describe a surface that does not fit the market; by rule (A4) the row is reported but not read as a clean result.

Table 5.5 is this section's central result, and the ordering of its first two columns matters. The graph operator has the *best hold-out fit in the entire thesis*, 1.26 vp in R3, and is simultaneously materially butterfly-violating on 21.75% of the independent audit domain, blind on 355 of 386 days, with a worst audit value of $\min g = -32.6$. Its own penalty-grid audit reports near-zero. It passes the fit gate on *every* day, so the violations cannot be dismissed as the artefacts of a model that failed to converge.

The same blind spot now appears in a second dimension. In Section 5.2 the penalty was blind *in space*, satisfied at the nodes and violated between them. Here it is blind in space *and in time*: the penalty was minimised on the training days' surfaces, and nothing transports that property to test days, so the constraint does not generalise with the fit. The contrast between the two architectures is the one predicted in Section 3.3. The DeepONet is essentially immune in the butterfly direction when trained on real data, its outputs lying in the span of a global smooth trunk basis (Proposition 3.7); its residual weakness is calendar, on 49 blind days. The locally assembled graph operator, through the same inductive bias that improves its accuracy (Remark 3.8), produces fine-scale curvature that a 22×9 finite-difference penalty cannot see and a 61×31 audit can.

One result was not anticipated: synthetic pretraining does not clean the DeepONet, it *moves* its failure. Real-only training leaves 49 calendar-blind days and a single butterfly-blind day; pretraining followed by finetuning leaves 25 butterfly-blind days and no calendar-blind day. The direction of the failure inverts while the fit improves only mildly (2.69 to 2.20 vp), a warning against reading any single violation column in isolation: improving one constraint during training can worsen the other.

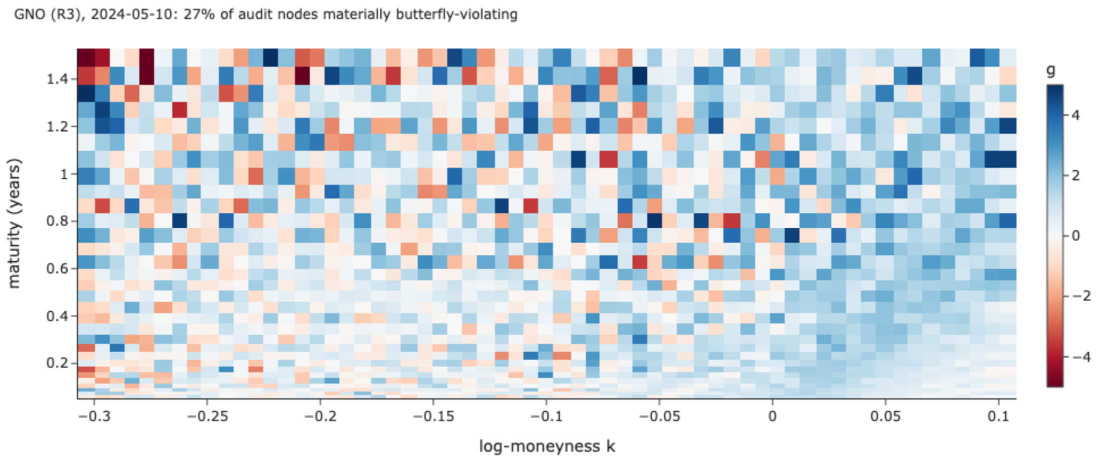

Figure 5.14: Independent butterfly audit of the graph operator on the worst of its 355 blind days (2024-05-10). The Durrleman function is evaluated on audit nodes disjoint from the 22×9 penalty grid; red marks materially negative cells, roughly a quarter of the domain, concentrated in the put wing and at the long end where quotes are sparse. The same day's penalty-grid audit reports materially clean. The colour scale is clipped at ± 5, and the worst value is not read off this figure: $\min g$ is an extreme order statistic carried by a single node, and is quoted in Table 5.5 at the audit's own resolution. Source: Notebook 04, cell 12.

Figure 5.15: The best operator on a genuinely unseen day (2025-08-29): the fit is reasonable but visibly coarser than the per-day smoother, which is the cost of amortisation. Source: Notebook 04, cell 12.

Finding 5.3 (The blind spot is transversal and two-dimensional)**.** *Soft no-arbitrage constraints fail to bind (i)* between collocation nodes *and, for operators, (ii)* across days*, since enforcing them on training surfaces does not constrain test surfaces. Reported violation rates are meaningful only relative to an audit grid independent of the penalty grid* and *a data split independent of training. Under that standard, the most accurate operator in this study is also the one with the highest violation rates, blind on* 355 *to* 356 *of* 386 *unseen days, an inversion that its own training log does not show.*

5.4 Closing the gap: embedding the certified prior in the operator

The previous section identified the problem; this section tests a possible solution, and in doing so resolves an inconsistency in the thesis as it stands: the deep smoother of Section 5.2 uses a certified prior that demonstrably carries the protection (Table 5.3), while the operators map the quote cloud *directly* to the surface with no prior anywhere. That was deliberate: the purpose of Section 5.3 was to audit the *published* design [29], and modifying the architecture first would have made the violation rates a statement about our variant rather than about the reference. Having completed that audit, we now ask whether the protective mechanism transfers.

Design. For every day, an SSVI prior is fitted from that day's own quotes inside the Gatheral–Jacquier region, so by Theorem 2.25 it carries a printed certificate. The operator then learns only a bounded multiplicative correction,

$$\widehat{\sigma}_{\mathrm{IV}}(k,\tau) \;=\; \underbrace{\sigma_{\mathrm{SSVI}}(k,\tau)}_{\text{certified, per day}} \times \exp\Bigl(\underbrace{s\,\tanh\bigl(\mathcal{G}_\vartheta(a)(k,\tau)\bigr)}_{\text{correction, } |c|\le s}\Bigr), \qquad s=0.35, \tag{5.2}$$

and the quote cloud enters through the log-residual $\log(\sigma_i^{\mathrm{mkt}}/\sigma_{\mathrm{SSVI}}(k_i,\tau_i))$, so the operator sees directly what the prior fails to explain. Four properties follow, each addressing something Section 5.3 exposed.

(i) **Training starts at the prior.** The correction head is zero-initialised, so $c \equiv 0$ at initialisation and the operator *is* the certified SSVI surface, asserted numerically before every run to 10^{-8}.

(ii) **Positivity becomes structural.** The softplus head whose saturation caused the collapse is no longer needed: positivity comes from a positive prior times an exponential.

(iii) **The correction is bounded**, $|c| \le s$: a curvature budget in the sense of Theorem 3.2. It does not prove $g \ge 0$, but it bounds how far the operator can travel from a surface that satisfies it.

(iv) **Inheritance.** Wherever the correction is flat, all derivatives of the output are the prior's, and the prior carries a theorem.

Prior-times-correction is a strong inductive bias, *not* a guarantee: a correction with enough curvature can still drive g negative within its budget. The result below is therefore reported as a measured reduction under an identical audit, never as a certificate.

The comparison is like for like: same regimes, split, loss weights, penalty grid and audit grid; only the architecture differs. One caveat is disclosed: the synthetic bank is generated *from* SSVI, so a fitted prior recovers it almost exactly and leaves the corrector nothing to learn. The prior-embedded families therefore pretrain on a perturbed synthetic bank, and the headline comparison is R1 against R1, which is unaffected by that choice.

Table 5.6: Effect of embedding the certified prior, real-only regime (R1), on the 386 out-of-sample days. Audit columns are material rates on the independent 61×31 grid. *(Notebook 04, figure "operator summary table, prior families".)*

Model	Hold-out (vp)	Audit bfly (mat)	Audit cal (mat)	Blind days (bfly/cal)	$\min g$ (audit)	Fit gate /386
DeepONet	2.69	0.000%	0.056%	1 / 49	−0.0028	383
DeepONet + prior	**1.32**	**0.000%**	**0.000%**	**0 / 0**	**+0.071**	386
GNO (R1)	1.62	21.17%	0.033%	356 / 79	−30.6	386
GNO (R3, best fit)	1.26	21.75%	0.067%	355 / 86	−32.6	386
GNO + prior	*not available: run terminated, see note*					

The `gno_prior` family did not yield a usable model: its training was terminated by a mid-run process kill on the shared cluster and the state dictionary was never checkpointed, so the 2×2 matrix is completed on three of its four cells. This does not weaken the constructive claim, which is carried by the DeepONet row: the critique it answers, namely that soft penalties do not transport across days, is architecture-agnostic.

Reading the result. Read against Table 5.5, the prior-embedded DeepONet resolves the inversion rather than merely improving one column. Its hold-out error falls from 2.69 to 1.32 vp, halving the gap to the accuracy leader, while both audit columns go to exactly zero and $\min g$ crosses from −0.0028 to +0.071: not a single materially violating cell on any of the 386 days, against 49 calendar-blind days for the plain DeepONet. The butterfly direction was already essentially clean without the prior, so what the prior removes here is the *calendar* blind spot, the one direction in which the trunk basis of Proposition 3.7 offers no protection.

The comparison that matters is with the graph operator. At 1.26 vp the GNO fits marginally better, by 0.06 vp, and pays for it with 21.75% of its audit domain materially violating and 355 blind days. The difference in fit is only 0.06 vp, while the prior-embedded model is audit-clean on every out-of-sample day. Among operators that fit the market at all, in the sense of rule (A4), it is the only such surface: the synthetic-only DeepONet of Table 5.5 is audit-clean too, but on a fit that clears the gate on two days out of 386.

Finding 5.4 (The certified prior transfers from the smoother to the operator)**.** *Embedding a per-day certified SSVI prior and learning only a bounded correction removes the DeepONet operator's remaining blind spot entirely: its* 49 *calendar-blind days and single butterfly-blind day fall to zero, both material audit rates to* 0.000%*, and the worst audit g from* −0.0028 *to* +0.071*, across all* 386 *out-of-sample days. It does so at* negative *cost in fit, halving the hold-out error from* 2.69 *to* 1.32 *vp, for a per-day prior calibration of roughly one second. The resulting model sits within* 0.06 *vp of the study's accuracy leader while being the only operator that is simultaneously audit-clean and usable as a fit. The mechanism that protects the deep smoother therefore transfers to the operator family. It remains an inductive bias rather than a guarantee, but on this data it removes the pathology entirely rather than merely reducing it.*

Chapter 6

The Downstream Economics of Arbitrage: Pricing the Audit

Chapter 5 established which fitted surfaces violate the no-arbitrage conditions and quantified the extent of those violations. The natural next question is whether they matter economically. Dupire's formula provides the link, $\sigma^2_{\text{loc}} = \partial_\tau w/g$ (Proposition 2.12), so the quantities audited in the previous chapter enter directly into downstream pricing, as anticipated in Remark 2.13. We examine that connection through five complementary exercises: Dupire local volatility and Monte-Carlo repricing (§6.2), Breeden–Litzenberger densities (§6.3), a data-level static-arbitrage scanner on bid/ask quotes (§6.4), a residual relative-value study and backtest with an arbitrage-dirty counterfactual (§6.5), and a detection-power experiment measuring what signal this market *could* monetise (§6.6). Section 6.1 comes first and fixes the distinction on which all five rest, between a surface that is internally inconsistent and a trade that makes money.

The inputs are exported surface packs: dense grids of $(w, \partial_\tau w, g)$, carrying exact-autodiff fields for the Section-5.2 smoother and w alone for the operators. For the operators the derivative fields are reconstructed by finite differences and validated against an analytic SSVI reference at the export resolution, to 1.23×10^{-4} in $|g|$ and 3.38×10^{-5} in $|\partial_\tau w|$, both an order of magnitude below tol_{mat}. The hybrid exp∪uniform τ export grid that achieves this reflects the lesson of Finding 5.1: accurate numerical differentiation requires sufficient resolution precisely in the regions where curvature is greatest.

The staging pack set holds 22 surfaces over 11 dates: 2018-01-08 (the ex-ante stress candidate; deep smoother at $\lambda \in \{10, 0\}$, exact fields) and the last ten out-of-sample days 2025-08-18 → 2025-08-29 (DeepONet R1 and GNO R3, FD fields). The parametric baseline (the GJ-certified SSVI) is refit in place from the quotes on every day, and a uniform crc32 hold-out applies to every family. All machinery is self-tested (Black round-trip 10^{-7}; flat-BS variance-swap 0.1997 vs 0.200). Out-of-domain queries are handled by `in_domain()` filtering, never by clipping: clipping out-of-domain log-moneyness to pack boundaries manufactures large artificial residuals with no economic meaning.

One experiment is not reported here. Every evaluation in this chapter is terminal or static, in the sense that it depends on the fitted surface only through a marginal distribution at a single maturity. That is the mildest possible test of an arbitrage violation, because the calibration objective is itself a marginal one. The sharper test is a path-dependent payoff, whose price depends on the local volatility $\partial_\tau w/g$ at every point a simulated path visits, and therefore on exactly the regions between the quotes where the audits of Chapter 5 found the violations. We built that experiment, a discretely monitored down-and-out call priced under an independent Heston ground truth, and it produced results consistent with everything reported below. We do not report it as a result of this thesis: its numerical components, a finite-difference PDE cross-check and a correction for discrete barrier monitoring, could not be verified to the standard applied to the rest of this work within the time available. The experiment and its indicative findings are described in the conclusion, where they motivate the first item of future work.

6.1 Two questions that must not be conflated

Chapters 2 to 5 established that several of our fitted surfaces violate the static-arbitrage conditions. Before pricing those violations, it is essential to separate two claims that the word "arbitrage" invites one to merge, and the separation organises this entire chapter.

Model arbitrage. A surface with $g(k,\tau) < 0$ on part of its domain is *internally inconsistent*: the option prices it manufactures there cannot all arise from a single probability distribution, so the model is not a valid pricing object on that region. This is a statement about the internal consistency of the model, not about the existence of an executable trade in the observed market. It is what the audits of Chapter 5 measure, and it is already sufficient to disqualify a surface from downstream use, because everything computed from it, densities, Greeks, local volatilities, inherits the inconsistency.

Executable market arbitrage. A trader making money requires something strictly stronger: a portfolio of *actually tradable* contracts, transacted at the quoted bid and ask, whose entry cost is negative and whose payoff is nonnegative in every state. For an equally weighted butterfly at strikes $K_1 < K_2 < K_3$ this means buying the wings at the offer and selling the body at the bid,

$$\text{cost} \;=\; \text{Ask}(K_1) - 2\,\text{Bid}(K_2) + \text{Ask}(K_3) \;<\; 0, \tag{6.1}$$

with the analogous inequality across the spread for calendar spreads. A model violation implies nothing about (6.1): the violating region may lie where nothing is quoted, the implied mispricing may be smaller than the bid-ask spread, or the surface may be inconsistent precisely because it is wrong rather than because the market is.

Consequence for this chapter. The two questions are answered separately and never combined. Sections 6.2 and 6.3 price the *model* violation, asking what a desk loses by using an internally inconsistent surface, which is the question the audit poses. Section 6.4 asks the *market* question directly, scanning the raw bid/ask quotes for portfolios satisfying (6.1), and thereby establishes the data-level baseline against which model violation rates must be read: a model cannot be faulted for reproducing an arbitrage that the quotes themselves already contain. Section 6.5 addresses a distinct question again: relative-value predictability, rather than either model inconsistency or executable arbitrage.

6.2 Dupire local volatility: the repair rate

For each surface pack we compute $v_{\text{loc}} = \partial_\tau w / g$ on the dense grid. Cells with $\partial_\tau w < 0$, $g \leq 0$, or a non-finite ratio are invalid and must be repaired (clipped into $[10^{-4}, 4]$) before any Monte-Carlo path can be drawn. Their share, the repair rate, measures how much of the surface is unusable as a pricing model. We follow [4, §7]: validity, day-to-day stability, and MC repricing against both the surface's own prices (round-trip) and held-out quotes (market).

Table 6.1: Local-volatility validity and repricing on the real staging days (2018-01-08 for deep/SSVI; ranges over the ten 2025-08 days for the operators and their SSVI baseline). Repair rate = share of the grid where (2.15) is invalid. Stability = mean $|\Delta\sigma_{\text{loc}}|$ between consecutive days (vp). Source: Notebook 05, §A.

Model	Repair rate	v_{loc} range (raw, worst)	MC round-trip (vp)	MC vs market (vp)	Stability (vp/day)
Deep (λ=10), 2018	0.0%	[+0.005, +4.5]	3.87	5.06	n/a
SSVI (certified), 2018	0.0%	[+0.005, +3.0]	3.59	5.08	n/a
DeepONet R1, 2025 (×10)	0.0–0.6%	[−0.019, +4.5]	0.40–0.48	0.43–0.76	0.40–1.13
GNO R3, 2025 (×10)	**48.4–50.1%**	[−4,645, +2,794]	0.94–2.78	1.19–3.22	**7.30–8.35**
SSVI (certified), 2025 (×10)	0.0%	[+0.001, +8.3]	2.62–6.91	4.26–8.31	n/a

Three observations follow from Table 6.1.

(1) The repair rate re-ranks the families. The certified-prior surfaces give fully valid local variance, and the DeepONet is valid on most days. The GNO, which leads on IV-RMSE, is invalid on half the surface, with local variance spanning four orders of magnitude and a day-over-day instability an order above the DeepONet's. This is Finding 5.3 in pricing units, not a finite-difference artefact: the reconstruction is validated to 10^{-4} on smooth references, so this is the fine-scale curvature the 22×9 penalty grid cannot resolve.

(2) Monte-Carlo repricing understates the extent of the violation. After repair, the GNO reprices within 0.9–3.2 vp. This apparently reasonable error is misleading, however: invalid cells have already been repaired before simulation, and the SDE further smooths their local effect. The repricing error therefore does not reveal the roughly 50% repair rate of the original surface, which is visible only in the repair and stability columns.

(3) The guarantee comes with a measurable fit cost. The DeepONet reprices its own market best (0.43–0.76 vp) while the rigid certified SSVI pays 4–8 vp. This trade-off motivates the certified-prior corrector considered in this thesis: it preserves the validity of the prior while recovering much of the flexibility needed for a competitive fit.

Monte-Carlo time grids are floored at $\tau_{\min}$, the boundary of the prior's certificate (Remark 2.26). This section fills a gap [4] leave open, the extraction of local volatility from implied volatilities, under their own §7 protocol.

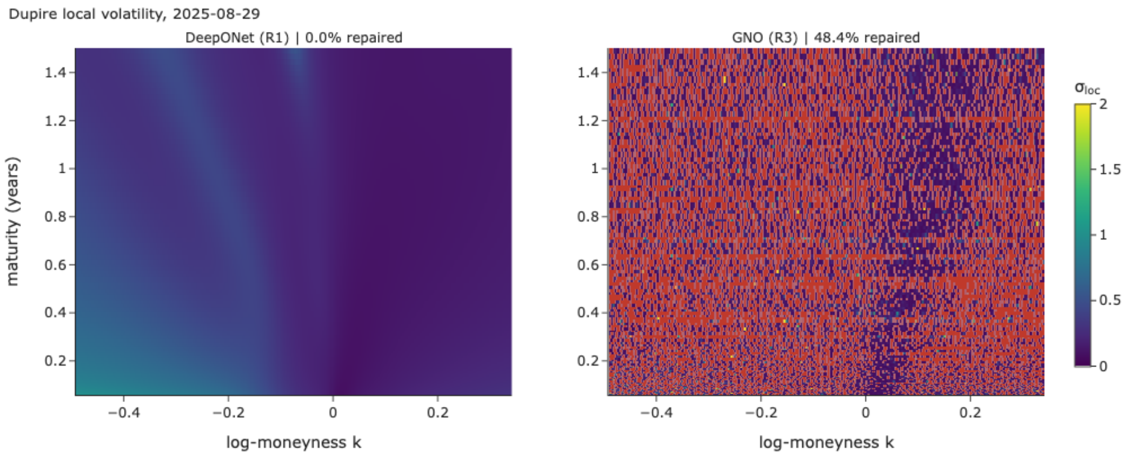

Figure 6.1: Dupire local volatility $\sigma_{\text{loc}} = \sqrt{\partial_\tau w / g}$ on 2025-08-29, log-moneyness on the horizontal axis and maturity in years on the vertical. Red cells are invalid and are clipped into $[10^{-4}, 4]$ before simulation. The DeepONet surface is valid everywhere on this day; the GNO, which fits the vanilla market better, is invalid on close to half its grid. Source: Notebook 05, §A.

6.3 Risk-neutral densities: butterfly in probability units

Proposition 2.5 restates each slice's butterfly audit as a density, $q = g\,\varphi(d_-)/\sqrt{w}$. Negative mass is the same violation expressed in the units a risk committee reads.

Table 6.2: Breeden–Litzenberger densities on real staging days (three quantile maturities per day; representative rows of the 63-row table). "Neg. mass" integrates $|q|$ over $\{q < 0\}$; total mass and the martingale check $\int e^k q$ deviate from 1 only through domain truncation. Source: Notebook 05, §C.

Model	τ	Neg. mass	Total mass	Martingale	$\min q$
Deep (λ=10), 2018-01-08	0.31	0.0%	1.010	1.010	3.5×10^{-7}
Deep (λ=10), 2018-01-08	0.98	0.0%	0.993	0.990	0.0004
Deep (λ=10), 2018-01-08	1.76	0.0%	0.980	0.974	0.002
DeepONet R1, 2025-08-29	0.13	0.0%	0.999	1.000	1.2×10^{-19}
DeepONet R1, 2025-08-29	1.27	0.0%	0.930	0.964	0.054
GNO R3, 2025-08-29	0.13	**77.8%**	0.995	0.997	−18.1
GNO R3, 2025-08-29	0.52	**120.0%**	1.053	1.038	−16.4
GNO R3, 2025-08-29	1.27	**229.9%**	0.986	1.005	−37.1

The deep smoother and the DeepONet give densities that are nonnegative everywhere, with unit mass up to truncation. The DeepONet's $\min q = 1.2 \times 10^{-19}$ at the short end is a smooth density pinned to zero in the far wing, not a violation. The GNO's densities, by contrast, carry negative mass of up to 2.3 times the total, produced by an oscillating g of amplitude 20 to 37. Its fine-scale convexity therefore has no economic interpretation, even though its IV-level fit is the best in the study. In other words, selection based on IV-RMSE alone would have favoured precisely the surface with the most severe density violations in this comparison.

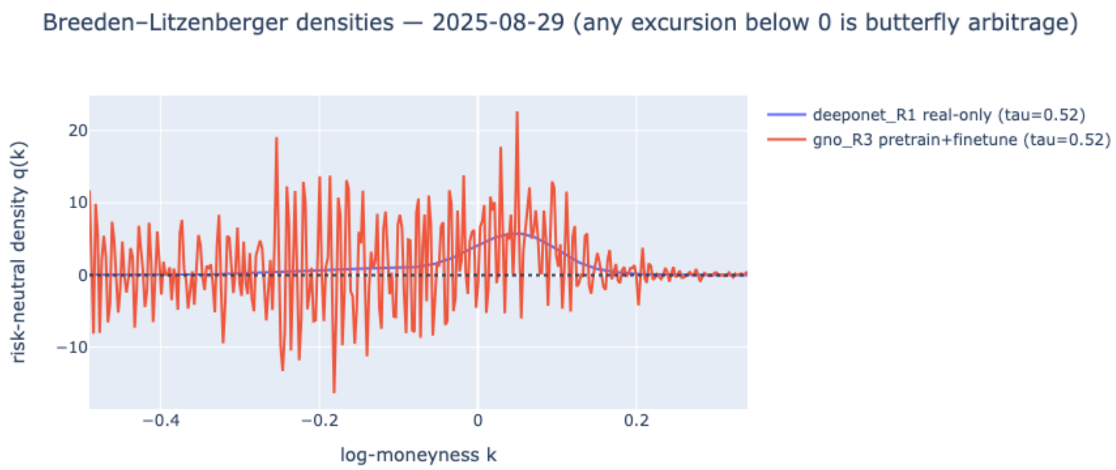

Figure 6.2: Breeden–Litzenberger densities on 2025-08-29 at $\tau = 0.52$. The DeepONet's density is smooth and nonnegative; the GNO's oscillates around it and crosses below zero repeatedly, giving the negative mass reported in Table 6.2. Any excursion below the dashed line is butterfly arbitrage. Source: Notebook 05, §C.

6.4 A static-arbitrage scanner on bid/ask quotes

The audits so far interrogate fitted surfaces. This section scans the raw quotes instead, asking per day and per (expiry, type) whether the quoted bid/ask system already contains executable static arbitrage, that is, whether the market itself violates Proposition 2.2 when priced at the touch. The same checks are run at mid prices alongside, and the gap between the two columns is the bid-ask spread absorbing the apparent violations. This gives us the data-level baseline against which model violation rates must be read, since a model cannot be blamed for reproducing an arbitrage the quotes contain, and it identifies the stress days used in Section 5.2.6.

Tests. Per (expiry, type) side, strikes ascending:

(S1) *Vertical (monotonicity):* calls must be non-increasing in K, puts non-decreasing; the executable flag is $C^{\text{bid}}(K_{i+1}) > C^{\text{ask}}(K_i)$ (sell the spread, collect a positive premium against a nonnegative payoff; Proposition 2.2(a), monotone part), and symmetrically for puts.

(S2) *Butterfly (convexity):* for consecutive strike triples $K_1 < K_2 < K_3$ with weights $\lambda_1 = (K_3 - K_2)/(K_3 - K_1)$, $\lambda_3 = (K_2 - K_1)/(K_3 - K_1)$, the executable flag is $\lambda_1 C^{\text{ask}}(K_1) - C^{\text{bid}}(K_2) + \lambda_3 C^{\text{ask}}(K_3) < 0$ (buy the wings at the ask, sell the body at the bid; Proposition 2.2(a), convex part).

Counts are reported per day, in absolute terms and as a share of checks, together with the premium locked by executable violations in index points, and stored in `nb05_static_arb.parquet` alongside the per-day model audit columns so that data-level and model-level rates can be compared directly.

Results. Over 1,926 trading days the scanner checks roughly 1.22×10^7 vertical pairs and 1.21×10^7 butterfly triples. At the touch it finds **62 executable violations** (6 vertical, 56 butterfly) on 43 **days** (2.2%), locking a cumulative premium of 18.3 index points over eight years. The SPX quote system is therefore, to first order, free of executable static arbitrage. The mid-price count is larger by orders of magnitude (2.5×10^4 vertical and 2.6×10^6 butterfly flags, with at least one on every single day), and the whole difference is the bid-ask spread: the violations a mid-quote surface would report are not transactable.

This is the baseline the model audits must be read against, and it is the counterpart of the detection-power result in §6.6. The same market whose residuals sit below the net-profitability frontier is the market whose quotes admit essentially no executable static arbitrage. The 43 flagged days concentrate in the ultra-short bucket, the same bucket where fitted-SVI violations concentrate (Table 5.2), which suggests that part of the short-end difficulty is in the data rather than in the smoother.

6.5 Residual relative value: economic significance rather than alpha

This is not an arbitrage. The strategy studied here is relative value, in the sense of Section 6.1. An arbitrage has a payoff structure guaranteeing no loss in any state, executable at quoted prices. What follows has no such structure: it ranks options by their residual $r_i = \sigma_i^{\text{mkt}} - \sigma_i^{\text{model}}$, goes long the cheap tail and short the rich tail, and profits only if residuals mean-revert on average. It can lose in any given period. We therefore test economic significance rather than alpha, and report the result as a comparison between a clean and a deliberately arbitrage-dirty surface on identical universes with identical costs. The question is whether the residuals carry economic information, and whether arbitrage-dirtiness degrades it.

Protocol. Per (day, family), residuals are computed on held-out, in-domain quotes and matched to $t + h$ by ($\texttt{exdate}$, strike). Two layers follow. The information layer reports

$$\mathrm{IC} = \mathrm{Spearman}(r_t, -\Delta\mathrm{IV}_{t\to t+1})$$

and a decile spread net of entry half-spreads. The backtest uses a common universe per day, made of held-out quotes inside every model's domain with finite cost capped at 5 vp, so that names and costs are identical across models and only the surface differs. Ranking is bucket-neutral (τ-terciles $\times$ $|k|$-terciles), with an entry filter that trades only quotes mispriced beyond their own half-spread. We sweep holding horizons $h \in \{1, 3, 5, 10\}$ and an execution ladder from touch (full spread paid) through half to mid (none), where mid bounds what a market maker could capture and touch is what a taker pays. P&L is first-order vega, one round trip per holding period, non-overlapping, fit-gated at 5 vp day RMSE, with real per-name half-spreads converted through Black vega. The counterfactual is the same day smoothed with $\lambda = 10$ and with $\lambda = 0$.

Information layer. On the controlled synthetic validation, where quote noise mean-reverts by construction, the pipeline behaves as designed: IC 0.712 for the clean surface against 0.455 for the dirty one, on the same day and quotes. On the real staging days the ICs are small and unstable in sign, as they must be at this sample size (DeepONet -0.62 to $+0.37$; GNO ≈ 0; SSVI -0.61 to $+0.36$), and the single real clean-versus-dirty day is statistically indistinguishable (IC 0.260 against 0.277). At $n = 10$ days this layer validates the pipeline; it is not evidence.

Table 6.3: Backtest, mean net P&L per holding period (vp), staging sample (deep rows: one period, 2018-01-08; operator and SSVI rows: the ten 2025-08 days). Sharpe, drawdown and hit rate are computed and stored (`nb05_backtest_stats.parquet`) but are meaningless at $n_{\text{periods}} \leq 10$; the production run populates them. Source: Notebook 05, §F2.

	execution = touch				execution = mid			
Model	h=1	3	5	10	h=1	3	5	10
Deep (λ=10)	−2.06	−1.36	−1.29	−0.58	+0.08	+0.45	+0.61	+1.12
Deep (λ=0, dirty)	−2.10	−1.57	−1.45	−1.23	+0.05	+0.30	+0.59	+0.75
DeepONet R1	−0.74	−0.79	−1.24	n/a	−0.05	−0.15	−0.69	n/a
GNO R3	−0.67	−0.81	−1.17	n/a	−0.06	−0.28	−0.67	n/a
SSVI	−0.75	−1.09	−1.53	−1.78	+0.01	−0.01	−0.14	−0.09

Despite the small sample, two descriptive patterns are consistent across the staging results. First, no model produces positive net P&L under touch execution at any holding horizon. Where a residual signal exists, it is therefore too small to compensate for the bid–ask spread. Section 6.6 shows that this is not simply a failure of the detection pipeline.

Second, the clean surface outperforms its arbitrage-dirty counterpart in all eight matched horizon–execution cells. At mid execution, for example, the ten-day result is +1.12 vp for the clean surface against +0.75 vp for the dirty one, with the universe and transaction-cost assumptions held fixed.

Finding 6.1 (Arbitrage-dirtiness weakens the residual signal)**.** *On matched counterfactuals (same day, quotes, universe, costs), the residual signal computed against an arbitrage-dirty surface is weaker than against its clean twin: IC* 0.455 *against* 0.712 *on the synthetic validation, and lower net P&L in* 8/8 *backtest cells on the real staging day. The violation is not only a pricing defect of the surface but degrades everything computed against it. Statistical confirmation requires the production run; the staging sample validates the pipeline.*

6.6 Detection power: the net-profitability frontier of this market

Section 6.5 showed that the observed residual signal is not profitable after paying the spread. That result alone does not tell us whether the signal is genuinely too weak or whether the backtest simply lacks the power to detect it. The purpose of this section is therefore to calibrate the smallest synthetic signal that the same pipeline could detect and monetise under the observed market conditions. Comparing the empirical residuals with that detection frontier allows us to distinguish between a limitation of the pipeline and a limitation imposed by transaction costs.

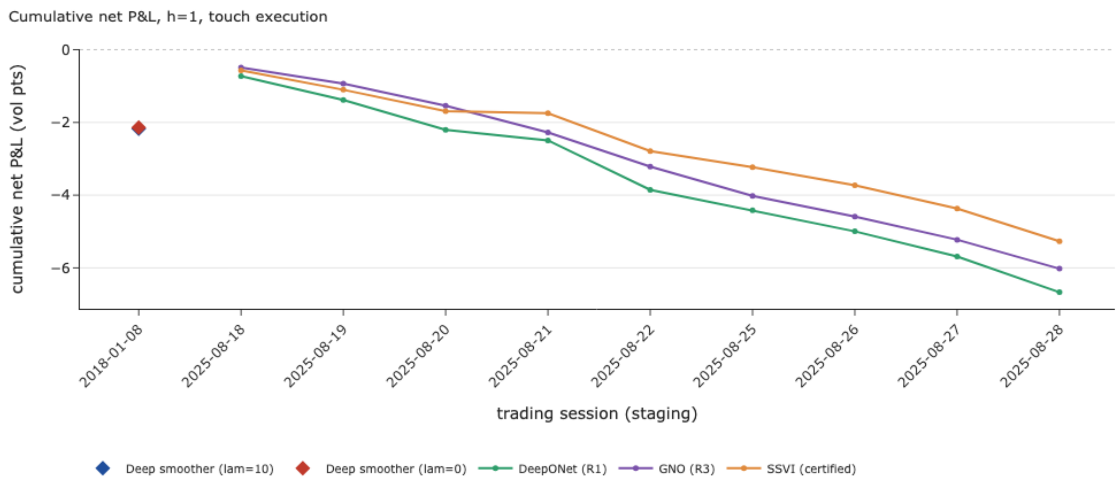

Figure 6.3: Cumulative net P&L at $h = 1$ with touch execution, cumulated over the nine trading sessions common to the three 2025 families. Every family loses steadily, which is result (1) in visual form. The two deep-smoother entries have a single staging period (2018-01-08) and appear as one overlapping marker on the left, their constrained and unconstrained P&L being almost identical on that day. Source: Notebook 05, §F2.

Protocol. Take a real day (2025-08-28; universe 1,760 names with real per-name half-spreads) and inject a synthetic mispricing of amplitude a (vol points, random sign) into 10% of names, decaying toward the surface with half-life h (trading days). The injected residuals go through the same backtest machinery as Section 6.5, with the same entry filter, bucket-neutral deciles and real costs, and we compute annualised net Sharpe at the touch over the best holding horizon, across 120 replications per (a, h) cell. Two calibrations keep the exercise honest. The observation noise through which the signal must be detected is the robust dispersion of actual residuals ($\sigma_{\text{obs}} = 3.92\,\text{vp}$), and the outcome noise is the robust dispersion of actual matched day-over-day IV moves net of per-expiry common shifts ($\sigma_\varepsilon = 0.09\,\text{vp/day}$, idiosyncratic).

Results. At the touch, a mispricing must have amplitude $\geq 3.0\,\text{vp}$ with a half-life of 5 days or less, rising to $\geq 5.0\,\text{vp}$ at half-lives of 8–12 days, to be net-profitable: longer persistence raises the amplitude required rather than lowering it. The pipeline validation passes: an injected signal with $a = 5\,\text{vp}$ and a 12-day half-life produces an annualised Sharpe of 18.4, so the machinery does detect large persistent signals and the empty real-data result of Section 6.5 is a property of the market rather than of the code. Against this frontier, the empirical residual distribution of the same day has 98.1% of quotes beyond their own half-spread, but a median exceedance of only 2.55 vp and a day-1 residual autocorrelation of +0.90 (half-life $\approx$ 6.9 trading days). The empirical point therefore sits below the 3–5 vp breakeven contour. The margin is narrow, roughly half a volatility point at the measured persistence, so the conclusion is that this market prices the residual signal out, not that it does so by a wide margin.

Finding 6.2 (The spread prices out the residual signal)**.** *On real SPX spreads and measured noise, a residual mispricing must exceed ≈ 3 vp with multi-day persistence to clear touch costs, and the market's empirical residuals (median exceedance 2.55 vp, half-life ≈ 7d) sit below that frontier, though by a narrow margin. The absence of net alpha at the touch is therefore a property of the* *market's spread structure, consistent with the scanner result of Section 6.4, and not an artefact of the pipeline, which does detect signals above the frontier.*

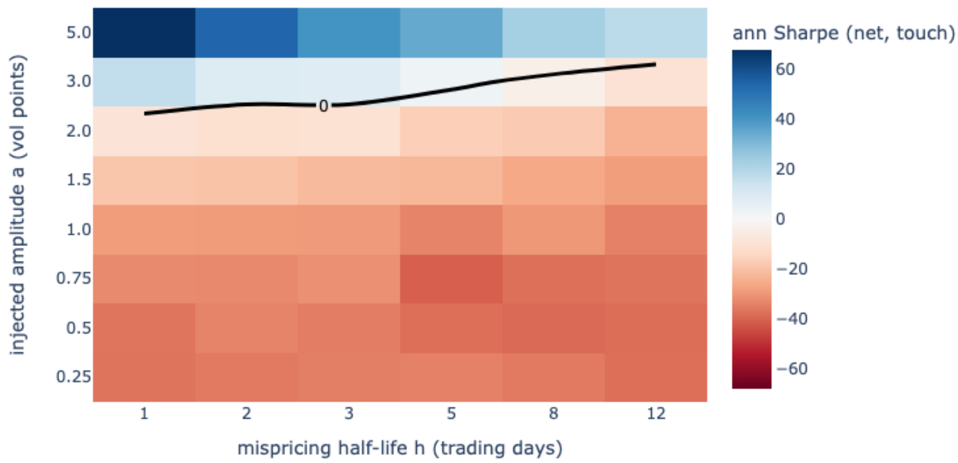

Figure 6.4: Annualised net Sharpe at the touch as a function of the injected amplitude a and half-life h. The black line is the breakeven contour, labelled with the Sharpe value it traces, namely zero; the region above it is net-profitable. Both axes are categorical, so their columns are equally spaced rather than to scale. The market's empirical residuals for this day, with median exceedance 2.55 vp and half-life ≈ 7 days, fall below the contour, as discussed in the text. Source: Notebook 05, §G.

6.7 Unified evaluation and discussion

6.7.1 One table, four questions

Table 6.4: The unified view. Fit is hold-out IV RMSE under the shared crc32 protocol (NB02: 8-day staging; NB03: staging days; NB04: 386 chronological out-of-sample days). Audit columns are *material* rates on grids independent of any penalty grid. Downstream columns from this chapter (staging days). Cost is per new day on CPU.

	Fit, hold-out (vp)	Guarantee	Material viol. (audit)	LV repair	Density neg. mass	Cost / day
SVI (per slice)	0.64	none (audited ex post)	12% of slices	n/a (no $\partial_\tau w$)	slice-level $g < 0$	∼1 s
SSVI (certified)	1.91	**theorem** (Thm. 2.25)	0%	0%	0%	∼1 s
Deep (λ=10)	1.7–1.8	soft + certified prior	0% (staging)	0%	0%	80–100 s
DeepONet (best)	2.20	soft, train days only	0.008%	0.3–0.6%	≈ 0	<1 ms
DeepONet + prior	**1.32**	soft + certified prior	**0.0**%	0%	0%	<1 ms+1 s
GNO (best)	1.26	soft, train days only	**21.8% bfly**	**49–52%**	up to **230%**	40–400 ms

Table 6.4 answers the four questions this thesis asks of any smoother.

How well does it fit? Per-slice SVI still leads on the staging sample, and the deep smoother and the GNO both land inside the SVI–SSVI window that Section 5.1 defined. The operator does so at amortised cost, which is the whole point of learning the smoothing map rather than refitting it every day.

What does it guarantee, and on what set? Only SSVI, and therefore the deep smoother's prior, carries a theorem. A soft penalty guarantees nothing beyond the grid it was evaluated on, and for an operator it guarantees nothing beyond the days it was trained on. That distinction, rather than the size of the penalty weight, is what separates the two halves of the table.

What does a failure cost? This is what the present chapter measured. The GNO buys its fit advantage with a local-variance surface invalid on half its domain, risk-neutral densities carrying up to 230% negative mass, and day-to-day instability of around 8 vp. The unreported path-dependent experiment described in the conclusion points the same way, and more sharply, but is not counted here.

What does it cost to run? The operators are four orders of magnitude faster per day than the per-day smoother, and that advantage is real. The main advantage of the GNO is therefore computational speed. The results above show, however, that this speed comes with weaker control of the no-arbitrage constraints on previously unseen days. The prior-embedded operator avoids this trade-off in the present experiments, combining fast evaluation with clean audit outcomes.

6.7.2 What we would tell a desk

(i) **Do not rely on a model's in-sample violation report alone.** A soft-constraint model is naturally evaluated on the same grid on which its penalty was imposed, so a low reported violation rate there provides limited evidence about its behaviour away from that grid. The relevant comparison is between the training nodes and an independent audit grid, together with the corresponding material violation rate.

(ii) **Use a certified prior multiplicatively when one exists.** The prior provides the protection and the corrector adds the accuracy, and the two do not trade off against each other (Tables 5.3 and 6.1).

(iii) **Project an operator's output before using it downstream.** If its speed is needed, apply one certified-prior corrector step, or the repairs of Section 5.1, before any Greek or density is computed from it. Monte-Carlo repricing should not be used as the sole diagnostic, because the repair step applied before simulation can mask violations present in the original surface (§6.2).

(iv) **Score smoothers on more than IV RMSE.** Repair rate, density mass and day-over-day stability all rank the families differently from the fit column, and the ranking inverts more than once between them.

(v) **Do not expect smoothing residuals to pay the touch on SPX.** The breakeven frontier sits at 3 to 5 vp and the market's residuals live below it (Finding 6.2). Their value is in marking and market-making, where mid-execution capture makes surface cleanliness worth about +0.4 vp per ten-day period on the staging counterfactual.

(vi) **Treat cleanliness as a requirement for path-dependent books.** A payoff that depends on the whole trajectory integrates the surface where the quotes do not constrain it, which is where soft penalties are least reliable. A surface indistinguishable from a certified one on the vanilla objective can therefore still misprice and mis-hedge an exotic. Project the surface before it reaches the book, not after the P&L exposes the problem.

Chapter 7

Conclusion

This thesis set out to benchmark three families of implied-volatility smoothers, and ended by changing the question the benchmark asks.

The parametric study came first. Implementing Gatheral and Jacquier's theory and reproducing their published values gave us both a validated toolkit and a reference range: per-slice SVI fits held-out quotes to 0.64 vp but leaks butterfly arbitrage on 12% of slices, while certified SSVI is arbitrage-free by Theorem 2.25 and pays roughly three times the error for it. Any learning-based method therefore needs to operate within this range to offer a meaningful improvement over the parametric baselines.

The deep smoother landed inside it, and produced the observation around which the rest of the thesis was organised. On a stress designed to force a calendar violation, the calendar penalty read exactly zero at its collocation nodes while an independent audit found violations on 28% of the domain. Both numbers were correct. The cause was the grid: exponentially spaced maturities, which the literature recommends because the butterfly constraint bites at the short end, leave a gap of nearly half a year at the long end, and the optimiser satisfied the penalty at every node while letting the surface dip between them. Theorem 3.2 formalises this mechanism by bounding the size of a violation that can remain hidden between neighbouring nodes. Serving each constraint from its own grid reduced material violations from 34.4% to zero on the same stress.

The operator study then showed that the problem is not confined to one architecture or to one kind of grid. A soft penalty only controls the locations at which it is evaluated. For an operator, that limitation extends across days as well, since satisfying the penalty on training surfaces does not guarantee the same behaviour on unseen ones. The most accurate model in the study, the GNO, is materially butterfly-violating on 355 of 386 out-of-sample days while its own training-grid audit reads clean.

Chapter 6 asked what that costs. Through $\sigma^2_{\text{loc}} = \partial_\tau w / g$, these violations translate directly into local-volatility defects: roughly half of the GNO surface requires repair, and the associated risk-neutral densities carry up to 230% negative mass. The same deterioration is visible in the residual study, where the clean surface outperforms its arbitrage-dirty counterpart in every matched backtest cell. A separate experiment showed that the market's own spread structure prices out residual mispricings below about three volatility points, so the value of a clean surface is in marking and market-making rather than in taker-side trading.

Taken together, these results expose a clear trade-off: the model with the best vanilla fit is also the one with the weakest downstream validity. The prior-embedded operator of Chapter 5 addresses it. Fitting a certified SSVI prior per day, and letting the operator learn only a bounded multiplicative correction, removes every material violation across all 386 test days while improving the fit from 2.69 to 1.32 vp. In this experiment the added protection therefore comes without a loss of fit.

Main implications. Three points seem to us to generalise beyond this data set. First, a soft no-arbitrage constraint is only as informative as the set on which it was evaluated, so independent auditing must extend beyond both the collocation nodes and the training dates. Second, where a closed-form certificate exists, incorporating it directly into the model provides stronger protection than approximating it with a penalty. Finally, the audit should be evaluated in downstream pricing quantities rather than only through violation counts, because the same derivatives controlled by the no-arbitrage conditions enter directly into densities and local volatility.

The economic reading is more specific than we expected at the outset. Arbitrage violations in a fitted surface are not a trading opportunity the market failed to remove: our scanner found essentially no executable static arbitrage in eight years of SPX quotes. They are better understood as a source of model risk, one that becomes most visible when the surface is used for path-dependent products, because such payoffs depend on the surface in regions that vanilla quotes constrain only weakly.

Future work. The results suggest three directions for further work.

The first priority is to complete and verify the experiments that are already in place. Notebook 06 contains a path-dependent experiment: a down-and-out call priced under an independent Heston ground truth, with a finite-difference cross-check and a paired hedging simulation. Its preliminary results are consistent with the findings reported above and suggest that the downstream effects may be larger for path-dependent products. Its numerical components could not be verified to the standard applied to the rest of this thesis in the time available, so we have not reported them as findings, and completing that verification is the immediate next step.

The same applies to several larger experiments that were begun but not finished. These include the five-year production run, which is intended to replace the staging tables and to test the "butterfly binds in stress" hypothesis on the Volmageddon and COVID weeks, the full residual-economics sample, and the VIX variance-swap replication started in Notebook 05. The last of these is particularly useful, since it would provide an external benchmark for the behaviour of the fitted wings.

The second direction is to test the framework in other markets. Much of the pipeline, including the cleaning, the independent audit and the certified-prior architecture, can be carried over to bitcoin options. That market provides a useful contrast with SPX: quotes are generally sparser, spreads wider, maturities shorter, and static-arbitrage violations potentially more prevalent. It may also change the economics of the frontier result, since wider spreads raise the breakeven while larger and more persistent mispricings may clear it. A particularly interesting question is whether the prior-embedded architecture retains the same protection where SSVI provides a weaker description of the underlying smile.

The third direction concerns new model architectures. Several extensions suggested by Dr Arthur Bizzi remain to be explored. One possibility is to transport operator constraints across days, either by penalising on generated query days or by introducing certified projection layers. Another is to complete the prior-embedded family with the graph operator, whose training run could not be retained in the present study. A third is to smooth local volatility directly, where Proposition 2.12 makes the constraint set explicit and Theorem 3.2 provides a natural starting point for adaptive collocation with a provable node-versus-audit bound. The recent non-parametric SANOS framework [28] offers another natural setting in which to test the independent-audit methodology developed here.

Bibliography

[1] D. Ackerer, N. Tagasovska, T. Vatter. *Deep Smoothing of the Implied Volatility Surface.* Advances in Neural Information Processing Systems (NeurIPS), 2020.

[2] A. G. Baydin, B. A. Pearlmutter, A. A. Radul, J. M. Siskind. *Automatic Differentiation in Machine Learning: a Survey.* Journal of Machine Learning Research 18(153), 2018.

[3] D. T. Breeden, R. H. Litzenberger. *Prices of State-Contingent Claims Implicit in Option Prices.* Journal of Business 51(4), 1978.

[4] M. Chataigner, S. Crépey, M. Dixon. *Deep Local Volatility.* Risks 8(3), 2020.

[5] T. Chen, H. Chen. *Universal Approximation to Nonlinear Operators by Neural Networks with Arbitrary Activation Functions and Its Application to Dynamical Systems.* IEEE Transactions on Neural Networks 6(4), 1995.

[6] A. M. G. Cox, D. Hobson. *Local Martingales, Bubbles and Option Prices.* Finance and Stochastics 9(4), 2005.

[7] X. Duan, Q. Liu, Z. Xu, Z. Ying, X. Zhang. *Option Implied Volatility and Trading Strategies Based on Neural Network Correction.* Journal of Futures Markets 46(1), 2026.

[8] C. Dugas, Y. Bengio, F. Bélisle, C. Nadeau, R. Garcia. *Incorporating Second-Order Functional Knowledge for Better Option Pricing.* Advances in Neural Information Processing Systems (NIPS), 2001.

[9] B. Dupire. *Pricing with a Smile.* Risk 7(1), 1994.

[10] V. Durrleman. *From Implied to Spot Volatilities.* Finance and Stochastics 14(2), 2010.

[11] M. R. Fengler. *Arbitrage-Free Smoothing of the Implied Volatility Surface.* Quantitative Finance 9(4), 2009.

[12] J. Gatheral. *A Parsimonious Arbitrage-Free Implied Volatility Parameterization with Application to the Valuation of Volatility Derivatives.* Global Derivatives & Risk Management, 2004.

[13] J. Gatheral. *The Volatility Surface: A Practitioner's Guide.* Wiley Finance, 2006.

[14] J. Gatheral, A. Jacquier. *Arbitrage-Free SVI Volatility Surfaces.* Quantitative Finance 14(1), 2014.

[15] K. Hornik. *Approximation Capabilities of Multilayer Feedforward Networks.* Neural Networks 4(2), 1991.

[16] K. Hoshisashi, C. E. Phelan, P. Barucca. *No-Arbitrage Deep Calibration for Volatility Smile and Skewness.* arXiv:2310.16703, 2023.

[17] J. M. Hutchinson, A. W. Lo, T. Poggio. *A Nonparametric Approach to Pricing and Hedging Derivative Securities via Learning Networks.* Journal of Finance 49(3), 1994.

[18] N. Kahale. *An Arbitrage-Free Interpolation of Volatilities.* Risk 17(5), 2004.

[19] H. G. Kellerer. *Markov-Komposition und eine Anwendung auf Martingale.* Mathematische Annalen 198, 1972.

[20] N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, A. Anandkumar. *Neural Operator: Learning Maps Between Function Spaces with Applications to PDEs.* Journal of Machine Learning Research 24(89), 2023.

[21] I. E. Lagaris, A. Likas, D. I. Fotiadis. *Artificial Neural Networks for Solving Ordinary and Partial Differential Equations.* IEEE Transactions on Neural Networks 9(5), 1998.

[22] R. W. Lee. *The Moment Formula for Implied Volatility at Extreme Strikes.* Mathematical Finance 14(3), 2004.

[23] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar. *Neural Operator: Graph Kernel Network for Partial Differential Equations.* arXiv:2003.03485, 2020.

[24] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, A. Anandkumar. *Fourier Neural Operator for Parametric Partial Differential Equations.* International Conference on Learning Representations (ICLR), 2021.

[25] L. Lu, P. Jin, G. Pang, Z. Zhang, G. E. Karniadakis. *Learning Nonlinear Operators via DeepONet Based on the Universal Approximation Theorem of Operators.* Nature Machine Intelligence 3, 2021.

[26] M. Raissi, P. Perdikaris, G. E. Karniadakis. *Physics-Informed Neural Networks: A Deep Learning Framework for Solving Forward and Inverse Problems Involving Nonlinear Partial Differential Equations.* Journal of Computational Physics 378, 2019.

[27] M. Roper. *Arbitrage Free Implied Volatility Surfaces.* Preprint, University of Sydney, 2010.

[28] H. Buehler, B. Horvath, A. Kratsios, Y. Limmer, R. Saqur. *SANOS: Smooth Strictly Arbitrage-Free Non-Parametric Option Surfaces.* arXiv:2601.11209, 2026.

[29] R. Wiedemann, A. Jacquier, L. Gonon. *Operator Deep Smoothing for Implied Volatility.* International Conference on Learning Representations (ICLR), 2025.

[30] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. Salakhutdinov, A. J. Smola. *Deep Sets.* Advances in Neural Information Processing Systems (NIPS), 2017.

Appendix A

Deferred proofs

This appendix collects the proofs deferred from Chapter 2. The division follows the policy stated there: proofs carrying an idea the reader needs in order to follow the argument, together with the two results proved in this thesis (Theorem 2.25 and Theorem 3.2), are given in the body of the text; proofs consisting of verification are reproduced here in full. Nothing is omitted, and each proof below is hyperlinked to its statement.

A.1 Static arbitrage and the audit quantities

A.1.1 Derivation of the Black formula (2.2)

Statement: equation (2.2), page 13.

Proof. $C/D = F\,\mathbb{E}[(M_T - e^k)^+]$ with $k = \log(K/F)$. On the event $\{M_T > e^k\} = \{Z > (k + v/2)/\sqrt{v}\} = \{Z > -d_-\}$ (with $v = \sigma^2\tau$),

$$\mathbb{E}[M_T \mathbf{1}_{\{Z>-d_-\}}] = \int_{-d_-}^{\infty} e^{\sqrt{v}z - v/2}\varphi(z)\,\mathrm{d}z = \int_{-d_-}^{\infty} \varphi(z - \sqrt{v})\,\mathrm{d}z = N(d_- + \sqrt{v}) = N(d_+),$$

by completing the square, while $\mathbb{E}[e^k \mathbf{1}_{\{Z>-d_-\}}] = e^k N(d_-)$. Multiplying by F and noting $Fe^k = K$ gives (2.2). $\square$

A.1.2 Proof of Lemma 2.3 (Black partials in (k, w))

Statement: Lemma 2.3, page 13.

Proof. Throughout, $\partial_k d_\pm = -1/\sqrt{w}$ and $\partial_w d_\pm = \dfrac{k}{2w^{3/2}} \pm \dfrac{1}{4\sqrt{w}}$.

(i) $c_k = \varphi(d_+)\partial_k d_+ - e^k N(d_-) - e^k\varphi(d_-)\partial_k d_- = -e^k N(d_-) - \dfrac{1}{\sqrt{w}}\big(\varphi(d_+) - e^k\varphi(d_-)\big) = -e^k N(d_-)$, the last step by the identity (2.5).

(ii) $c_w = \varphi(d_+)\partial_w d_+ - e^k\varphi(d_-)\partial_w d_- = e^k\varphi(d_-)\big(\partial_w d_+ - \partial_w d_-\big) = \dfrac{e^k\varphi(d_-)}{2\sqrt{w}} > 0$.
Since $\partial w/\partial\sigma = 2\sigma\tau$, this recovers the classical Black vega $DF\varphi(d_+)\sqrt{\tau}$.

(iii) Differentiating (2.6) in k gives $c_{kk} = -e^k N(d_-) - e^k\varphi(d_-)\partial_k d_- = c_k + \dfrac{e^k\varphi(d_-)}{\sqrt{w}}$.

(iv) Differentiating (2.6) in w gives $c_{kw} = -e^k\varphi(d_-)\partial_w d_- = -e^k\varphi(d_-)\big(\dfrac{k}{2w^{3/2}} - \dfrac{1}{4\sqrt{w}}\big) = \dfrac{e^k\varphi(d_-)}{2\sqrt{w}}\big(\dfrac{1}{2} - \dfrac{k}{w}\big) = \big(\dfrac{1}{2} - \dfrac{k}{w}\big)c_w$.

(v) Differentiating (2.7) in w and using $\varphi'(x) = -x\varphi(x)$ gives

$$c_{ww} = e^k\left[\frac{-d_-\varphi(d_-)\partial_w d_-}{2\sqrt{w}} - \frac{\varphi(d_-)}{4w^{3/2}}\right] = c_w\left[(-d_-)\partial_w d_- - \frac{1}{2w}\right].$$

Since $-d_- = \dfrac{k}{\sqrt{w}} + \dfrac{\sqrt{w}}{2}$, then $(-d_-)\partial_w d_- = \left(\frac{k}{\sqrt{w}} + \frac{\sqrt{w}}{2}\right)\left(\frac{k}{2w^{3/2}} - \frac{1}{4\sqrt{w}}\right) = \frac{k^2}{2w^2} - \frac{1}{8}$.

□

A.1.3 Proof of Proposition 2.5 (implied density in log-moneyness)

Statement: Proposition 2.5, page 15. Step 1 is also given in the body of the text, where it produces equation (2.13); it is repeated here so that the proof is self-contained.

Proof. **Step 1: Density in log-moneyness.** By Proposition 2.2(c), the undiscounted density of S_T at K is $\partial_{KK}(C/D)$. Since $K = Fe^k$, then $q(k) = K\,\partial_{KK}(C/D)$. Using $C/D = Fc(k)$ and $\partial_K = K^{-1}\partial_k$,

$$\partial_{KK}(C/D) = \frac{1}{K}\partial_k\left[\frac{F}{K}\partial_k c\right] = \frac{F}{K^2}\big(\partial_{kk}c - \partial_k c\big),$$

hence $q(k) = e^{-k}\big(\partial_{kk}c - \partial_k c\big)$, which is (2.13).

Step 2: Total derivatives. For $c(k) = c(k, w(k))$,

$$\partial_k c = c_k + c_w w', \qquad \partial_{kk}c = c_{kk} + 2c_{kw}w' + c_{ww}(w')^2 + c_w w''.$$

Step 3: Substitution. Using Lemma 2.3 in (2.13),

$$q(k) = e^{-k}c_w\left[2\sqrt{w} - \frac{2kw'}{w} + \left(\frac{k^2}{2w^2} - \frac{1}{2w} - \frac{1}{8}\right)(w')^2 + w''\right].$$

Since $e^{-k}c_w = \varphi(d_-)/(2\sqrt{w})$,

$$q(k) = \frac{\varphi(d_-)}{\sqrt{w}}\left[1 - \frac{kw'}{w} + \left(\frac{k^2}{4w^2} - \frac{1}{4w} - \frac{1}{16}\right)(w')^2 + \frac{w''}{2}\right].$$

The bracket is precisely $g(k)$ from (2.11), so

$$q(k) = g(k)\frac{\varphi(d_-(k))}{\sqrt{w(k)}},$$

which is (2.12). Since $\varphi(d_-)/\sqrt{w} > 0$, the sign of q is the sign of g. □

A.1.4 Proof of Lemma 2.6 (butterfly characterisation)

Statement: Lemma 2.6, page 15.

Proof. By Proposition 2.5, $g \geq 0$ is equivalent to the convexity of the call slice in strike, i.e. absence of interior butterfly arbitrage. This is (i).

For (ii): using (2.4) and (2.5) with the Mill's-ratio asymptotic $N(x) \sim \varphi(x)/|x|$ as $x \to -\infty$,

$$c = N(d_+) - e^k N(d_-) \quad \text{with} \quad e^k N(d_-) \sim e^k \frac{\varphi(d_-)}{|d_-|} = \frac{\varphi(d_+)}{|d_-|}$$

whenever $d_- \to -\infty$. If $d_+ \to -\infty$ then both $N(d_+) \to 0$ and $\varphi(d_+)/|d_-| \to 0$, so $c \to 0$: the total mass of q is 1 and no mass escapes to $+\infty$. Conversely, if $d_+ \not\to -\infty$ along some subsequence then $N(d_+)$ stays bounded away from 0 faster than $e^k N(d_-)$ compensates and $c(k)$ does not vanish; a nonvanishing call price at infinite strike is the classical arbitrage of selling calls of arbitrarily large strike at a price bounded away from zero (see [27] and [14, §2] for the complete argument, including the local-martingale subtleties of [6]). $\square$

A.1.5 Proof of Proposition 2.9 (calendar characterisation)

Statement: Proposition 2.9, page 16. The equivalence $(a) \iff (b)$ is also proved in the body of the text; it is repeated here so that the proof is self-contained.

Proof. (i) **Proof of $(a) \iff (b)$.** By Lemma 2.3, $c_w > 0$, so at fixed k the Black price is strictly increasing in total variance. Hence $\tau \mapsto w(k,\tau)$ is non-decreasing if and only if $\tau \mapsto c(k,\tau)$ is non-decreasing.

(ii) **Proof of $(c) \Rightarrow (b)$.** Let (M_u) be a nonnegative martingale with $M_t = 1$ representing the forward-normalised underlying, so that $c(k,\tau_i) = \mathbb{E}\big[(M_{T_i} - e^k)^+\big]$. For $\tau_1 < \tau_2$, conditional Jensen's inequality applied to the convex function $x \mapsto (x - e^k)^+$ gives

$$\begin{aligned}
\mathbb{E}\big[(M_{T_2} - e^k)^+\big] &= \mathbb{E}\Big[\mathbb{E}\big[(M_{T_2} - e^k)^+ \mid \mathcal{F}_{T_1}\big]\Big] \\
&\geq \mathbb{E}\Big[\big(\mathbb{E}[M_{T_2} \mid \mathcal{F}_{T_1}] - e^k\big)^+\Big] \\
&= \mathbb{E}\big[(M_{T_1} - e^k)^+\big].
\end{aligned}$$

Therefore, $c(k,\tau_2) \geq c(k,\tau_1)$.

(iii) **Proof of $(b) \Rightarrow (c)$.** Assume that (b) holds and that every slice is free of butterfly arbitrage. The marginal laws implied by the slices are then increasing in convex order. Kellerer's theorem [19] yields a martingale with these marginals, so no static portfolio can arbitrage the surface.

Conversely, suppose that (b) fails for some $\tau_1 < \tau_2$ and k, so that $c(k,\tau_2) - c(k,\tau_1) < 0$. Consider a long T_2-call and a short T_1-call with the same forward moneyness k. The portfolio has strictly negative initial cost. At T_1, the short leg pays $-(M_{T_1} - e^k)^+$, while the value of the long leg is at least its intrinsic value by Proposition 2.2(b). The portfolio therefore has nonnegative value at T_1 and defines a static arbitrage. The comparison uses equal moneyness rather than equal strike; in discounted forward units, the strikes coincide.

□

A.1.6 Proof of Proposition 2.11 (Dupire's equation)

Statement: Proposition 2.11, page 17.

Proof. The Kolmogorov forward (Fokker–Planck) equation for the density of the driftless diffusion is

$$\partial_T p(T,K) = \tfrac{1}{2}\,\partial_{KK}\big[\sigma_{\text{loc}}^2(T,K)\,K^2\,p(T,K)\big].$$

Write $C(K,T) = \int_0^\infty (s-K)^+ p(T,s)\,\mathrm{d}s = \int_K^\infty (s-K)p(T,s)\,\mathrm{d}s$ and differentiate under the integral:

$$\partial_T C = \int_K^\infty (s-K)\,\partial_T p(T,s)\,\mathrm{d}s = \tfrac{1}{2}\int_K^\infty (s-K)\,\partial_{ss}\big[\sigma_{\text{loc}}^2 s^2 p\big]\,\mathrm{d}s.$$

Integrating by parts twice, with the boundary terms at $s=\infty$ vanishing by assumption and those at $s=K$ vanishing because $(s-K)$ and its derivative are evaluated where $(s-K)=0$ (first integration) and the surviving term is picked up at $s=K$ (second integration):

$$\int_K^\infty (s-K)\,\partial_{ss}\big[\sigma^2 s^2 p\big]\,\mathrm{d}s = \Big[(s-K)\,\partial_s(\sigma^2 s^2 p)\Big]_K^\infty - \int_K^\infty \partial_s(\sigma^2 s^2 p)\,\mathrm{d}s = 0 - \Big[\sigma^2 s^2 p\Big]_K^\infty = \sigma_{\text{loc}}^2(T,K)K^2 p(T,K).$$

Since $p(T,K) = \partial_{KK}C(K,T)$ (Proposition 2.2(c)), (2.14) follows. □

A.2 SVI and SSVI

A.2.1 Proof of Lemma 2.16 (Jump-Wings from raw)

Statement: Lemma 2.16, page 20.

Proof. $v_t t = w(0)$ follows by direct evaluation of (2.16) at $k=0$. ψ_t is by definition $w'(0)/(2\sqrt{w_t})$; insert (2.17) at $k=0$. p_t and c_t are the normalised wing slopes $\lim_{k\to\mp\infty}|w'(k)|/\sqrt{w_t}$; by (2.18) these are $b(1\mp\rho)/\sqrt{w_t}$ with the stated sign convention. For $\tilde{v}_t$: minimise $u \mapsto \rho u + \sqrt{u^2+\sigma^2}$ $(u = k-m)$. The first-order condition $\rho + u/\sqrt{u^2+\sigma^2} = 0$ gives $u^\star = -\rho\sigma/\sqrt{1-\rho^2}$, whence $\sqrt{(u^\star)^2+\sigma^2} = \sigma/\sqrt{1-\rho^2}$ and the minimum value $\rho u^\star + \sigma/\sqrt{1-\rho^2} = \sigma\sqrt{1-\rho^2}$; therefore $\min_k w = a + b\sigma\sqrt{1-\rho^2} = \tilde{v}_t\,t$. □

A.2.2 Proof of Lemma 2.17 (raw from Jump-Wings)

Statement: Lemma 2.17, page 20.

Proof sketch. Adding and subtracting the wing equations of (2.23) gives b and ρ directly. The skew equation then determines $m/\sqrt{m^2+\sigma^2} = \rho - 2\psi_t\sqrt{w_t}/b = \beta$, i.e. $\sigma = \alpha m$ with α as stated (the sign bookkeeping selects the branch). Finally $v_t - \tilde{v}_t$ measures the gap between ATM and minimum variance, which in raw parameters is $b\{-\rho m + \sqrt{m^2+\sigma^2} - \sigma\sqrt{1-\rho^2}\}$; substituting $\sigma = \alpha m$ and solving for m yields (2.25), and a follows from $\tilde{v}_t t = a + b\sigma\sqrt{1-\rho^2}$ (Lemma 2.16). ☐

A.2.3 Proof of Lemma 2.19 (SSVI slices are SVI slices)

Statement: Lemma 2.19, page 22.

Proof. With the candidate parameters (2.27), $k - m = k + \rho/\varphi$, so

$$b\,\rho(k-m) = \frac{\theta\varphi}{2}\,\rho\Big(k + \frac{\rho}{\varphi}\Big) = \frac{\theta}{2}\big(\rho\varphi k + \rho^2\big), \qquad b\sqrt{(k-m)^2+\sigma^2} = \frac{\theta}{2}\sqrt{(\varphi k+\rho)^2 + 1 - \rho^2},$$

and adding $a = \theta(1-\rho^2)/2$ reproduces (2.26) exactly. ☐

A.2.4 Proof of Proposition 2.20 (calendar certificate)

Statement: Proposition 2.20, page 23.

Proof. By Proposition 2.9 and the chain rule $\partial_\tau w = \partial_\theta w \cdot \partial_\tau \theta_\tau$, calendar-freeness is equivalent to (i) together with $\partial_\theta w(k,\theta) \geq 0$ for all (k,θ). Write $\psi(\theta) := \theta\varphi(\theta)$ and multiply (2.26) by 2:

$$2w(k,\theta) = \theta + \rho\,\psi(\theta)\,k + \sqrt{\big(\psi(\theta)k + \rho\theta\big)^2 + (1-\rho^2)\theta^2}.$$

Differentiating in θ, with $R := \sqrt{(\psi k + \rho\theta)^2 + (1-\rho^2)\theta^2}$, we obtain

$$2\,\partial_\theta w = 1 + \rho\,\psi'(\theta)\,k + \frac{(\psi k + \rho\theta)\big(\psi'(\theta)k + \rho\big) + (1-\rho^2)\theta}{R}. \tag{A.1}$$

Necessity of $\psi' \geq 0$. As $k \to +\infty$, $R \sim \psi k$ and (A.1) $\sim \rho\psi' k + \psi' k = (1+\rho)\,\psi'\,k$; as $k \to -\infty$, $R \sim \psi|k|$ and (A.1) $\sim \rho\psi' k + \psi'|k| = (1-\rho)\,\psi'\,|k|$. Since $1 \pm \rho > 0$, nonnegativity of $\partial_\theta w$ in both wings forces $\psi'(\theta) \geq 0$; this is the left inequality of (ii).

The sharp upper bound. Assume $\psi' \geq 0$ and set $x := \psi(\theta)k + \rho\theta$, which defines a bijection in k for fixed θ. A direct computation reduces the problem to minimising a one-variable function of x. The minimiser is obtained by setting the x-derivative of (A.1) to zero, which yields a quadratic equation. Requiring the resulting minimum to be nonnegative shows that $\inf_k \partial_\theta w \geq 0$ holds precisely when

$$\theta\,\frac{\psi'(\theta)}{\psi(\theta)} \;\leq\; \frac{1+\sqrt{1-\rho^2}}{\rho^2}, \qquad \text{i.e.} \quad \psi'(\theta) \leq \frac{1}{\rho^2}\big(1 + \sqrt{1-\rho^2}\big)\varphi(\theta),$$

the right inequality of (ii); the bound is attained (the constant $(1+\sqrt{1-\rho^2})/\rho^2$ is sharp), so (ii) is also necessary. The complete algebra of the minimisation is in the proof of [14, Thm. 4.1]; the boundary cases $\rho = \pm 1$ are treated there by a separate limiting argument, and $\rho = 0$ makes the upper bound vacuous ($+\infty$), consistent with (A.1) being manifestly nonnegative when $\rho = 0$ and $\psi' \geq 0$. $\square$

A.2.5 Proof of Proposition 2.22 (butterfly certificate)

Statement: Proposition 2.22, page 23.

Proof. Fix θ and use Lemma 2.19: the slice is raw SVI with $b = \theta\varphi/2$ and correlation ρ.

(i) **The first condition is the wing condition.** By (2.18), the wing slopes are $b(1 \pm \rho)$, so

$$b(1+|\rho|) = \frac{\theta\varphi(\theta)}{2}(1+|\rho|) < 2 \iff \theta\varphi(\theta)(1+|\rho|) < 4.$$

Thus the first condition in (2.28) is exactly the strict Lee wing bound (2.19). It yields condition (ii) of Lemma 2.6, namely $d_+ \to -\infty$ and the vanishing of call prices at extreme strikes.

(ii) **The second condition ensures $g \geq 0$.** Substituting the slice parameters (2.27) into the Durrleman function (2.11) and setting $z := \varphi(\theta)k$ reduces $g \geq 0$ on $\mathbb{R}$ to the nonnegativity of an explicit rational-algebraic function of z. Its coefficients depend only on $\theta\varphi^2$ and ρ. As shown in [14, Thm. 4.2], the condition $\theta\varphi(\theta)^2(1+|\rho|) \leq 4$ together with the wing bound from step (i) ensures that this function is nonnegative for every z.
The quantity $\theta\varphi^2$ is the natural curvature scale of the slice. Indeed, from (2.27),

$$\sigma = \frac{\sqrt{1-\rho^2}}{\varphi}, \qquad w''(m) = \frac{b}{\sigma} = \frac{\theta\varphi^2}{2\sqrt{1-\rho^2}}.$$

The second condition therefore controls the curvature near the money relative to the level θ, which is precisely the combination through which the terms amplified by $1/w$ in (2.11) may make g negative.

The full algebraic computation of step (ii) is carried out line by line in the proof of [14, Thm. 4.2, pp. 14–15] and is not reproduced here. In this thesis it is instead encoded directly: Propositions 2.20–2.22 are implemented as executable certificates and verified against the paper's own condition values (Section 5.1). $\square$